

# Validation of a Hybrid Multiphase Model for Electrolysis

Validierung eines hybriden Mehrphasenmodells für die Elektrolyse

Master Thesis of

**Alexandre SIGIRIDIS-SCHLEPP**

At the Department of Mechanical Engineering,  
Institute of Fluid Mechanics

Advisors: Prof. Dr.-Ing. Bettina Frohnapfel  
Dr.-Ing. Alexander Stroh  
M.Sc. Ludwig Bossle

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I declare that I have developed and written the enclosed thesis completely by myself, and have not used sources or means without declaration in the text.

**Karlsruhe, 08. 09. 2026**

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(Alexandre SIGIRIDIS-SCHLEPP)



# **Declaration of the Use of Generative Artificial Intelligence**

Generative artificial intelligence (ChatGPT and Claude) was used as a supporting tool to translate and improve the English formulation of text written by the author, in French or in English. All scientific content, modelling choices, simulations, analysis, and conclusions are the author's own work. References were selected and inserted by the author, and all scientific claims were verified independently.



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# Abstract

This thesis evaluates the MultiMorph hybrid multiphase model for confined gas–liquid flows relevant to PEM water electrolysis. The model combines an unresolved Euler–Euler representation of small gas structures with a continuous, VOF-like representation of larger interfaces. Its ability to predict pressure drop, gas morphology, phase distribution, and near-wall flow behaviour is assessed using three experimental reference configurations from Cubaud and Ho, Lafmejani et al., and Kim and Jung. The results show that MultiMorph reproduces the main changes in gas morphology over a broad range of conditions. In the Lafmejani configuration, gas introduced through the dispersed field can accumulate and form larger continuous structures after morphology transfer. The location of this transition remains sensitive to the dispersed bubble diameter, the continuous threshold, and the morphology-transfer time scale. The pressure-drop results are more restrictive. The Cubaud calculations agree well with the experimental trend while the liquid still occupies a substantial part of the channel. At low liquid fractions, the agreement deteriorates as the remaining liquid becomes confined to thin wall films and corner regions. Near-wall mesh refinement changes the predicted pressure drop significantly, but further refinement becomes numerically unstable before a grid-converged film solution can be obtained. A modified wall-wet boundary condition changes the near-wall phase distribution but does not recover the missing film-scale momentum transport. The main limitation identified in this work is therefore associated with the liquid scale rather than with the formation of large gas structures. MultiMorph can represent gas structures that span many computational cells while the much thinner liquid layer controlling the near-wall hydraulic response remains under-resolved. An unresolved liquid-film treatment is proposed as a possible extension of the present model. Such an extension would be particularly relevant for PEM electrolyzer simulations, where gas generated at the porous transport layer can accumulate into large channel-scale structures while the remaining liquid is confined to thin near-wall regions.



# Kurzzusammenfassung

Diese Arbeit untersucht das hybride Mehrphasenmodell MultiMorph für eingeschlossene Gas-Flüssigkeitsströmungen, wie sie bei der PEM-Wasserelektrolyse auftreten. Das Modell kombiniert eine nicht aufgelöste Euler-Euler-Beschreibung kleiner Gasstrukturen mit einer kontinuierlichen, VOF-ähnlichen Darstellung größerer Grenzflächen. Die Vorhersage von Druckverlust, Gasmorphologie, Phasenverteilung und wandnahe Strömung wird anhand von drei experimentellen Referenzkonfigurationen nach Cubaud und Ho, Lafmejani et al. sowie Kim und Jung untersucht. Die Ergebnisse zeigen, dass MultiMorph die wesentlichen Änderungen der Gasmorphologie über einen großen Bereich der untersuchten Bedingungen wiedergeben kann. In der Lafmejani-Konfiguration kann das über das disperse Feld eingebrachte Gas akkumulieren und nach dem Morphologietransfer größere kontinuierliche Strukturen bilden. Die Position dieses Übergangs reagiert jedoch empfindlich auf den angenommenen Blasendurchmesser, den Continuous Threshold und die Zeitskala des Morphologietransfers. Bei der Vorhersage des Druckverlusts treten deutlich stärkere Einschränkungen auf. Für die Cubaud-Konfiguration stimmen die Simulationen gut mit dem experimentellen Trend überein, solange noch ein wesentlicher Flüssigkeitsanteil im Kanal vorhanden ist. Bei kleinen Flüssigkeitsanteilen verschlechtert sich die Übereinstimmung, wenn die verbleibende Flüssigkeit hauptsächlich in dünnen Wandfilmen und in den Kanalecken transportiert wird. Eine Verfeinerung des Gitters in Wandnähe verändert den berechneten Druckverlust deutlich. Eine weitere Verfeinerung führt jedoch zu numerischen Instabilitäten, bevor eine gitterunabhängige Lösung des Flüssigkeitsfilms erreicht wird. Eine modifizierte Wall-Wet-Randbedingung verändert zwar die wandnahe Phasenverteilung, stellt jedoch den fehlenden Impulstransport auf der Skala des Flüssigkeitsfilms nicht wieder her. Die in dieser Arbeit identifizierte wesentliche Einschränkung betrifft damit eher die Flüssigkeitsskala als die Bildung großer Gasstrukturen. MultiMorph kann Gasstrukturen darstellen, die sich über viele Rechenzellen erstrecken, während die deutlich dünnere Flüssigkeitsschicht, welche die wandnahe hydraulische Antwort beeinflusst, weiterhin unteraufgelöst bleibt. Als mögliche Erweiterung wird daher ein nicht aufgelöstes Flüssigkeitsfilmmodell vorgeschlagen.



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# 1. Introduction

The decarbonization of energy systems requires both the expansion of renewable electricity and the development of energy carriers suitable for sectors that cannot directly be electrified (such as heavy transport, high-temperature industrial processes, steel production etc.). Under the 2015 Paris Agreement, the global objective is to keep the temperature increase lower than 2°C in comparison with the preindustrial era. However, fossil CO<sub>2</sub> emissions remain at historically high levels, which highlights the need for additional low-carbon energy conversion and storage technologies. [19]

## 1.1. Green hydrogen and water electrolysis

The most discussed and studied alternative to fossil fuel is hydrogen. Hydrogen is often considered a promising energy carrier and feedstock for sectors where direct electrification is difficult. Hydrogen can be used as a chemical feedstock, as a fuel, and also as a long-term storage medium for renewable electricity [7]. Nevertheless, hydrogen is not automatically low-carbon. Most of the current global hydrogen supply is still produced from fossil resources, reducing its positive effect, whereas low-emissions hydrogen accounts for only a small fraction of total production [20]. This motivates the development of green hydrogen production routes, in particular water electrolysis powered by low-carbon electricity [1]. Among water electrolysis technologies, proton exchange membrane water electrolysis is especially attractive because of its compactness, high current density, and dynamic response. In a PEM electrolyzer, liquid water is supplied to the anode side, where oxygen is produced, while protons are transported through the membrane and hydrogen is produced at the cathode. The anode side therefore contains a gas-liquid two-phase flow: water must be transported towards the porous transport layer and catalyst layer, while oxygen bubbles must be removed through the flow channels. Figure 1.1 summarizes the main transport paths in a PEM water electrolyzer. Water is supplied on the anode side, oxygen is removed through the anode flow field, protons cross the membrane, and hydrogen is produced at the cathode. The present work focuses on the resulting gas-liquid transport on the anode side.

This two-phase flow is not only a secondary detail of the process. Gas accumulation can obstruct the liquid path, reduce the effective water supply to the reaction sites, and lead to non-uniform operating conditions [29]. At the channel scale, the presence of gas modifies several phenomena in the flow, as discussed in the following chapters. Those phenomena lead to a higher operating cost and a decrease in efficiency.

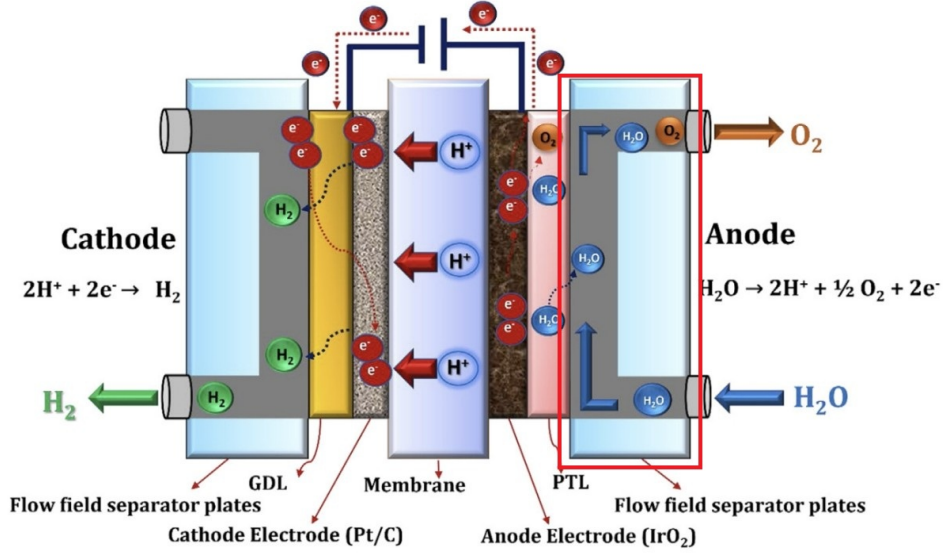


Figure 1.1: Schematic representation of a proton exchange membrane water electrolyzer. [40]

## 1.2. Numerical modelling challenge

The gas-liquid flow described above is difficult to predict numerically because the relevant phenomena occur at different spatial scales. In an electrolyzer channel, oxygen may appear as small bubbles, larger gas structures or locally accumulated gas regions. These structures influence the liquid distribution, the pressure losses, and the flow close to the walls. A numerical model must therefore not only represent the presence of gas and liquid, but also reproduce how their spatial distribution affects the hydraulic behavior of the channel. An inaccurate description of the gas distribution can lead to wrong predictions of the liquid pathways, the wall shear stress, and the additional pressure losses caused by the two-phase flow. Therefore, the ability of a model to reproduce these effects is crucial before it can be used for the analysis or design of electrolyzer flow channels.

## 1.3. Objectives of the thesis

The objective of this thesis is to evaluate whether the hybrid multiphase model MultiMorph [30] can predict the main gas-liquid flow features relevant to water electrolysis. The focus is placed on pressure-drop prediction, gas-liquid morphology, void fraction, and near-wall effects such as liquid films, corner flow, and wall shear stress.

For this purpose, numerical simulations are compared with selected reference cases from the literature. In addition, the influence of selected model parameters and boundary conditions is investigated to identify the strengths and limitations of the following numerical approach. The study does not aim to model the electrochemical reaction itself. Instead, it focuses on the hydraulic and multiphase-flow behavior in channel configurations relevant to electrolyzer applications.

## 1.4. Thesis structure

Chapter 2 introduces the gas-liquid flow mechanisms relevant to the study and the MultiMorph model. Chapter 3 describes the three reference experiments, the numerical setups, and the quantities used for comparison. The simulation results are presented in Chapter 4 for the Cubaud, Lafmejani, and Kim configurations. Chapter 5 discusses the pressure-drop

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prediction, morphology transitions, wall treatment, and numerical limitations identified from these cases. The main conclusions and possible further development of the model are given in Chapter 6.



## 2. Theory and fundamentals

The multiphase problem considered in this thesis originates from gas production inside a water-fed PEM electrolyzer. On the anode side, liquid water reaches the reaction region while oxygen is generated and transported towards the flow channel. Once both phases are present in the channel, their distribution is controlled not only by the imposed gas and liquid flow rates, but also by surface tension, phase slip, wall wetting, and the geometry of the channel [47, 14].

Experimental observations of PEM electrolyzers show that the gas is not distributed as a homogeneous mixture [36, 29]. Bubbles can accumulate, merge into larger structures, and occupy a significant part of the channel while liquid remains connected along the walls and through other available flow paths [18, 13]. The resulting two-phase flow pattern can also influence electrolyzer operation and performance. These structures modify the hydraulic behavior of the channel even when the electrochemical reaction itself is not modelled [22].

The present chapter therefore focuses on the gas–liquid flow after both phases are present. The first part introduces the confined two-phase-flow mechanisms used later to interpret the simulations. Resolved and unresolved numerical descriptions are then discussed before introducing the MultiMorph model and its treatment of the different gas morphologies.

### 2.1. Gas-liquid two-phase flow in confined channels

Gas–liquid flow in channels whose characteristic dimension approaches the capillary length differs from conventional large-scale pipe flow. The relevant confinement is determined by the balance between the channel size and the characteristic force scales. In particular, decreasing the hydraulic diameter reduces the Bond number,  $\text{Bo} \propto D_h^2$ , and, for a given velocity, the Weber number,  $\text{We} \propto D_h$  [51, 52]. Gravity and inertial deformation therefore become weaker relative to surface tension. The balance between viscous and capillary stresses, however, is governed by the capillary number,  $\text{Ca} = \mu_l U_c / \sigma$ , and does not depend directly on the channel size [39]. Gas–liquid flows can adopt several distinct phase arrangements, commonly referred to as flow regimes or flow patterns. The regime affects the phase distribution, the relative phase velocities, the wetted wall area, and the pressure gradient [52, 11].

#### 2.1.1. Flow regimes and morphology

Here, *morphology* means the local shape and connectivity of the gas and liquid phases. A phase may appear as separate bubbles, as elongated structures of about the channel

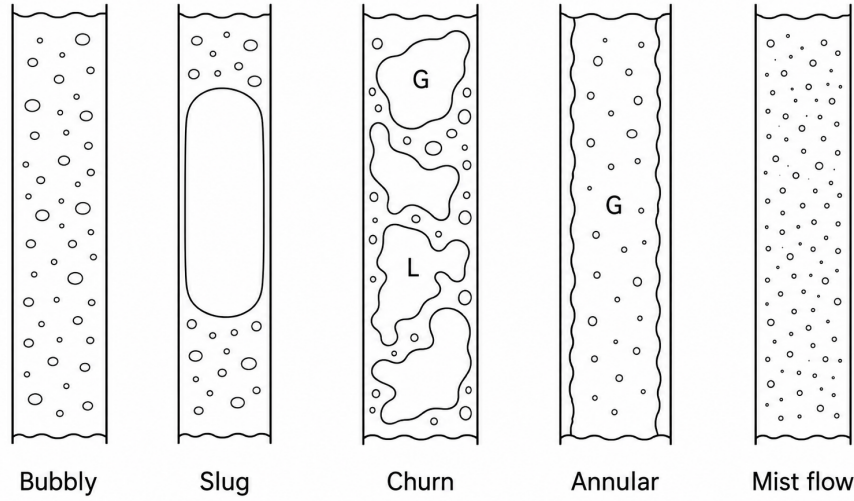


Figure 2.1: Schematic representation of the principal confined gas–liquid flow regimes. Author’s schematic, inspired by [51, 44].

width, or as a continuous region [51, 53]. A flow regime is the corresponding large-scale classification of the flow in space and time. Figure 2.1 summarizes the principal gas–liquid morphologies considered in this work. The sequence illustrates the progressive increase in gas connectivity, from dispersed bubbles to channel-confined and eventually continuous gas structures. Flow-regime maps are usually based on the gas and liquid superficial velocities, although void fraction is also used in some classification [54, 8]. For phase  $k$ , the superficial velocity is

$$J_k = \frac{Q_k}{A}, \quad (2.1)$$

where  $Q_k$  is the volumetric flow rate of phase  $k$  and  $A$  is the full channel cross-section. The definition uses the total area, including the part occupied by the other phase [51]. The mixture superficial velocity is

$$J_m = J_g + J_l. \quad (2.2)$$

The superficial velocity differs from the actual mean phase velocity because each phase occupies only part of the cross-section.

At low gas flow rates, the gas is often carried as separate bubbles in a continuous liquid. This is *bubbly flow*. If the bubbles remain small compared with the channel, the gas may be treated as a dispersed phase. The bubbles can still deform because of viscous stress, surface tension, buoyancy, and interaction with the walls [50, 11].

As the gas flow rate or local gas content rises, bubbles grow, merge, and become confined by the channel. When their width is close to the channel width, they can fill most of the cross-section. This regime is called *slug flow*, *plug flow*, or *Taylor flow* [11, 42, 24]. Liquid slugs separate the elongated bubbles, while thin liquid films usually remain between the gas and the walls. In a non-circular channel, more liquid can remain in the corners because the interface cannot follow the flat walls and fill the sharp corners at the same time [16, 48].

Liquid-filled corners arise from the capillary geometry of the gas–liquid interface near the channel edges [10]. Wall films and corner liquid pathways influence the bubble motion and the associated viscous pressure loss [24, 46]. Two Taylor flows with similar gas contents may still behave differently if their film thicknesses, meniscus curvatures, or liquid bypass rates differ [16, 25].

At higher gas flow rates, elongated bubbles may lose their regular shape and interact with nearby gas structures. This irregular, intermittent regime is often called *churn flow*. Unlike regular Taylor flow, it shows large time variations, strong interface deformation, and repeated breakup and coalescence. There is no universal boundary between slug and churn flow; it depends on channel size and shape, fluid properties, and the chosen classification method [44, 39].

When the gas becomes continuous along the channel axis, *annular flow* may form. The gas occupies the core, while most of the liquid moves through wall films and, in non-circular channels, through the corners. A low liquid volume fraction does not necessarily mean that the liquid is dispersed: a small amount of liquid can remain connected through thin films [44, 48].

In a sufficiently large horizontal channel, gravity can produce *stratified flow*, with gas above the liquid. This regime is less likely under strong capillary confinement, because surface tension can maintain interfaces and wall films regardless of channel orientation [51, 52].

No single gas or liquid volume fraction fixes these transitions. The superficial velocities, channel size and shape, density and viscosity ratios, surface tension, wettability, and inlet conditions all enter the problem [44, 9, 39]. This limits the use of a flow-regime map outside the channel and fluid pair for which it was established.

### 2.1.2. Flow fractions, void fraction, and phase slip

Several quantities describe how much gas and liquid are present in the channel. Their notation is often similar, although they do not mean the same thing. The inlet gas flow fraction is defined from the imposed flow rates as

$$\alpha_{G,h} = \frac{Q_g}{Q_g + Q_l} = \frac{J_g}{J_g + J_l}, \quad (2.3)$$

The liquid flow fraction is

$$\alpha_{L,h} = \frac{Q_l}{Q_g + Q_l} = \frac{J_l}{J_g + J_l}, \quad (2.4)$$

The subscript  $h$  refers to the homogeneous-flow definition. These fractions depend only on the inlet flow rates. They say nothing about the actual phase distribution or relative velocity. Cubaud and Ho write the homogeneous liquid fraction simply as  $\alpha_L$ ; the same notation is used later when their correlations are discussed.

The in-situ gas volume fraction, or void fraction, is the actual fraction of the channel occupied by gas. Its cross-sectional and time average can be written as

$$\bar{\varepsilon}_g = \frac{1}{AT} \int_{t_0}^{t_0+T} \int_A \chi_g(\mathbf{x}, t) dA dt, \quad (2.5)$$

where  $\chi_g = 1$  in the gas and  $\chi_g = 0$  in the liquid. The mean liquid volume fraction is

$$\bar{\varepsilon}_l = 1 - \bar{\varepsilon}_g. \quad (2.6)$$

Using intrinsic phase-averaged velocities, the superficial velocities can be written as

$$J_g = \bar{\varepsilon}_g \bar{U}_g, \quad J_l = (1 - \bar{\varepsilon}_g) \bar{U}_l. \quad (2.7)$$

The phase slip ratio is defined as

$$S = \frac{\bar{U}_g}{\bar{U}_l}. \quad (2.8)$$

The slip ratio compares the mean phase velocities. Combining Eqs. (2.7) and (2.8) gives [55, 51]

$$\bar{\varepsilon}_g = \frac{J_g}{J_g + SJ_l}. \quad (2.9)$$

The actual void fraction equals the inlet gas flow fraction only for homogeneous flow, with  $S = 1$ :

$$\bar{\varepsilon}_g = \alpha_{G,h}. \quad (2.10)$$

This equality does not generally hold in confined flow because the phases can travel at different velocities. A Taylor bubble, for example, may move faster than the mean liquid while some liquid bypasses it through wall films and corners [16, 25]. Contact-line motion and partial wall drying can additionally modify the hydraulic resistance and the local wetting state [11, 43]. As a result, the inlet flow fractions do not uniquely specify the local morphology or wetting state.

### 2.1.3. Relevant dimensionless numbers

Several dimensionless numbers are used to compare inertia, viscosity, gravity, and surface tension. For a non-circular channel, the characteristic length is usually the hydraulic diameter

$$D_h = \frac{4A}{P_w}, \quad (2.11)$$

where  $P_w$  is the wetted perimeter. For a square channel of side length  $h$ ,  $D_h = h$ . The phase Reynolds number is

$$\text{Re}_k = \frac{\rho_k J_k D_h}{\mu_k}. \quad (2.12)$$

It compares inertia with viscosity for phase  $k$ . Equation (2.12) uses the superficial velocity [51].

The capillary number compares viscous stress with surface tension:

$$\text{Ca} = \frac{\mu_l U_c}{\sigma}, \quad (2.13)$$

where  $\mu_l$  is the liquid dynamic viscosity,  $\sigma$  is the gas–liquid surface tension, and  $U_c$  is a characteristic velocity. For an elongated bubble,  $U_c$  is usually the bubble velocity or the mixture superficial velocity.

The ratio between characteristic viscous stress and capillary pressure gives

$$\frac{\tau_\mu}{\Delta p_\sigma} \sim \frac{\mu_l U_c}{\sigma} = \text{Ca}. \quad (2.14)$$

The ratio is the capillary number. It affects both the meniscus shape and the thickness of the deposited film. In the limit

$$\text{Ca} \ll 1, \quad (2.15)$$

surface tension dominates the viscous action on the interface. Menisci remain strongly curved and the film behind an elongated bubble is thin. Bretherton [6] obtained the corresponding asymptotic relation

$$\frac{\delta_f}{R_t} = 1.34 \text{Ca}^{2/3}, \quad (2.16)$$

where  $R_t$  is the tube radius. The derivation assumes  $\delta_f \ll R_t$ . Aussillous and Quéré [3] later proposed an extended relation for larger capillary numbers,

$$\frac{\delta_f}{R_t} = \frac{1.34 \text{Ca}^{2/3}}{1 + 3.35 \text{Ca}^{2/3}}. \quad (2.17)$$

For a circular channel, for which  $D_h = 2R_t$ , this relation can equivalently be written as

$$\frac{\delta_f}{D_h} = \frac{0.67 \text{Ca}^{2/3}}{1 + 3.35 \text{Ca}^{2/3}}. \quad (2.18)$$

At low Ca, Eq. (2.17) recovers Bretherton's  $\text{Ca}^{2/3}$  law. Its normalized thickness approaches a finite value at higher capillary numbers:

$$\frac{\delta_f}{R_t} \rightarrow \frac{1.34}{3.35} \simeq 0.40. \quad (2.19)$$

The saturation is an empirical correction for the range in which  $\delta_f$  is no longer small compared with  $R_t$  [3].

Balestra et al. later extended this form to confined droplets with a finite viscosity ratio,

$$\lambda = \frac{\mu_d}{\mu_c}, \quad (2.20)$$

using fitting coefficients that depend on  $\lambda$ . Their model covers both the low-Ca limit and saturation at higher Ca. For gas bubbles in water,  $\lambda \ll 1$ , so the Aussillous–Quéré equation remains a useful first approximation [4]. The liquid film deposited behind a moving confined gas structure is not an independently prescribed geometrical quantity. Its thickness results from the coupled viscous–capillary and wetting dynamics of the moving interface. The capillary number controls the balance between viscous stresses and surface tension and therefore strongly influences the deposited-film thickness, while the wall wettability and contact-line dynamics determine how the interface approaches and moves along the solid surface [3, 38].

For a moving contact line, the apparent contact angle may differ from the static equilibrium value and may also differ between advancing and receding motion. The resulting near-wall morphology therefore depends on the bubble or slug velocity, liquid viscosity, surface tension, contact-angle behaviour, and channel geometry. In the present work, the contact angle is consequently treated as a wetting condition rather than as a prescribed liquid-film thickness. The coefficients were fitted for circular capillaries. Square and rectangular channels also involve corner flow and a non-uniform transverse curvature, so the same equation is only an approximation there. Wetting and the choice of characteristic length introduce further differences [16]. The capillary number does not compare gravity with surface tension. This is the role of the Bond number:

$$\text{Bo} = \frac{(\rho_l - \rho_g) g D_h^2}{\sigma}, \quad (2.21)$$

It compares the hydrostatic pressure change over a distance  $D_h$ , with the capillary pressure scale

$$\frac{\Delta p_g}{\Delta p_\sigma} \sim \text{Bo}. \quad (2.22)$$

The Bond number in Eq. (2.21) uses the hydraulic diameter as the channel-scale length [51, 53].

A small Bond number means that gravity produces little deformation at the channel scale. Bubbles do not necessarily collect at the upper wall, and liquid can remain connected through films and corners even in a horizontal channel. Gravity becomes able to deform the interface and promote stratification once Bo is large.

The same comparison can be expressed using the capillary length

$$\ell_c = \sqrt{\frac{\sigma}{(\rho_l - \rho_g)g}}. \quad (2.23)$$

Combining Eqs. (2.21) and (2.23) gives

$$\text{Bo} = \left(\frac{D_h}{\ell_c}\right)^2. \quad (2.24)$$

The capillary length gives the scale below which gravity weakly deforms the interface. For air and water near ambient conditions,  $\ell_c$  is a few millimetres. In a channel only a few hundred micrometres wide, Bo is usually much smaller than one and surface tension and wettability dominate [11, 51].

The competition between inertia and surface tension is quantified by the Weber number,

$$\text{We} = \frac{\rho_l U_c^2 D_h}{\sigma}. \quad (2.25)$$

It compares the dynamic pressure  $\rho_l U_c^2$  with the capillary pressure  $\sigma/D_h$ . For  $\text{We} \ll 1$ , surface tension limits deformation caused by inertia. When We is of order one or higher, strong deformation, waves, breakup, and droplet entrainment become more likely. Their onset also depends on viscosity, geometry, and morphology [51, 53].

When the same characteristic velocity and length are used, the Reynolds, capillary, and Weber numbers satisfy

$$\text{We} = \text{Re}_l \text{Ca}. \quad (2.26)$$

A low capillary number does not mean that inertia is negligible. A flow may have both  $\text{Ca} \ll 1$  and a moderate Reynolds number: surface tension then dominates viscous stress at the interface, while inertia still matters in the bulk.

No single number describes the balance of forces. In the confined flows studied here, the common combination

$$\text{Bo} \ll 1, \quad \text{Ca} \ll 1, \quad (2.27)$$

corresponds to strong capillary confinement and thin wall films. The Reynolds and Weber numbers are still needed to assess the role of inertia in the bulk flow and in interface deformation.

#### 2.1.4. Pressure-drop mechanisms

The pressure drop between two cross-sections has several contributions. For a channel segment from  $x_1$  to  $x_2$ , it may be written schematically as

$$\Delta p = \Delta p_{\text{fr}} + \Delta p_{\text{acc}} + \Delta p_{\text{grav}} + \Delta p_{\text{cap}} + \Delta p_{\text{loc}}, \quad (2.28)$$

where the terms represent wall friction, streamwise acceleration, gravity, capillary effects, and local losses [21].

This split is not exact in a general unsteady two-phase flow. A passing meniscus changes the curvature, velocity field, and wall shear at the same time. Equation (2.28) is mainly used here to organise the discussion of the different mechanisms.

##### Wall-friction contribution

For steady, fully developed single-phase flow in a straight channel, the axial pressure force balances the wall shear. For a slice of length  $dx$ ,

$$-A \frac{dp}{dx} = \int_{P_w} \tau_{w,x} ds, \quad (2.29)$$

where  $\tau_{w,x}$  is the magnitude of the resistive streamwise wall shear stress. The perimeter average is

$$\bar{\tau}_w = \frac{1}{P_w} \int_{P_w} \tau_{w,x} ds, \quad (2.30)$$

Using  $D_h = 4A/P_w$  then gives

$$-\frac{dp}{dx} = \frac{4\bar{\tau}_w}{D_h}. \quad (2.31)$$

The same balance can be written with the Darcy friction factor,

$$-\frac{dp}{dx} = f_D \frac{\rho U_b^2}{2D_h}, \quad (2.32)$$

or with the Fanning friction factor,

$$\bar{\tau}_w = f_F \frac{\rho U_b^2}{2}, \quad f_D = 4f_F, \quad (2.33)$$

where  $U_b$  is the bulk velocity [37].

For fully developed laminar flow in a square channel of side length  $h$ , the pressure gradient and flow rate are related by

$$Q = -\frac{h^4}{C_{sq}\mu} \frac{dp}{dx}, \quad C_{sq} \simeq 28.4. \quad (2.34)$$

The corresponding hydraulic resistance is

$$R = \frac{\Delta p}{Q} = C_{sq} \frac{\mu L}{h^4}. \quad (2.35)$$

The  $h^{-4}$  dependence makes a single-phase square duct very sensitive to its size [37]. A wall film follows a different scaling, derived below.

The same control-volume balance can be used for gas-liquid flow, but Eq. (2.31) is recovered only if the mean momentum-flux change, gravity, capillarity, and other axial surface forces are negligible. Otherwise, these terms must be added to the wall traction [21]. Wall shear is also non-uniform: different parts of the wall may be in contact with liquid, gas, a moving contact line, or a thin film. Since the liquid is usually much more viscous than the gas, liquid-covered regions often account for most of the viscous loss in wetting microflows [24, 49].

A thin liquid film adjacent to the wall requires a more local description than a homogeneous mixture model. In annular gas-liquid flow, the wall is covered by the liquid phase while the gas occupies the channel core. The wall shear stress is therefore determined by the liquid velocity gradient at the solid boundary,

$$\tau_w = \mu_l \left. \frac{\partial u_l}{\partial y} \right|_{y=0}. \quad (2.36)$$

Detailed measurements of annular liquid films confirm that very large velocity gradients occur in the thin near-wall liquid layer. Ashwood et al. reconstructed complete mean

liquid-film velocity profiles using thin-film PIV and obtained the local wall shear directly from the slope of the viscous near-wall region. Their wall-shear estimates agreed within about 20% with values inferred independently from pressure-drop measurements [2], supporting the use of the near-wall liquid velocity gradient to estimate the frictional wall stress.

Consider locally a liquid film of thickness  $\delta_f$  adjacent to a stationary wall,

$$0 < y < \delta_f,$$

with a gas core above the interface. The liquid satisfies the no-slip condition

$$u_l(0) = 0, \quad (2.37)$$

while continuity of velocity at a sharp gas–liquid interface gives

$$u_l(\delta_f) = u_g(\delta_f) = U_i. \quad (2.38)$$

The tangential viscous stress is also continuous,

$$\mu_l \left. \frac{\partial u_l}{\partial y} \right|_{y=\delta_f} = \mu_g \left. \frac{\partial u_g}{\partial y} \right|_{y=\delta_f} = \tau_i. \quad (2.39)$$

These sharp-interface conditions are the same as those used in the layered-Poiseuille formulation of Lai et al. [28]. The difference in the present interpretation is that the liquid-film resistance is considered locally, rather than imposing one fully developed pressure gradient on two infinite parallel layers and using the resulting phase flow rates.

The local liquid-film momentum equation may be written as

$$\mu_l \frac{\partial^2 u_l}{\partial y^2} = -G_l, \quad G_l = -\frac{\partial p_l}{\partial x}, \quad (2.40)$$

where  $G_l$  represents the local streamwise pressure-gradient contribution acting on the film. Integrating Eq. (2.40) and using the interfacial-stress condition gives

$$u_l(y) = \frac{\tau_i}{\mu_l} y + \frac{G_l}{2\mu_l} (2\delta_f y - y^2). \quad (2.41)$$

The no-slip condition requires the liquid velocity to rise from zero at the wall to a finite value of order  $U_i$  over only the film thickness. The characteristic gradient therefore scales as

$$\left| \frac{\partial u_l}{\partial y} \right| \sim \frac{U_i}{\delta_f}, \quad (2.42)$$

with the corresponding wall-shear scale

$$\tau_w \sim \mu_l \frac{U_i}{\delta_f}. \quad (2.43)$$

Hence, even a moderate velocity difference can generate a very large wall-normal gradient when it is accommodated over a micrometre-scale film. Ashwood et al. [2] observed a similar near-wall behaviour, with an approximately linear viscous region from which  $\tau_w$  was obtained directly.

The present film-resistance argument differs from the fully developed layered-Poiseuille problem in its starting assumptions. In the latter, both phases are parallel, the interface is flat, one streamwise pressure gradient acts in both fluids, and the phase flow rates follow from that imposed gradient. For an air–water viscosity ratio, such a construction can

predict a much stronger velocity variation in the gas than in the liquid. That solution is mathematically valid for its boundary conditions, but it is not the closure used here. The present argument instead starts from the existence of a finite liquid-film flow and a no-slip wall condition and asks what wall-normal gradient and shear are required across the physical film thickness. Lai et al. remain useful for the sharp-interface continuity conditions and for understanding how a diffuse numerical interface modifies the velocity-gradient transition [28].

The strong dependence of wall shear on film thickness also makes thin liquid films demanding to describe numerically. As  $\delta_f$  decreases, sufficient spatial resolution is required to represent both the liquid distribution and the steep near-wall velocity gradient. The consequences of insufficient resolution are discussed later together with the numerical modelling and mesh-refinement results.

The present description is local. In square and rectangular channels, part of the liquid can bypass the gas core through the corner regions [48, 49]. The amount of liquid transported in the flat-wall film therefore depends on the morphology and on the available corner flow [16, 25]. On the other hand, when the wall film carries a significant fraction of the liquid transport, its thickness can become a controlling parameter for the near-wall shear and pressure loss [11, 24].

### Acceleration and gravitational contributions

Acceleration contributes when the streamwise momentum flux changes between two cross-sections. Define the phase-resolved momentum flux as

$$\mathcal{M}_x = \frac{1}{A} \int_A \sum_k r_k \rho_k u_{k,x}^2 dA. \quad (2.44)$$

The acceleration contribution is then approximately

$$\Delta p_{\text{acc}} = \mathcal{M}_x(x_2) - \mathcal{M}_x(x_1) \quad (2.45)$$

for a constant-area control volume, provided the other axial forces are treated separately [21].

In a constant-area channel with incompressible fluids and a statistically developed morphology, the mean momentum flux is nearly constant and  $\Delta p_{\text{acc}}$  is small. This assumption is less appropriate near inlets, contractions, bends, gas injection, bubble formation, or regime transitions. Gas compressibility can produce a streamwise change in gas velocity and void fraction as well.

The gravitational contribution depends on channel orientation and on the in-situ phase distribution. For a channel inclined by an angle  $\theta$  from the horizontal,

$$\Delta p_{\text{grav}} = \int_{x_1}^{x_2} \bar{\rho}_m(x) g \sin \theta dx, \quad (2.46)$$

where

$$\bar{\rho}_m = \frac{1}{A} \int_A \sum_k r_k \rho_k dA \quad (2.47)$$

is the cross-sectional mean mixture density.

For a straight horizontal channel,  $\sin \theta = 0$ , so there is no streamwise hydrostatic contribution. Gravity may still affect the morphology if the Bond number is not small. In a vertical channel, Eq. (2.46) must be included when measurements are compared with

simulations, especially if the reported pressure difference contains both hydrostatic and frictional parts.

The mixture density depends on the actual void fraction rather than only on the inlet flow fraction. When available, the in-situ phase distribution should be used for the hydrostatic correction. Substitution of the inlet fraction amounts to assuming homogeneous motion,  $S = 1$ , and can cause a large error when gas–liquid slip is strong. This is the usual distinction between homogeneous and separated-flow corrections [21].

### Capillary pressure contribution

Surface tension creates a pressure jump across a curved gas–liquid interface. For a static interface, the Young–Laplace relation is

$$p_g - p_l = \sigma \kappa = \sigma \left( \frac{1}{R_1} + \frac{1}{R_2} \right), \quad (2.48)$$

where  $R_1$  and  $R_2$  are the principal radii of curvature, and  $\kappa$  is the total curvature [51].

An elongated bubble has a front and a rear meniscus. Its net capillary contribution may be written schematically as

$$\Delta p_{\text{cap},b} = \sigma (\kappa_f - \kappa_r), \quad (2.49)$$

where the sign depends on the interface-normal convention. For a train of  $N_b$  bubbles,

$$\Delta p_{\text{cap}} \simeq \sum_{b=1}^{N_b} \sigma (\kappa_{f,b} - \kappa_{r,b}). \quad (2.50)$$

Pressure jumps at the front and rear menisci would cancel if both interfaces had the same equilibrium curvature. This is generally not the case for a moving bubble. Viscous bending, dynamic contact angles, wetting hysteresis, film deposition, and contact-line motion can leave a finite capillary contribution [6, 24, 49].

Capillarity can cause large oscillations in the instantaneous pressure signal. The local curvature and phase distribution change quickly when a meniscus passes a measurement section, and a short recording may differ greatly from the long-time mean.

The number of bubbles also enters the pressure balance. In slug flow, increasing the gas flow rate may produce longer bubbles and fewer menisci within a fixed channel length. Under these conditions, the capillary contribution can fall while the gas flow rate rises. This mechanism may contribute to a non-monotonic pressure drop across a regime transition, but it is not a general law [11, 24].

In a square channel, the transverse curvature varies around the perimeter. Liquid-filled corners, wall films, and dry patches give the meniscus a three-dimensional shape that cannot be reduced to one radius in a two-dimensional section. Its numerical prediction depends on the resolved interface geometry and on the treatment of wettability and contact angles.

## 2.2. Numerical modelling of gas–liquid flows

Small bubbles, elongated bubbles, wall films, and a continuous gas core may all occur in the same channel. Their size relative to the mesh determines how they can be represented numerically.

Let  $d_s$  denote the characteristic size of a gas structure and  $\Delta$  the local mesh size. If  $d_s$  is much larger than  $\Delta$ , its shape and motion can be represented on the grid. The

reconstructed geometry becomes sensitive to the grid when the two scales are comparable. A structure with  $d_s < \Delta$  is below the grid scale and has to be modelled.

Resolution is a local issue: the outer shape of a Taylor bubble may cover many cells while the liquid film between the bubble and the wall remains thinner than the first cell. In that case the bubble is resolved, but the film and its wall-normal velocity gradient are not. Small unresolved bubbles may also coexist with a resolved gas region. The terms *resolved* and *unresolved* are used here in this mesh-dependent sense.

### 2.2.1. Resolved-interface methods

Resolved-interface methods represent the position and shape of the gas–liquid boundary on the mesh. Volume of Fluid, level set, and front tracking are common examples [17, 41, 45].

In a Volume of Fluid (VOF) method, each cell contains a gas volume fraction  $r_g$ . For a two-phase gas–liquid system,

$$r_g = \begin{cases} 1, & \text{inside the gas,} \\ 0, & \text{inside the liquid,} \\ 0 < r_g < 1, & \text{in an interface cell.} \end{cases} \quad (2.51)$$

The liquid fraction is  $r_l = 1 - r_g$ . For incompressible phases without phase change, the gas fraction follows the flow. Its simplest transport equation is

$$\frac{\partial r_g}{\partial t} + \nabla \cdot (r_g \mathbf{u}) = 0, \quad (2.52)$$

where  $\mathbf{u}$  is the velocity used in the one-fluid formulation. In practice, interface reconstruction or compression is added to keep the transition between  $r_g = 0$  and  $r_g = 1$  narrow [17].

The phase fraction is also used to calculate the material properties. A common interpolation is

$$\rho = r_g \rho_g + (1 - r_g) \rho_l, \quad \mu = r_g \mu_g + (1 - r_g) \mu_l. \quad (2.53)$$

The one-fluid formulation solves a single momentum equation. Pressure and velocity are shared in interface cells, while the fraction field is used to distinguish the materials.

Surface tension is commonly included as a volumetric force with the continuum surface force (CSF) model [5]. With one common sign convention,

$$\mathbf{f}_\sigma = \sigma \kappa \nabla r_g, \quad (2.54)$$

with

$$\mathbf{n} = \frac{\nabla r_g}{|\nabla r_g|}, \quad \kappa = -\nabla \cdot \mathbf{n}. \quad (2.55)$$

The signs of  $\mathbf{n}$  and  $\kappa$  depend on the selected phase and on the normal direction. Close to a solid wall, the reconstructed interface normal can be modified to impose a prescribed contact-angle behaviour. In the present calculations, this role is fulfilled by an **alphaContactAngle**-type boundary condition for the continuous gas–liquid interface. The condition constrains the local orientation of the interface at the wall through the specified wetting angle; it does not prescribe a finite liquid-film thickness.

The near-wall phase distribution is therefore obtained from the coupled solution of interface advection, surface tension, viscous stresses, and the wall-wetting condition. Depending on the local interface motion, the gas structure may approach the wall closely or leave a liquid layer between the gas and the solid boundary. Any resulting film is thus an

outcome of the resolved wetting and interface dynamics, subject to the available numerical resolution. VOF does not require the shape of a large gas structure to be prescribed. Menisci, Taylor bubbles, annular gas cores, and interface waves can develop during the calculation, including breakup and coalescence. Its good phase-volume conservation is another reason for its frequent use in microchannel and electrolyzer simulations [15, 27, 35].

The cost rises quickly with the required interface resolution. Several cells are needed across a structure to calculate its curvature and deformation with reasonable accuracy. In three dimensions, halving  $\Delta$  increases the number of cells by about a factor of eight, and the time step generally has to be reduced as well. Resolving every small bubble in a complete channel soon becomes too expensive.

Wall films are a typical example of this limitation. A bubble may look well resolved while a film with  $\delta_f < \Delta$  occupies less than one cell. The numerical solution may contain some liquid next to the wall, but it cannot recover the velocity profile across the film or the gradient  $U_f/\delta_f$  from that mesh. Curvature errors, interface smearing, parasitic currents, and sensitivity to the contact-angle treatment also become more pronounced near the grid scale. The label “interface resolving” applies to the method, not automatically to every interface feature in a calculation.

Level-set and front-tracking methods use different interface descriptions. A level-set method places the interface at the zero contour of a signed-distance function. This smooth field is convenient for normals and curvature, although extra care is required to conserve phase volume [41]. Front tracking follows the interface with a separate set of markers. The geometry remains sharp, but the marker topology is harder to manage during breakup and coalescence [45]. Since MultiMorph uses a VOF-like continuous field, the following sections mainly use VOF terminology.

### 2.2.2. Unresolved-interface methods

Unresolved-interface methods do not reconstruct each bubble or droplet boundary. Instead, the equations are averaged over a volume larger than the individual structures. Euler–Euler, also called two-fluid, models represent each phase as a continuum with its own volume fraction and velocity [12, 21].

For phase  $k$ , the variables  $r_k$ ,  $\rho_k$ , and  $\mathbf{u}_k$  describe its local mean volume fraction, density, and velocity. The fractions satisfy

$$\sum_k r_k = 1. \quad (2.56)$$

Because the equations are volume-averaged, both phases may have a non-zero volume fraction in the same cell without being molecularly mixed. Their detailed arrangement inside the averaging volume is no longer available.

Closure models replace part of the missing geometrical information. For monodisperse spherical bubbles of diameter  $d_b$ , the interfacial area per unit mixture volume is

$$a_i = \frac{6r_g}{d_b}. \quad (2.57)$$

The assumed bubble diameter sets the interface area and enters the interfacial forces. Several bubble-size groups or a population-balance equation can be introduced when one diameter is not sufficient.

Momentum exchange between the phases is introduced through source terms. Drag is usually the main contribution and can be written in the general form

$$\mathbf{f}_{D,g} = K_D (\mathbf{u}_l - \mathbf{u}_g), \quad \mathbf{f}_{D,l} = -\mathbf{f}_{D,g}, \quad (2.58)$$

where  $K_D$  depends on the bubble size and shape, the relative Reynolds number, and the local phase fraction. Depending on the flow, further closures may be needed for lift, virtual mass, turbulent dispersion, wall lubrication, coalescence, and breakup [21].

Because the mesh does not follow every bubble, many sub-grid bubbles can be represented by smooth averaged fields. This makes Euler–Euler models practical for bubbly flows in large domains, particularly when the mean behaviour of the bubble population matters more than individual bubble shapes.

Averaging removes the detailed topology. For example,  $r_g = 0.6$  does not say whether the cell contains many bubbles, part of an elongated bubble, or a connected gas core. The phase-fraction field also contains no meniscus shape, contact line, wall film, or capillary pressure jump. Any model for these effects carries assumptions about the morphology.

The loss of topology becomes important during transitions from dispersed bubbles to Taylor or annular flow. A dispersed-bubble drag law can still return a numerical solution at high gas fraction, although the geometry behind the law is no longer appropriate. Likewise, a wall-lubrication model can alter bubble migration near a wall but cannot recover the velocity profile or shear stress in a continuous liquid film. Closures must be chosen according to the morphology they are meant to represent.

### 2.2.3. Hybrid resolved–unresolved approaches

An electrolyzer channel can contain both sub-grid bubbles and large connected gas structures. Resolving all small bubbles over the complete domain is generally not affordable. Treating a Taylor bubble or annular core as dispersed gas avoids this cost, but also removes the interface geometry needed for capillary pressure, wall wetting, and liquid bypass. Hybrid methods use both descriptions in one calculation [33].

Large gas structures are assigned to a resolved continuous field and small bubbles to an unresolved dispersed field. The total gas fraction is the sum of both. Gas is moved between the fields as the local morphology changes. A continuous structure that becomes too small for the mesh is transferred to the dispersed field. Accumulated dispersed gas can be transferred in the opposite direction when it forms a larger structure.

During this operation, the same physical gas is redistributed between numerical fields. Mass removed from one gas field is added to the other, together with its momentum, preserving total gas mass and mixture momentum.

Deciding which representation to use is not straightforward. A volume-fraction threshold, for instance, may assign the same class to a thin connected structure and to a group of separate bubbles with a similar local fraction. Local length scale, mesh resolution, neighbouring cells, and phase connectivity can provide additional information.

The transfer time scale affects both the field separation and numerical stability. Short transfer times remove gas rapidly from an unsuitable field, but can make the source terms stiff. Longer times are usually easier to stabilise and allow both representations to coexist for longer.

Momentum coupling also depends on the interface representation. At a well-resolved interface, strong normal coupling prevents the phases from passing through each other, while tangential slip may remain between a gas core and a slower liquid film. Relative motion of dispersed bubbles is instead controlled by drag and other averaged forces. A coarsely resolved interface lies between these limits and needs an intermediate treatment [32].

MultiMorph follows this multifield approach. Its resolved and unresolved fields use phase-averaged balance equations of the same form. Separate models transfer gas between the representations and adapt momentum exchange to the interface resolution [31, 32, 30].

### 2.3. The MultiMorph hybrid multiphase model

MultiMorph is a morphology-adaptive multifield two-fluid framework developed at the Helmholtz-Zentrum Dresden-Rossendorf (HZDR) and implemented within the `multi-phaseEulerFoam` framework of the OpenFOAM Foundation code base [32, 30]. Originally referred to as OpenFOAM-Hybrid, the model combines VOF-like treatment of large continuous interfaces with an Euler–Euler representation of unresolved dispersed structures.

The numerical methods introduced in the previous section describe two different limits. A resolved-interface method keeps the geometry of large gas structures on the mesh, whereas an unresolved Euler–Euler description represents sub-grid bubbles through averaged fields and closure models. The gas flow considered here can move between these two situations as bubbles accumulate, merge, or form larger connected structures.

MultiMorph combines both representations in a multifield formulation [31, 32, 30]. The physical gas phase is divided between a continuous field and a dispersed field, with transfer between the two representations as the local morphology changes.

Three parts of the model are particularly relevant for the present work. The first is the decomposition of the physical phases into numerical fields. The second is the morphology-transfer model, which determines when gas moves between the continuous and dispersed representations. The third is the momentum exchange between the fields, which must account for the fact that a resolved gas–liquid interface and a dispersed bubble population do not have the same relative-motion behaviour.

#### 2.3.1. Physical phases and numerical field decomposition

The gas is decomposed into a continuous field, denoted by  $g_C$ , and a dispersed field, denoted by  $g_D$ . Water is represented by the continuous field  $l$ . The field index is

$$k \in \mathcal{K} = \{g_C, g_D, l\}.$$

The total gas volume fraction is

$$r_g = r_{g_C} + r_{g_D}.$$

The fraction  $r_{g_C}$  is VOF-like, whereas  $r_{g_D}$  is Euler–Euler-like. Together with the liquid fraction, they satisfy

$$\sum_{k \in \mathcal{K}} r_k = 1.$$

#### 2.3.2. Governing equations

MultiMorph solves phase-averaged mass and momentum equations for each field  $k$  [31, 30]. Its volume fraction, density, and velocity are denoted by  $r_k$ ,  $\rho_k$ , and  $\mathbf{u}_k$ . The mass balance is

$$\frac{\partial r_k \rho_k}{\partial t} + \nabla \cdot (r_k \rho_k \mathbf{u}_k) = \sum_{j \neq k} \Gamma_{kj}^{PT}. \quad (2.59)$$

The term  $\Gamma_{kj}^{PT}$  transfers mass between numerical fields. By convention,  $\Gamma_{kj}^{PT} > 0$  denotes transfer from  $j$  to  $k$ . Here,  $\Gamma_{kj}^{PT}$  is an inter-field transfer term rather than a physical mass source.

Following Meller et al. [30], the conservative momentum equation for field  $k$  is

$$\begin{aligned} \frac{\partial r_k \rho_k \mathbf{u}_k}{\partial t} + \nabla \cdot (r_k \rho_k \mathbf{u}_k \otimes \mathbf{u}_k) = & -r_k \nabla p + \nabla \cdot (2r_k \mu_k \mathbf{S}_k) + r_k \rho_k \mathbf{g} \\ & + \sum_{j \neq k} r_k \sigma_{kj} \kappa_{kj} \mathbf{n}_{kj} + \mathbf{f}_k^{MT} + \sum_{j \neq k} \Gamma_{kj}^{PT} \mathbf{u}_j. \end{aligned} \quad (2.60)$$

The left-hand side contains the local and convective changes in field momentum. On the right,  $-r_k \nabla p$  is the phase-weighted pressure force, with one pressure  $p$  shared by all fields. The viscous term uses the strain-rate tensor

$$\mathbf{S}_k = \frac{1}{2} (\nabla \mathbf{u}_k + \nabla \mathbf{u}_k^T) - \frac{1}{3} (\nabla \cdot \mathbf{u}_k) \mathbf{I}. \quad (2.61)$$

Gravity is given by  $r_k \rho_k \mathbf{g}$ . In the capillary term  $\sum_{j \neq k} r_k \sigma_{kj} \kappa_{kj} \mathbf{n}_{kj}$ ,  $\sigma_{kj}$  is surface tension and  $\kappa_{kj}$  is curvature. The vector  $\mathbf{n}_{kj}$  contains the interface direction and the interfacial area density. Its unit is inverse length; using only a unit normal would not give a force per unit volume. The vector  $\mathbf{f}_k^{MT}$  collects momentum-transfer models such as drag, lift, virtual mass, turbulent dispersion, and wall lubrication. Finally,

$$\sum_{j \neq k} \Gamma_{kj}^{PT} \mathbf{u}_j,$$

is the momentum carried by mass transferred from  $j$  to  $k$ . This term is required for mixture-momentum conservation during morphology transfer [30].

### 2.3.3. Continuous-structure identification

The transfer from a continuous to a dispersed representation requires a criterion that distinguishes a resolved structure from continuous-phase material that has become too small for the local grid. A criterion based only on the local continuous-phase fraction  $r_c$  is not sufficient. Cells located inside the numerically smeared interface of a resolved bubble can contain relatively small values of  $r_c$ , although they still belong to the same connected gas structure.

MultiMorph introduces the continuous-structure indicator  $I_c$  for this purpose [30]. The construction starts from cells in which the continuous phase occupies more than 74% of the cell volume,

$$r_c > 0.74. \quad (2.62)$$

These cells are considered to belong to the core of a continuous structure and are assigned

$$I_c = 1. \quad (2.63)$$

The fixed value 0.74, used to identify the core of a continuous structure, is independent of the `continuousThreshold` introduced later in the morphology-transfer criterion.

Identifying only the core cells would still leave the interface of the same structure unprotected. In a VOF-like representation, the transition from  $r_c = 1$  to  $r_c = 0$  extends over several computational cells because of the finite numerical interface thickness. Meller et al. therefore extend the region for which  $I_c = 1$  by an additional layer of

$$N_c = 2 \quad (2.64)$$

cells in the direction of decreasing continuous-phase fraction, approximately following

$$-\nabla r_c.$$

The resulting mask covers both the high-volume-fraction core and the surrounding interface region of a sufficiently resolved continuous structure. All remaining cells have

$$I_c = 0.$$

The local mesh size enters this criterion implicitly. Consider the same physical bubble represented on two different grids. On a sufficiently fine grid, several cells satisfy  $r_c > 0.74$ ,

and the additional protected layers cover the resolved interface around them. When the bubble becomes comparable to the grid spacing, either because the bubble becomes smaller or because the mesh becomes coarser, fewer cells form such a continuous core. Parts of the structure can then fall outside the  $I_c = 1$  region and become available for transfer to the dispersed field.

The indicator does not modify the phase fraction directly. It appears in the transfer source through

$$1 - I_c. \quad (2.65)$$

For  $I_c = 1$ , the source vanishes and the identified continuous structure is protected from `disintegrationToDisperse`. For  $I_c = 0$ , transfer can occur if the additional phase-fraction conditions described in Section 2.3.4 are satisfied.

The volume fraction  $r_c$  and the indicator  $I_c$  contain different information. The volume fraction remains a local geometrical quantity: it gives the fraction of the cell occupied by the continuous field. The indicator instead contains information about whether this material is attached to a sufficiently resolved continuous structure. Two cells with identical  $r_c$  may therefore have different values of  $I_c$ .

The transfer formulation also uses a non-linear dependence on the donor phase fraction,

$$\Gamma_{dc}^{PT} \propto \sqrt{r_c}.$$

Meller et al. chose this form because a source proportional to  $r_c$  itself would lead to an exponential decay, for which  $r_c = 0$  is reached only asymptotically. A source proportional to  $\sqrt{r_c}$  produces a polynomial evolution and allows the continuous donor field to vanish completely within finite time [30].

### 2.3.4. Morphology transfer

The numerical transfer between  $g_C$  and  $g_D$  introduced above appears in the governing equations through  $\Gamma_{kj}^{PT}$  [30]. Its sign convention is

$$\Gamma_{kj}^{PT} > 0 \quad \Longleftrightarrow \quad \text{mass is transferred from field } j \text{ to field } k.$$

$$[\Gamma_{kj}^{PT}] = \text{kg m}^{-3} \text{s}^{-1}.$$

For an ordered pair of fields  $(k, j)$ , the transfer term can be written as

$$\Gamma_{kj}^{PT} = \rho_j \frac{\mathcal{F}_{kj}}{\tau_{PT}}, \quad (2.66)$$

where  $\tau_{PT}$  is the transfer time scale and  $\mathcal{F}_{kj}$  depends on the chosen transfer model [30]. The transfer becomes faster as  $\tau_{PT}$  decreases. Very short values can make the source term stiff, particularly when they approach the numerical time step. Increasing  $\tau_{PT}$  improves this behaviour but allows gas to remain longer in a field that does not match the local morphology.

The morphology-transfer time scale  $\tau_{PT}$  controls conversion between fields, whereas the resolved-interface drag relaxation time controls their velocity coupling.

#### Transfer from continuous to dispersed morphology

Transfer from  $g_C$  to  $g_D$  uses the `disintegrationToDisperse` model. It follows Meller et al. [30], with two implementation details: a configurable threshold and a numerical lower bound on the transfer time scale.

The implemented source is

$$\Gamma_{g_D g_C}^{PT, \text{dis}} = \begin{cases} (1 - I_c) \frac{\rho_c \sqrt{r_c}}{\max(\tau_{PT}, 2\Delta t)}, & \sum_{\alpha \in C} r_\alpha > r_{C, \text{th}} \quad \text{and} \quad r_{\max, C} > r_c, \\ 0, & \text{otherwise.} \end{cases} \quad (2.67)$$

In Eq. (2.67),  $r_c$  and  $\rho_c$  are the volume fraction and density of the continuous donor field. The denominator  $\max(\tau_{PT}, 2\Delta t)$  prevents transfer on a time scale shorter than twice the numerical time step.

The continuous-structure indicator  $I_c$  is one in a protected continuous structure and zero elsewhere. Multiplication by  $(1 - I_c)$  blocks the transfer inside a structure that is still considered resolved.

The largest volume fraction among the continuous fields is

$$r_{\max, C} = \max_{\alpha \in C}(r_\alpha). \quad (2.68)$$

The condition

$$r_{\max, C} > r_c \quad (2.69)$$

prevents the locally dominant continuous phase from becoming dispersed.

The other condition,

$$\sum_{\alpha \in C} r_\alpha > r_{C, \text{th}}. \quad (2.70)$$

requires a minimum amount of continuous-phase material in the cell. The original formulation uses a threshold of 0.5; in the present implementation it can be changed through `continuousThreshold` in the `disintegrationToDisperse` model.

For the present air–water case, the continuous fields are  $g_C$  and  $l$ , so

$$\sum_{\alpha \in C} r_\alpha = r_{g_C} + r_l, \quad (2.71)$$

and

$$r_{\max, C} = \max(r_{g_C}, r_l). \quad (2.72)$$

The transfer from continuous gas to dispersed gas is then

$$\Gamma_{g_D g_C}^{PT, \text{dis}} = \begin{cases} (1 - I_{g_C}) \frac{\rho_g \sqrt{r_{g_C}}}{\max(\tau_{PT}, 2\Delta t)}, & r_{g_C} + r_l > r_{C, \text{th}} \quad \text{and} \quad \max(r_{g_C}, r_l) > r_{g_C}, \\ 0, & \text{otherwise.} \end{cases} \quad (2.73)$$

Since

$$\max(r_{g_C}, r_l) > r_{g_C} \quad (2.74)$$

means that a continuous field other than  $g_C$  is locally dominant, it reduces here to

$$r_l > r_{g_C}. \quad (2.75)$$

In this form,  $g_C \rightarrow g_D$  transfer occurs mainly in liquid-dominated cells in which  $I_{g_C}$  does not protect the gas structure.

### Transfer from dispersed to continuous morphology

The reverse transfer uses `accumulationToContinuous`. It converts dispersed gas into continuous gas once the total local gas fraction exceeds a threshold.

For donor  $g_D$  and receiver  $g_C$ , the source term is

$$\Gamma_{g_C g_D}^{PT,acc} = \begin{cases} \frac{\rho_d \sqrt{r_d}}{\tau_{PT}}, & r_c + r_d \geq \alpha_{th}, \\ 0, & \text{otherwise.} \end{cases} \quad (2.76)$$

Here,  $r_d$  and  $\rho_d$  are the volume fraction and density of the dispersed donor field. The threshold  $\alpha_{th}$  is set by `alphaThreshold`. If this entry is absent, the model uses the `continuousThreshold` of the receiving continuous phase.

The corresponding source in the dispersed-gas equation has the opposite sign:

$$\Gamma_{g_D g_C}^{PT,acc} = -\Gamma_{g_C g_D}^{PT,acc}. \quad (2.77)$$

The two terms cancel in the total gas-mass balance. The operation replaces Euler–Euler-like gas by the same amount of VOF-like gas.

### 2.3.5. Morphology-adaptive momentum exchange

The morphology-transfer model determines whether gas is represented by the continuous field  $g_C$  or by the dispersed field  $g_D$ . The momentum closures associated with these two representations are different. Changing morphology affects more than the phase-fraction distribution: it also changes the momentum closure used for the gas–liquid interaction [32, 30].

For a generic pair of fields  $\alpha$  and  $\beta$ , the drag contribution has the form

$$\mathbf{f}_{\alpha\beta}^D = K_{\alpha\beta}^D (\mathbf{u}_\beta - \mathbf{u}_\alpha). \quad (2.78)$$

The required coefficient  $K_{\alpha\beta}^D$  depends on the morphology represented by the two fields.

#### Dispersed–continuous momentum exchange

For a dispersed gas field interacting with continuous liquid, the interface of each individual bubble is below the grid scale. The numerical fields then have an Euler–Euler interpretation: gas and liquid are treated as interpenetrating continua and a non-zero relative velocity

$$\mathbf{u}_{g_D} - \mathbf{u}_l \neq \mathbf{0} \quad (2.79)$$

is generally expected.

The detailed geometry of the individual bubbles is no longer available. Interfacial forces are therefore represented by closure models. The baseline bubbly-flow model set described for MultiMorph by Meller et al. contains drag, lift, virtual mass, wall lubrication, and turbulent dispersion contributions [30]. In this formulation, drag is evaluated with the Ishii–Zuber correlation, lift with a shear-dependent Tomiyama correlation, virtual mass is added to account for acceleration of the surrounding liquid, and wall lubrication acts on bubbles approaching a solid boundary. Turbulent dispersion represents the redistribution of dispersed gas caused by liquid turbulence.

The dispersed-bubble diameter  $d_b$  enters these closure relations because the physical interface is not reconstructed directly. For spherical bubbles, the corresponding interfacial area density scales as

$$a_i = \frac{6r_{g_D}}{d_b}, \quad (2.80)$$

as introduced in Eq. (2.57). The selected value of  $d_b$  therefore affects the momentum exchange even when the local dispersed-gas volume fraction is unchanged.

In the baseline formulation reported by Meller et al., turbulence of the continuous liquid is described with the  $k$ - $\omega$  SST model, while the dispersed gas field itself is treated as laminar. Turbulent dispersion and bubble-induced turbulence are included through additional closures [30]. The precise closure combination used in a simulation remains a modelling choice, whereas the important distinction here is that the Euler–Euler representation allows phase slip and replaces the unresolved interface geometry by averaged interfacial-force models. In the following work, both phases will be treated as laminar.

### Continuous–continuous momentum exchange

The interaction between  $g_C$  and  $l$  has a different interpretation. Both fields describe continuous phases and their common interface is represented on the computational grid. In the limit of a well-resolved interface, MultiMorph should recover VOF-like behaviour. This is achieved by applying a very strong momentum coupling between the two continuous fields [32].

For a well-resolved interface, Meller et al. use the resolving drag coefficient

$$K_{\alpha\beta}^{D,R} = r_\alpha r_\beta \frac{\rho_{\alpha\beta}}{\tau_r}, \quad (2.81)$$

with

$$\rho_{\alpha\beta} = \frac{r_\alpha \rho_\alpha + r_\beta \rho_\beta}{r_\alpha + r_\beta}. \quad (2.82)$$

The relaxation time is chosen much smaller than the physical numerical time step,

$$\tau_r = 10^{-8} \Delta t. \quad (2.83)$$

The resulting value of  $K_{\alpha\beta}^{D,R}$  is very large. The relative velocity is then rapidly suppressed,

$$\mathbf{u}_\alpha \simeq \mathbf{u}_\beta,$$

and the multifield formulation approaches the homogeneous, VOF-like limit [32, 30].

Using this coupling everywhere is suitable only when the velocity variation around the interface is sufficiently resolved. A continuous gas structure can still occupy several cells while the associated interfacial shear layer is represented too coarsely. Imposing the same very large resolving drag in this situation forces both velocity fields to follow each other too closely and can suppress physical interfacial slip.

MultiMorph therefore introduces a second closure for an under-resolved interface,

$$K_{\alpha\beta}^{D,UR} = r_\alpha \rho_{\alpha\beta} C_d A_d |\mathbf{u}_\beta - \mathbf{u}_\alpha|, \quad (2.84)$$

where

$$A_d = \frac{2 |\nabla r_\alpha| |\nabla r_\beta|}{|\nabla r_\alpha| + |\nabla r_\beta|}. \quad (2.85)$$

Here,  $C_d$  is a model coefficient and  $A_d$  provides a measure of interfacial area density from the phase-fraction gradients [32]. Compared with  $K_{\alpha\beta}^{D,R}$ , this coefficient remains much smaller and allows a finite relative velocity between the two continuous fields.

The question is then where the resolving and under-resolving closures should be used. The model does not decide this only from the number of cells across the complete gas structure. It evaluates the local flow around the interface and compares the numerical interface thickness with the characteristic length scale of the interfacial shear.

### Identification of interfacial shear

For two continuous fields  $\alpha$  and  $\beta$ , Meller et al. define an interface vector

$$\mathbf{i}_{\alpha\beta} = r_\beta \nabla r_\alpha - r_\alpha \nabla r_\beta. \quad (2.86)$$

The vector is normal to the local interface and is not normalized. Its magnitude therefore has the unit of inverse length.

The phase velocities are physically meaningful mainly on their respective sides of the diffuse numerical interface, whereas  $\mathbf{i}_{\alpha\beta}$  is meaningful inside the interface region. The adaptive model requires all three quantities at the same location. Meller et al. therefore transport information from both sides of the interface towards the interface region using an iterative cell-to-cell procedure [32].

The information associated with  $\mathbf{u}_\alpha$ ,  $\mathbf{u}_\beta$ , and  $\mathbf{i}_{\alpha\beta}$  is propagated along the negative gradients of the corresponding measures of meaningfulness,

$$-\nabla r_\alpha, \quad -\nabla r_\beta, \quad -\nabla |\mathbf{i}_{\alpha\beta}|.$$

The operation is repeated over a prescribed maximum number  $N^I$  of neighbouring cells. Quantities reconstructed in this way are denoted by the superscript  $I$ . The purpose is to evaluate the relative velocity using information originating from the two sides of the interface rather than directly from the already mixed interface cells.

The local flow orientation is then characterized by the shear-flow indicator

$$\psi_{\alpha\beta}^I = \frac{\left| (\mathbf{u}_\alpha^I - \mathbf{u}_\beta^I) \times \mathbf{i}_{\alpha\beta} \right|}{\left| \mathbf{u}_\alpha^I - \mathbf{u}_\beta^I \right| |\mathbf{i}_{\alpha\beta}|}. \quad (2.87)$$

For a relative velocity normal to the interface,

$$\psi_{\alpha\beta}^I = 0,$$

which corresponds to a stagnation-type configuration. If the relative velocity is tangential to the interface,

$$\psi_{\alpha\beta}^I = 1.$$

Intermediate orientations give values between zero and one. If the denominator of Eq. (2.87) vanishes, the indicator is set to zero [32].

At the front and rear of a moving bubble, the surrounding flow can approach a stagnation-type configuration and both phases must remain strongly coupled in the normal direction to preserve the kinematic interface condition. Along the lateral part of the bubble, relative motion is mostly tangential and an interfacial shear layer develops. If this shear layer cannot be resolved on the grid, some relative slip has to be permitted instead of forcing both continuous-phase velocities to be identical.

### Measure of interface resolution

The interface thickness is estimated from the magnitude of the transported interface vector as

$$\Delta_{\alpha\beta}^I \simeq \frac{1}{|\mathbf{i}_{\alpha\beta}^I|}. \quad (2.88)$$

This follows from the approximately linear variation of phase fraction across the numerically smeared interface.

Meller et al. compare this numerical thickness with a viscous shear length scale, in a form similar to the wall coordinate  $y^+$ . An approximate interfacial shear stress is first calculated from

$$\tau_{\alpha\beta}^I = \tilde{\mu}_{\alpha\beta} |(\mathbf{u}_\alpha^I - \mathbf{u}_\beta^I) \times \mathbf{i}_{\alpha\beta}^I|, \quad (2.89)$$

where the material properties of the phase pair are evaluated with arithmetic means. For a generic property  $\varphi$ ,

$$\tilde{\varphi}_{\alpha\beta} = \frac{\varphi_\alpha + \varphi_\beta}{2}. \quad (2.90)$$

An interfacial shear velocity follows as

$$u_{\alpha\beta}^{*I} = \sqrt{\frac{\tau_{\alpha\beta}^I}{\tilde{\rho}_{\alpha\beta}}}, \quad (2.91)$$

and the corresponding viscous shear length is

$$L_{\alpha\beta}^{*I} = \frac{\tilde{\nu}_{\alpha\beta}}{u_{\alpha\beta}^{*I}}. \quad (2.92)$$

The dimensionless interface thickness is then

$$y_{\alpha\beta}^{+I} = \frac{\Delta_{\alpha\beta}^I}{L_{\alpha\beta}^{*I}}. \quad (2.93)$$

The interpretation is similar to a viscous wall scale. For

$$y_{\alpha\beta}^{+I} < 1,$$

the velocity variation associated with the interfacial shear is considered resolved. Values larger than one indicate that the numerical interface thickness is larger than the characteristic shear length and that the interfacial shear is increasingly under-resolved [32].

The shear orientation and the resolution measure are finally combined in the under-resolution indicator

$$\xi_{\alpha\beta}^{UR} = \begin{cases} \psi_{\alpha\beta}^I \left(1 - \exp\left(1 - y_{\alpha\beta}^{+I}\right)\right), & y_{\alpha\beta}^{+I} > 1, \\ 0, & y_{\alpha\beta}^{+I} \leq 1. \end{cases} \quad (2.94)$$

The indicator approaches zero for a resolved interface or for stagnation-type flow. It approaches one when a strongly tangential interfacial shear flow is far below the available grid resolution.

### Blending between resolving and under-resolving drag

The final continuous-continuous drag coefficient is obtained from

$$K_{\alpha\beta}^D = \frac{K_{\alpha\beta}^{D,UR} K_{\alpha\beta}^{D,R}}{K_{\alpha\beta}^{D,UR} + \xi_{\alpha\beta}^{UR} K_{\alpha\beta}^{D,R}}. \quad (2.95)$$

A harmonic rather than linear interpolation is used because  $K_{\alpha\beta}^{D,R}$  and  $K_{\alpha\beta}^{D,UR}$  can differ by several orders of magnitude [32].

For

$$\xi_{\alpha\beta}^{UR} = 0, \quad (2.96)$$

Eq. (2.95) gives

$$K_{\alpha\beta}^D = K_{\alpha\beta}^{D,R},$$

so that the continuous fields are strongly coupled and the solution approaches the VOF-like limit.

For a fully under-resolved shear region,

$$\xi_{\alpha\beta}^{UR} \rightarrow 1,$$

and because

$$K_{\alpha\beta}^{D,R} \gg K_{\alpha\beta}^{D,UR},$$

the adaptive coefficient approaches

$$K_{\alpha\beta}^D \simeq K_{\alpha\beta}^{D,UR}.$$

Interfacial slip is then permitted.

MultiMorph switches between VOF-like and Euler–Euler-like behaviour at two distinct levels. Morphology transfer changes the numerical field in which the gas is represented, from  $g_C$  to  $g_D$  or in the opposite direction. The resolution-adaptive drag acts before such a complete change of representation is necessarily required: a continuous interface can still be present on the mesh while its local shear layer is already under-resolved. In this intermediate situation the interface geometry is retained, but the momentum coupling is relaxed locally.

A gas core can remain sufficiently large to be represented by  $g_C$ , while a thin liquid film or the shear layer next to the gas–liquid interface can have a much smaller characteristic length scale than the gas structure itself. The large-scale morphology can then remain continuous even though the local momentum transfer is already operating partly in the under-resolved regime.

### 2.3.6. Liquid films and corner flow

A continuous gas structure can be resolved over several cells while the remaining liquid occupies much smaller regions near the wall. In a square channel, this liquid is distributed between flat-wall films and larger corner regions [11, 16, 48].

The corner regions are not equivalent to a uniform film around the gas core. Their curvature and available cross-sectional area are different, and part of the liquid can bypass an elongated gas structure through these regions [48, 49]. The liquid flow rate carried by a flat-wall film is therefore only a part of the total liquid flow whenever corner transport is present.

The numerical requirements of the two structures also differ. A corner liquid region can become resolved when the transverse mesh is sufficiently fine to represent its geometry. A wall film may remain thinner than the first near-wall cell. Mesh convergence of the large-scale morphology does not automatically imply that the wall film is resolved.

For a film of physical thickness  $\delta_f$ , the velocity variation between the wall and the interface occurs over this distance. If the film becomes smaller than the local mesh scale, liquid may still appear next to the wall in the phase-fraction field while the velocity variation across the physical film remains unresolved.

## 3. Methodology

This chapter defines the reference configurations, numerical setups, and quantities used for comparison with the experiments.

### 3.1. Verification and validation strategy

The assessment is based mainly on pressure drop and gas morphology. The in-situ gas fraction is also used where required, in particular for the hydrostatic correction of the vertical Kim configuration. Wall shear stress is not available experimentally and is therefore used only as a numerical diagnostic of the near-wall momentum transfer.

#### 3.1.1. Numerical post-processing

The initial transient is excluded from all time averages. For quasi-periodic cases, the averaging interval is chosen to contain complete cycles whenever this can be identified from the pressure signal. Less regular cases are averaged over a statistically stationary interval for which extending the sampling window does not produce a relevant change in the mean.

Figure 3.1 shows the procedure used for the pressure signal.

For conservation and phase-distribution analyses, the physical gas fraction is reconstructed as

$$r_g = r_{gC} + r_{gD}. \quad (3.1)$$

Morphology transfer changes the distribution between the two numerical gas fields but not the total gas quantity.

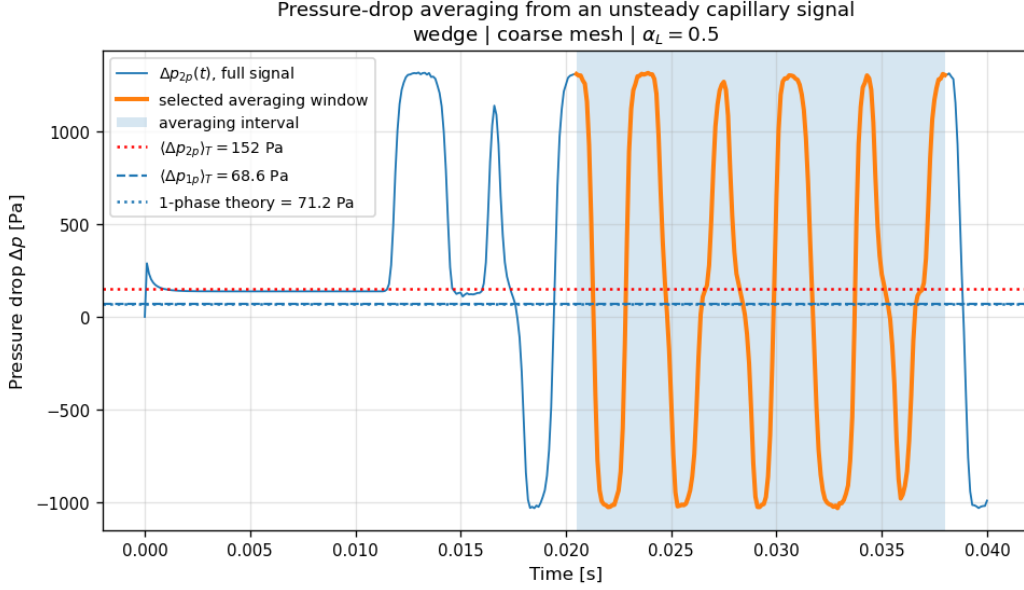


Figure 3.1: Illustration of the pressure-drop averaging strategy.

### 3.1.2. Validation quantities

The validation is based on pressure drop, in-situ gas fraction, and wall shear stress, complemented by error metrics. The corresponding quantities and averaging procedures are defined below.

#### Pressure drop

The pressure difference is not evaluated from individual cells because the instantaneous pressure field can exhibit strong local variations close to gas–liquid interfaces. Instead, pressure is first spatially averaged over two channel cross-sections. For a section  $A_i$ , the area-averaged pressure is

$$\langle p \rangle_{A_i}(t) = \frac{1}{A_i} \int_{A_i} p(\mathbf{x}, t) dA. \quad (3.2)$$

The instantaneous pressure drop between the upstream section  $A_1$  and the downstream section  $A_2$  is then

$$\Delta p(t) = \langle p \rangle_{A_1}(t) - \langle p \rangle_{A_2}(t). \quad (3.3)$$

The value used for validation is the temporal average over the statistically stationary interval,

$$\overline{\Delta p} = \frac{1}{T_{\text{avg}}} \int_{t_0}^{t_0 + T_{\text{avg}}} \Delta p(t) dt. \quad (3.4)$$

The averaging interval is selected according to the procedure described in Section 3.1.1.

Several of the reference results are expressed through the increase in hydraulic resistance caused by the presence of the gas phase. For this purpose, the normalized two-phase pressure drop is defined as

$$\Pi_{\Delta p} = \frac{\overline{\Delta p_{2P}}}{\overline{\Delta p_{1P}}}, \quad (3.5)$$

where  $\overline{\Delta p_{2P}}$  is the time-averaged two-phase pressure drop and  $\overline{\Delta p_{1P}}$  is the corresponding single-phase reference value. This normalization reduces the direct influence of channel length, channel size, and liquid viscosity and emphasizes the additional hydraulic resistance associated with the two-phase morphology. The precise definition of the single-phase reference condition is chosen consistently with each experimental dataset and is specified in the corresponding reference-case subsection.

For vertical configurations, the quantity reported experimentally may contain both hydraulic losses and a hydrostatic contribution. When required by the reference experiment, the gravitational contribution is removed consistently before the numerical and experimental pressure losses are compared. The corresponding correction is described for the relevant reference case in Section 3.2.

### Gas Volume Fraction

The second validation quantity is the in-situ gas volume fraction. In contrast to the homogeneous inlet gas fraction, it depends on the gas distribution and on the relative velocity between the phases.

Because the physical gas is represented by two numerical fields in MultiMorph, the local gas fraction used for post-processing is

$$r_g(\mathbf{x}, t) = r_{gC}(\mathbf{x}, t) + r_{gD}(\mathbf{x}, t). \quad (3.6)$$

For a measurement volume  $V_m$ , the instantaneous mean void fraction is calculated from the volume integral

$$\varepsilon_g(t) = \frac{1}{V_m} \int_{V_m} (r_{gC} + r_{gD}) \, dV. \quad (3.7)$$

Its time-averaged value is

$$\bar{\varepsilon}_g = \frac{1}{T_{\text{avg}}} \int_{t_0}^{t_0 + T_{\text{avg}}} \varepsilon_g(t) \, dt. \quad (3.8)$$

The integration volume is selected to reproduce, as closely as possible, the spatial region represented by the corresponding reference measurement. If the reference quantity represents the complete numerical channel,  $V_m$  is taken as the complete fluid volume.

The validation requires a clear separation between inlet gas fraction and in-situ void fraction. Two cases with identical inlet flow fractions can exhibit different values of  $\bar{\varepsilon}_g$  if the gas and liquid phases have different mean velocities.

### Wall Shear Stress

Wall shear stress is used as an additional diagnostic quantity for the near-wall momentum transport. Unlike pressure drop and void fraction, it is not treated as a direct experimental validation quantity when no corresponding wall-shear measurement is available. Its role is instead to help interpret the hydraulic response of the model and, in particular, the influence of the liquid-film treatment.

In the present wall treatment, the liquid field occupies the solid boundary. The wall shear stress is consequently reconstructed from the velocity gradient of the water field. Using the liquid viscous stress tensor, the tangential wall traction can be written as

$$\boldsymbol{\tau}_w = (\mathbf{I} - \mathbf{n}_w \mathbf{n}_w) \cdot (2\mu_l \mathbf{S}_l) \cdot \mathbf{n}_w, \quad (3.9)$$

where  $\mathbf{n}_w$  is the wall-normal unit vector and  $\mathbf{S}_l$  is the strain-rate tensor of the liquid field.

For the streamwise analysis, its axial component is

$$\tau_{w,x} = \mathbf{e}_x \cdot \boldsymbol{\tau}_w. \quad (3.10)$$

The spatially averaged wall shear stress is evaluated over all solid channel walls,

$$\langle \tau_{w,x} \rangle_W(t) = \frac{1}{A_W} \int_{A_W} \tau_{w,x}(\mathbf{x}, t) \, dA, \quad (3.11)$$

where  $A_W$  is the total wall area included in the analysis. Its long-time mean is

$$\bar{\tau}_{w,x} = \frac{1}{T_{\text{avg}}} \int_{t_0}^{t_0+T_{\text{avg}}} \langle \tau_{w,x} \rangle_W(t) \, dt. \quad (3.12)$$

The signed streamwise component is retained in the averaging procedure so that a local reversal of the near-wall flow is not interpreted as positive friction. Local wall-shear distributions and magnitudes are considered separately when the spatial structure of the near-wall flow is analysed.

## 3.2. Reference Experiments

The three reference configurations are used for different parts of the assessment. Cubaud and Ho provide the main quantitative pressure-drop benchmark in a simple square microchannel [11]. Sadeghi Lafmejani et al. provide detailed visualisations of the morphology produced by distributed gas injection through a porous wall [26]. The Kim and Jung configuration is closest to the intended PEM-electrolyzer application and is used to examine gas-rich wall-injected flow [23].

### 3.2.1. Cubaud and Ho

Cubaud and Ho studied horizontal air–water flow in square microchannels with side lengths of 200  $\mu\text{m}$  and 525  $\mu\text{m}$  [11]. Gas and liquid were mixed in a cross-shaped inlet before entering the test section. The channel was made from silicon and glass and was partially wetting. The experimental arrangement is shown in Fig. 3.2. The cross-shaped inlet allows the gas and liquid flow rates to be imposed independently before the resulting two-phase flow enters the square test section where the pressure drop is measured.

The reported advancing contact angles were approximately 25° on glass and 9° on the DRIE silicon surface. The receding angles were close to zero. At these dimensions, the Bond number is small and gravity has little influence on the shape of the gas structures. The channel is also horizontal, so gravity does not contribute directly to the streamwise pressure difference. The measured pressure drop can then be compared with the hydraulic response of the numerical two-phase flow without applying a hydrostatic correction.

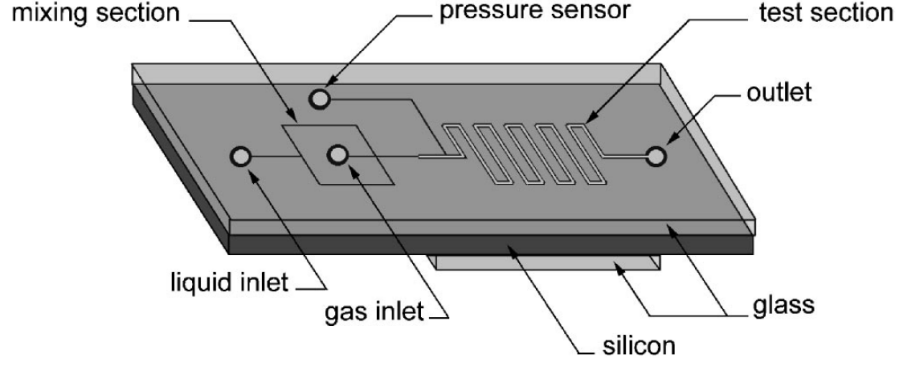


Figure 3.2: Experimental microchannel configuration used by Cubaud and Ho, showing the gas–liquid mixing section, pressure measurement, and test section [11].

Five flow regimes were identified experimentally: bubbly, wedging, slug, annular, and dry flow. Cubaud and Ho found the transitions at approximately

$$\alpha_L \simeq 0.75, \quad 0.20, \quad 0.04, \quad 0.005, \quad (3.13)$$

for bubbly–wedging, wedging–slug, slug–annular, and annular–dry flow, respectively [11]. The homogeneous liquid fraction is

$$\alpha_L = \frac{Q_l}{Q_l + Q_g}. \quad (3.14)$$

Figure 3.3 places the operating points selected for the numerical study on the experimental flow-regime map. The selected cases cover the transition from bubbly and wedging flow towards slug and gas-dominated conditions, allowing the model to be assessed across substantially different gas morphologies.

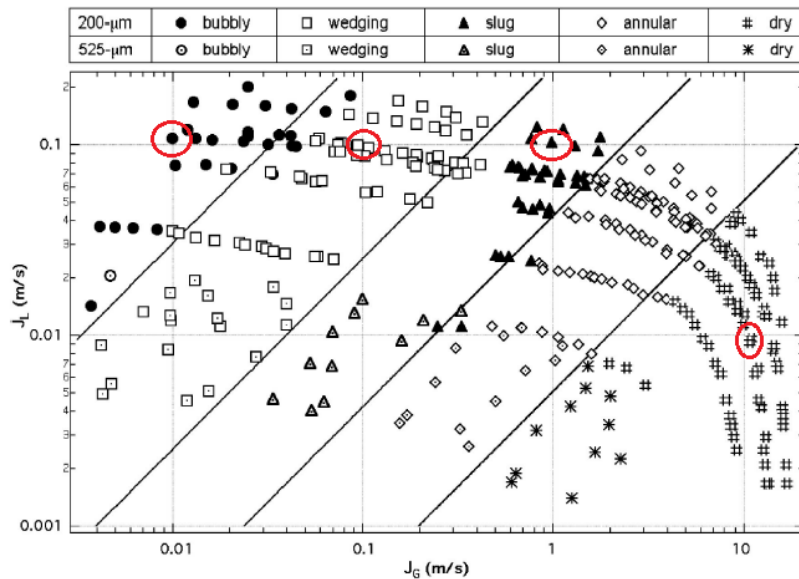


Figure 3.3: Experimental air–water flow-regime map of Cubaud and Ho [11]. The operating conditions selected for the numerical validation are highlighted in red

The transition values are specific to these square air–water microchannels and to their wetting conditions. They are used here as experimental reference points rather than as general flow-regime limits.

Pressure was measured only after the experiment had reached a steady state in terms of flow rates, bubble distribution, and pressure signal. The experimental value reported by Cubaud and Ho corresponds to an average over 1000 samples acquired during 20 s [11]. The numerical pressure signal is treated in the same way conceptually: the transient part of the calculation is removed and the pressure drop is averaged only once a statistically stationary state has been reached, as described in Section 3.1.1.

The single-phase liquid flow provides a first check independently of the multiphase model. For a square channel, the hydraulic resistance can be calculated from Eq. (2.35). The numerical single-phase pressure gradient is compared with both this analytical value and the experimental resistance before analysing the two-phase calculations.

Cubaud and Ho scaled the two-phase pressure drop with the pressure drop of a single liquid flow at the same liquid flow rate. For the bubbly and wedging regimes, their measurements are approximately described by

$$\frac{\Delta p_{2P}}{\Delta p_L} \simeq \alpha_L^{-1}. \quad (3.15)$$

At lower liquid fraction, the two phases develop a much larger slip and the liquid is increasingly transported close to the walls and in the corners. The experimental trend becomes closer to

$$\frac{\Delta p_{2P}}{\Delta p_L} \simeq \alpha_L^{-1/2}. \quad (3.16)$$

Both expressions are empirical correlations obtained from the measurements. They are not used as closures in the simulations. The calculated pressure ratios are compared with them afterwards.

The 200  $\mu\text{m}$  channel is used as the main Cubaud configuration. The smaller cross-section gives the most restrictive case in terms of interface and near-wall resolution.

### Cubaud computational domain

The numerical Cubaud case represents the 200  $\mu\text{m}$  square channel. The computational domain is shown in Fig. 3.4. Gas is introduced through the lateral inlet, while liquid enters through the two opposite branches of the cross-shaped mixing section. This reproduces the injection principle of the experimental configuration of Cubaud and Ho while retaining only the part of the geometry required for the numerical investigation.

Downstream of the mixing region, the channel cross-section is gradually expanded numerically before reaching the constant 200  $\mu\text{m} \times 200 \mu\text{m}$  test section. The mesh shown in Fig. 3.4 corresponds to the baseline discretization. The wall-refined mesh used for the mesh-sensitivity assessment is introduced separately in Section 3.3.2.

The pressure drop is extracted inside the constant-section part of the domain rather than directly between the numerical inlet and outlet. Two cross-sectional sampling planes are located at

$$x_1 = 3 \text{ mm}, \quad x_2 = 4 \text{ mm}. \quad (3.17)$$

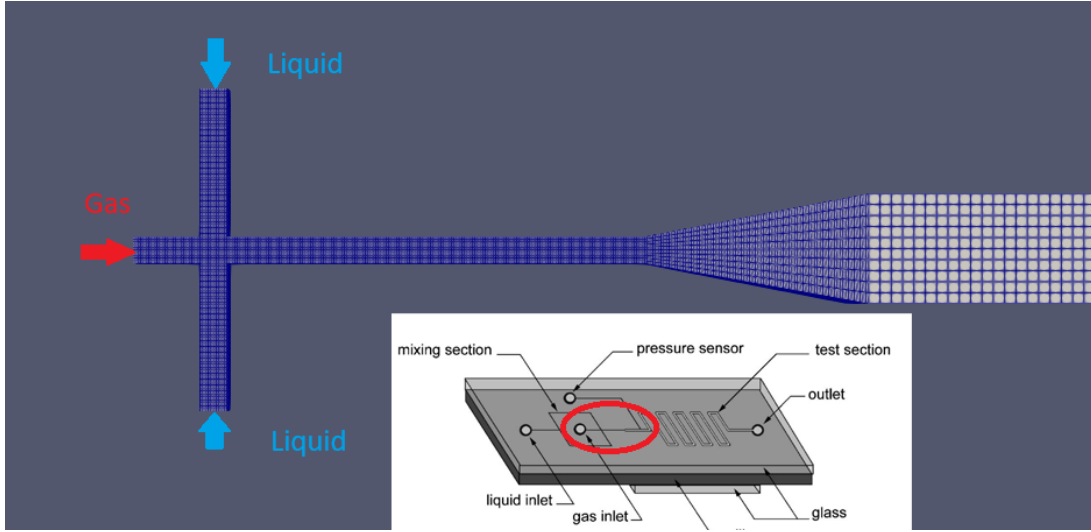


Figure 3.4: Computational domain and baseline mesh used for the Cubaud configuration. Cubaud and Ho; experimental schematic [11].

The pressure difference is therefore evaluated over

$$L_p = |x_2 - x_1| = 1 \text{ mm}. \quad (3.18)$$

The same two sections are used for the corresponding single-phase calculation. Using internal sampling planes avoids including regions directly affected by the inlet and outlet boundary conditions in the pressure-drop evaluation.

The field boundary conditions used for the Cubaud configuration are summarized in Tables 3.1 and 3.2. Only the boundary-condition types are reported here; the imposed inlet values depend on the selected operating point.

Table 3.1: Velocity and pressure boundary conditions used for the Cubaud configuration.

| Patch            | $\mathbf{u}_l$ | $\mathbf{u}_{gC}$ | $\mathbf{u}_{gD}$ | $p_{rgh}$         |
|------------------|----------------|-------------------|-------------------|-------------------|
| liquidInletLeft  | fixedValue     | zeroGradient      | zeroGradient      | fixedFluxPressure |
| liquidInletRight | fixedValue     | zeroGradient      | zeroGradient      | fixedFluxPressure |
| gasInlet         | zeroGradient   | fixedValue        | zeroGradient      | fixedFluxPressure |
| outlet           | zeroGradient   | zeroGradient      | zeroGradient      | fixedValue        |
| walls            | noSlip         | noSlip            | noSlip            | fixedFluxPressure |
| frontAndBack     | noSlip         | noSlip            | noSlip            | fixedFluxPressure |

Table 3.2: Phase-fraction boundary conditions used for the Cubaud configuration.

| Patch            | $r_l$             | $r_{gC}$          | $r_{gD}$     |
|------------------|-------------------|-------------------|--------------|
| liquidInletLeft  | fixedValue        | inletOutlet       | fixedValue   |
| liquidInletRight | fixedValue        | inletOutlet       | fixedValue   |
| gasInlet         | inletOutlet       | fixedValue        | fixedValue   |
| outlet           | inletOutlet       | zeroGradient      | inletOutlet  |
| walls            | alphaContactAngle | alphaContactAngle | zeroGradient |
| frontAndBack     | alphaContactAngle | alphaContactAngle | zeroGradient |

No dispersed gas is imposed at the boundaries in the Cubaud configuration. The gas enters directly through the continuous field  $g_C$ , while any gas subsequently appearing in  $g_D$  results from the internal morphology-transfer mechanism.

The Lafmejani configuration differs fundamentally from the Cubaud case in the gas-injection mechanism: gas enters progressively through a wall rather than through a dedicated channel inlet.

### 3.2.2. Lafmejani

The Lafmejani experiment provides detailed morphological observations of gas injection through a porous wall.

Lafmejani et al. [26] constructed an ex-situ experiment to study gas–liquid transport close to the porous transport layer of a PEM electrolyzer without including electrochemistry.

$$5 \text{ mm} \times 1 \text{ mm} \times 94 \text{ mm}. \quad (3.19)$$

One long side of the channel is covered by a 0.35 mm-thick titanium felt. Air crosses this porous layer and enters water flowing upward from the bottom of the channel. The Plexiglas test section gives optical access over nearly the complete channel length.

Unlike the Cubaud configuration, the gas flow is not introduced at a single inlet. Gas enters progressively through the porous wall, so the gas content changes with streamwise position.

The experiment also shows that the injection through the titanium felt is not spatially uniform. Air first penetrates at a few preferential positions. New paths through the porous material appear as the imposed gas flow is increased. Bubbles generated at these locations move upward and merge with other bubbles further downstream [26].

At  $Q_w = 5.3 \text{ mL min}^{-1}$ , the air supply was increased from 2.6 to  $31.2 \text{ L min}^{-1}$ . Other measurements were performed with

$$Q_w = 10.6 \text{ mL min}^{-1}. \quad (3.20)$$

The visualisations contain bubbly regions, mixed bubbly–Taylor flow, Taylor bubbles, and annular structures. The morphology can also change along one channel because the gas continues to enter through the porous wall.

The individual gas paths inside the experimental titanium felt are not represented in the numerical case. Gas injection is described at the boundary of the fluid domain instead. The comparison therefore does not attempt to reproduce the position of each experimental bubble. It is used to examine whether dispersed gas injected along the wall can accumulate and form larger continuous structures downstream.

#### Lafmejani computational domain

The Lafmejani domain follows the dimensions of the experimental test section and consists of a

$$5 \text{ mm} \times 1 \text{ mm} \times 94 \text{ mm} \quad (3.21)$$

vertical channel. One long wall represents the titanium felt and is used for gas injection.

Table 3.3: Velocity and pressure boundary conditions used for the Lafmejani configuration.

| Patch    | $\mathbf{u}_l$                  | $\mathbf{u}_{gC}$ | $\mathbf{u}_{gD}$ | $p_{rgh}$         |
|----------|---------------------------------|-------------------|-------------------|-------------------|
| inlet    | flowRateInletVelocityfixedValue |                   | fixedValue        | fixedFluxPressure |
| outlet   | zeroGradient                    | zeroGradient      | zeroGradient      | fixedValue        |
| walls    | noSlip                          | noSlip            | noSlip            | fixedFluxPressure |
| top      | noSlip                          | noSlip            | noSlip            | fixedFluxPressure |
| reaction | noSlip                          | noSlip            | fixedValue        | fixedFluxPressure |

Table 3.4: Phase-fraction boundary conditions used for the Lafmejani configuration.

| Patch    | $r_l$             | $r_{gC}$          | $r_{gD}$          |
|----------|-------------------|-------------------|-------------------|
| inlet    | fixedValue        | fixedValue        | fixedValue        |
| outlet   | inletOutlet       | inletOutlet       | inletOutlet       |
| walls    | fixedValue        | fixedValue        | fixedValue        |
| top      | fixedValue        | fixedValue        | fixedValue        |
| reaction | alphaContactAngle | alphaContactAngle | alphaContactAngle |

The field boundary conditions used for the Lafmejani configuration are summarized in Tables 3.3 and 3.4. The **reaction** patch represents the gas-injection surface corresponding to the porous titanium wall. Gas injection is imposed through the dispersed gas field  $g_D$ , while the continuous gas field  $g_C$  is not directly injected at this boundary.

Unlike the Cubaud configuration, the gas therefore enters the numerical domain through a wall boundary and is initially represented by the dispersed field  $g_D$ . The formation of larger continuous gas structures inside the channel consequently requires transfer from  $g_D$  to  $g_C$  through the morphology-transfer model.

The Kim configuration retains the principle of wall gas injection but uses a simplified channel segment derived directly from an operating PEM water electrolyzer.

### 3.2.3. Kim

The previous cases were not in-situ cases. The Kim case is a more application-oriented case, where Kim and Jung studied the two-phase flow inside an operating PEM water electrolyzer, which is shown in Fig. 3.5, with single-serpentine, double-serpentine, and parallel anode flow fields [23]. Oxygen was generated by the electrochemical reaction and entered the channel from the porous transport layer instead of through a separate gas inlet. Optical access is provided on the anode side, where the generated oxygen enters the flow field through the porous transport layer. The goal of this setup is to make the configuration closer to an operating electrolyzer than the preceding ex-situ reference cases.

The active area is 25 cm<sup>2</sup>. Each flow channel had a width of 3 mm, with a rib width of 2.2 mm. A titanium mesh with a thickness of 0.27 mm and a pore diameter of approximately 0.1 mm was used as porous transport layer. Gravity is opposed to the flow. Measurements were performed at 50°C and 70°C. The pressure-drop data considered here correspond to 70°C. The supplied water flow rate was calculated with an electro-stoichiometric ratio

$$\lambda_{st} = 50. \quad (3.22)$$

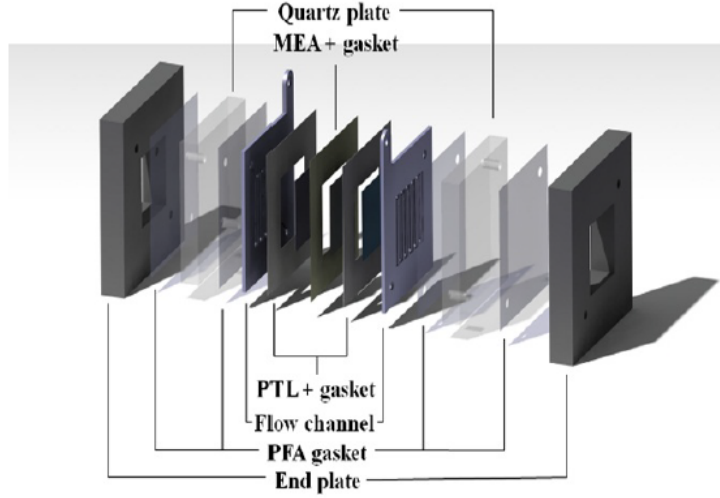


Figure 3.5: Exploded view of the PEM water electrolyzer used by Kim and Jung. Optical access to the anode flow channel was obtained through the quartz plate [23].

Kim and Jung measured the pressure difference between the inlet and outlet of the anode flow field and expressed the effect of the gas using the two-phase multiplier

$$\Phi_w^2 = \frac{\Delta p_{2P}}{\Delta p_w}, \quad (3.23)$$

where  $\Delta p_w$  is the pressure drop with only water flowing through the same channel. Values were reported at

$$j = 0.2, 0.52, 1.0, 2.0 \text{ A cm}^{-2}. \quad (3.24)$$

The observed morphology depends strongly on the flow-field geometry. At high current density, the parallel channels contain Taylor bubbles with a width close to the full channel width [23].

Electrochemistry is not included in the present simulations. The generated oxygen and supplied water are converted into equivalent gas and liquid flow rates and then imposed for the numerical case at the inlet for water and at the wall for gas. At  $j = 2 \text{ A cm}^{-2}$ , the current over the  $25 \text{ cm}^2$  active area is

$$I = j A_{\text{act}} = 50 \text{ A}. \quad (3.25)$$

The oxygen production follows Faraday's law,

$$\dot{n}_{\text{O}_2} = \frac{I}{4F}, \quad (3.26)$$

where  $F$  is Faraday's constant. After conversion to the operating conditions and distribution over the channel represented numerically, the corresponding gas and liquid flow rates are approximately

$$Q_g \simeq 3.64 \times 10^{-7} \text{ m}^3 \text{ s}^{-1}, \quad (3.27)$$

and

$$Q_l \simeq 2.39 \times 10^{-8} \text{ m}^3 \text{ s}^{-1}. \quad (3.28)$$

The homogeneous liquid fraction is then

$$\alpha_L = \frac{Q_l}{Q_l + Q_g} \simeq 0.062. \quad (3.29)$$

Only a small part of the imposed volumetric flow is therefore liquid. Even though the goal is to simulate an in-situ electrolyzer, this setup is useful for the wall-film problem because the gas slug occupies most of the channel while the remaining water has to be transported near the solid boundaries or between the long bubbles. So an under-resolution of the effect of the film will still be highlighted.

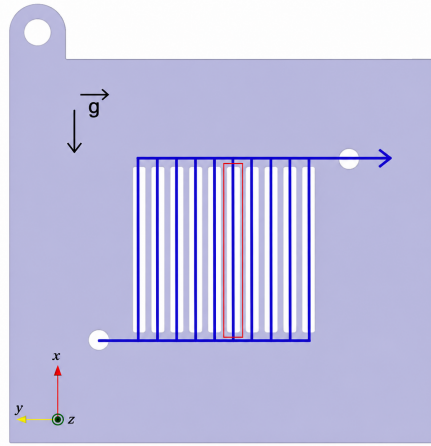


Figure 3.6: Selection of the straight channel segment used to construct the simplified Kim numerical configuration. The blue line indicates the experimental flow path, while the red region indicates the section represented numerically.

### Kim computational domain

The computational domain can be seen in Fig. 3.7. This domain consists of a vertical straight channel. Liquid water enters through the bottom boundary, while gas is introduced through the side boundary representing the porous transport layer side of the anode channel. The resulting gas-liquid mixture leaves the domain through the top outlet.

This simplified setup does not reproduce the complete experimental pressure-loss path of the electrolyzer, which also includes bends and the streamwise evolution of gas generation along the full serpentine. Its purpose is instead to isolate a representative local channel section in which gas accumulates, interacts with the liquid flow, and forms gas-rich morphologies comparable to those observed experimentally. The selected channel section can be seen in Fig. 3.6.

The field boundary conditions used for the Kim configuration are summarized in Tables 3.5 and 3.6. Water enters through the bottom **inlet**, whereas oxygen is introduced through the **reaction** patch representing the porous transport layer. The injected gas is assigned to the dispersed field  $g_D$ ; the continuous gas field  $g_C$  is not imposed through this boundary.

The gas injection is restricted to the active part of the **reaction** patch through the coded boundary conditions applied to  $r_{g_D}$  and  $\mathbf{u}_{g_D}$ . Gas therefore enters the fluid domain initially

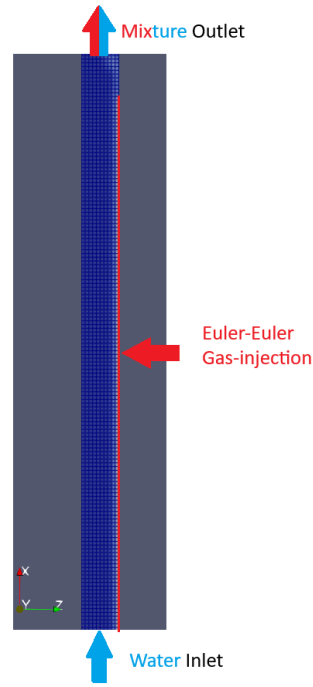


Figure 3.7: Simplified computational domain used for the Kim configuration.

Table 3.5: Velocity and pressure boundary conditions used for the Kim configuration.

| Patch    | $\mathbf{u}_l$        | $\mathbf{u}_{gC}$ | $\mathbf{u}_{gD}$ | $p_{rgh}$         |
|----------|-----------------------|-------------------|-------------------|-------------------|
| inlet    | flowRateInletVelocity | fixedValue        | fixedValue        | fixedFluxPressure |
| outlet   | zeroGradient          | zeroGradient      | zeroGradient      | fixedValue        |
| walls    | noSlip                | noSlip            | noSlip            | fixedFluxPressure |
| top      | noSlip                | noSlip            | noSlip            | fixedFluxPressure |
| reaction | noSlip                | noSlip            | codedFixedValue   | fixedFluxPressure |

Table 3.6: Phase-fraction boundary conditions used for the Kim configuration.

| Patch    | $r_l$             | $r_{gC}$          | $r_{gD}$        |
|----------|-------------------|-------------------|-----------------|
| inlet    | fixedValue        | fixedValue        | fixedValue      |
| outlet   | inletOutlet       | inletOutlet       | inletOutlet     |
| walls    | alphaContactAngle | alphaContactAngle | zeroGradient    |
| top      | alphaContactAngle | alphaContactAngle | zeroGradient    |
| reaction | alphaContactAngle | alphaContactAngle | codedFixedValue |

in the dispersed representation. Larger connected gas structures can subsequently appear only after accumulation and transfer from  $g_D$  to  $g_C$  through the MultiMorph morphology-transfer model. The coded boundary condition restricts gas injection to the active 50 mm-long section of the `reaction` patch, corresponding to  $0.05 < x < 0.10$  m.

### 3.3. Numerical setup

The numerical model remains the same for the three reference cases. The domain, physical properties, gravity direction, and imposed boundary conditions are changed according to the corresponding experiment. Components outside the test section, such as pumps, reservoirs, tubing, and pressure-sensor lines, are not part of the computational domains.

#### 3.3.1. Solver and physical properties

The calculations use the MultiMorph formulation introduced in Chapter 2. Gas is divided between the continuous field  $g_C$  and the dispersed field  $g_D$ , while water is represented by the continuous liquid field  $l$ . The total gas fraction used for post-processing is

$$r_g = r_{g_C} + r_{g_D}. \quad (3.30)$$

All multiphase simulations are transient. Even when the final quantity is a time-averaged pressure drop, the phase distribution changes with time as gas structures move, merge, and transfer between the two numerical gas representations.

The fluid properties are kept constant within each case. Cubaud and Lafmejani use air and water close to ambient conditions. For the Cubaud calculations, the liquid properties are  $\rho_l = 998 \text{ kg m}^{-3}$ ,  $\mu_l = 1.002 \times 10^{-3} \text{ Pa s}$ .

The Kim pressure measurements used for comparison were performed at  $70^\circ\text{C}$ . The corresponding water properties used in the numerical case are  $\rho_l = 977.63 \text{ kg m}^{-3}$ ,  $\mu_l = 4.035 \times 10^{-4} \text{ Pa s}$ .

Surface tension is retained in the multiphase calculations. Gravity follows the orientation of the experiment. For the horizontal Cubaud channel, it does not contribute to the stream-wise hydrostatic pressure difference. The vertical configurations retain the gravitational term.

An adaptive time step is used. Interface motion and morphology transfer can change the local velocity and phase-fraction gradients during a calculation, so one fixed time step would either be unnecessarily restrictive over part of the simulation or too large during the most demanding events. The time-step evolution is monitored together with the pressure signal, especially for the wall-refined cases where the smallest cells are located close to the solid boundary.

#### 3.3.2. Mesh strategy

The mesh must represent two different scales. Large gas structures need enough cells for the continuous field and the interface treatment, while the remaining liquid can be confined to much smaller wall and corner regions.

Structured hexahedral cells are used in the straight-channel regions. The cross-sectional resolution directly affects the way MultiMorph separates resolved and unresolved structures because the continuous-structure identification depends on the phase distribution over neighbouring cells.

The baseline mesh uses an approximately uniform resolution over most of the cross-section. A second mesh increases the resolution close to the walls. The change is concentrated in the wall-normal direction rather than being a uniform refinement of the complete domain.

The mesh comparison uses the time-averaged pressure drop, in-situ gas fraction, predicted morphology, liquid distribution close to the corners, and wall shear stress. These quantities do not necessarily converge at the same rate. A gas core can already be represented over several cells while the liquid film between this core and the wall remains smaller than one near-wall cell.

An estimate of the film resolution can be written as

$$N_f = \frac{\delta_f}{\Delta y_w}, \quad (3.31)$$

where  $\delta_f$  is the estimated physical film thickness and  $\Delta y_w$  is the characteristic wall-normal size of the first cell. When  $N_f$  is of order one or smaller, the calculation cannot reconstruct a full velocity profile across the film. This index is used in the following mesh study to classify the different discretizations with respect to their ability to represent the liquid film. For  $N_f < 1$ , the estimated film is thinner than one computational cell and is therefore clearly under-resolved. A value  $N_f \simeq 1$  indicates that approximately one cell is available across the film, whereas  $N_f > 1$  corresponds to several cells across the same physical length scale. The latter does not by itself guarantee a grid-converged film solution, but provides a useful measure for comparing the relative film resolution of the different meshes. For the Cubaud mesh sequence considered in this work, all meshes remain in the  $N_f < 1$  range except the finest stable mesh, which reaches approximately  $N_f = 1$ . Even the finest mesh therefore only reaches the onset of geometrical film resolution and cannot be considered sufficient to resolve the velocity profile within the film. Mesh refinement also changes the scale at which a gas structure is interpreted by the hybrid model. It is not only a reduction of the spatial discretisation error. A structure that is unresolved on one mesh can become part of the continuous representation after refinement.

### 3.3.3. Post-processing

The total gas fraction is reconstructed from

$$r_g = r_{gC} + r_{gD}. \quad (3.32)$$

The total fraction is used to visualise the physical gas distribution, while  $r_{gC}$  and  $r_{gD}$  are also inspected separately. The same total gas structure can contain different proportions of continuous and dispersed gas, especially in regions where morphology transfer is active.

Pressure is averaged over complete cross-sections using the procedure defined in Eq. (3.2). The instantaneous pressure drop is calculated from the difference between both section-averaged values and is then averaged over the stationary part of the simulation.

For Cubaud, the main pressure quantity is

$$\Pi_{\Delta p} = \frac{\overline{\Delta p_{2P}}}{\overline{\Delta p_L}}, \quad (3.33)$$

where the two pressure differences are evaluated between the same two cross-sections.

The single-phase result is also checked with the square-channel relation

$$\Delta p_{1P} = 28.43 \frac{\mu L Q}{h^4} \cdot [11] \quad (3.34)$$

A discrepancy already present in the single-phase calculation would then be separated from the additional error introduced by the multiphase model.

For a vertical channel, the measured pressure difference contains a hydrostatic contribution. The mixture density is estimated from the in-situ phase fractions,

$$\bar{\rho}_m = \bar{\varepsilon}_l \rho_l + \bar{\varepsilon}_g \rho_g. \quad (3.35)$$

The corrected pressure difference over a height  $H$  is

$$\Delta p_{\text{corr}} = \Delta p_{\text{meas}} - \bar{\rho}_m g H. \quad (3.36)$$

The in-situ fraction is used instead of the inlet homogeneous fraction because gas and liquid do not necessarily travel at the same mean velocity.

Wall shear stress is extracted on the solid surfaces using Eq. (3.9). Both the spatial distribution and the area-averaged value are retained. This makes it possible to distinguish a change in global pressure drop from a local redistribution of the wall shear.

Velocity profiles are also sampled normal to the wall at selected streamwise positions. The liquid and gas velocities are extracted together with the local phase fractions. For gas-dominated cases, these profiles show over which numerical distance the liquid velocity changes between the no-slip wall and the faster flow inside the channel.

### 3.4. Sensitivity analysis

Several MultiMorph parameters control the numerical representation of the morphology rather than a directly measurable fluid property. Their influence is assessed by changing one parameter while keeping the flow rates, physical properties, and mesh unchanged.

The comparison uses the time-averaged pressure drop, total gas fraction, distribution between  $g_C$  and  $g_D$ , observed morphology, and wall shear stress.

#### 3.4.1. Morphology-transfer time scale

The morphology-transfer time scale was varied over

$$\tau_{PT} = 0.50, 0.10, 0.05, 0.01 \text{ s}. \quad (3.37)$$

All other model parameters and the mesh were kept unchanged. The continuous and dispersed gas fields were analysed separately in addition to the total gas fraction.

#### 3.4.2. Continuous threshold

The parameter `continuousThreshold` is defined for the continuous gas field  $g_C$  in the `phaseProperties` dictionary. It is distinct from the value 0.74 used to identify the core of a continuous structure in Section 2.3.3. The two thresholds have different roles.

The tested values are

$$r_{C,\text{th}} = 0.20, 0.30, 0.40, 0.60, 0.75. \quad (3.38)$$

The value enters the morphology-transfer criteria associated with the continuous gas representation. In the `disintegrationToDisperse` model, it appears in the condition introduced in Eq. (2.67). Decreasing  $r_{C,\text{th}}$  makes this condition easier to satisfy, while increasing it restricts the transfer to cells containing a larger total fraction of continuous material.

The same parameter can also be used by the reverse `accumulationToContinuous` transfer when no separate `alphaThreshold` is specified, as described in Section 2.3.4. Varying `continuousThreshold` can therefore modify the distribution between  $g_C$  and  $g_D$  and the point at which the numerical representation changes.

The comparison focuses on the distribution between the continuous and dispersed gas fields, the resulting large-scale morphology, and the pressure drop. The threshold is expected to have its strongest effect in regions where the gas is close to the transition between the two numerical representations.

### 3.4.3. Bubble diameter

The Euler–Euler closures require a characteristic diameter for the dispersed gas structures. The bubble diameter changes the modelled interfacial area and the momentum exchange of  $g_D$ . Once gas has been transferred to the continuous field, this dispersed-bubble diameter no longer describes its geometry.

Four values are tested,

$$d_b = 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3} \text{ m.} \quad (3.39)$$

The sweep covers three orders of magnitude. The value  $d_b = 10^{-4} \text{ m}$  is of the same order as the pore scale used in the PEM-related configuration, while the other values test the sensitivity of the unresolved gas description to smaller and larger assumed dispersed structures.

For spherical bubbles, the corresponding interfacial area density is

$$a_i = \frac{6r_{g_D}}{d_b}. \quad (3.40)$$

At fixed dispersed gas fraction, the interfacial area density is inversely proportional to  $d_b$ . The difference between  $10^{-6}$  and  $10^{-3} \text{ m}$  therefore changes the modelled interfacial area by a factor of 1000. The resulting pressure drop, distribution between  $g_C$  and  $g_D$ , and large-scale morphology are compared to determine how strongly the assumed unresolved bubble size affects the complete flow.

### 3.4.4. Liquid-film boundary condition

The last sensitivity study concerns the wall treatment in gas-dominated flow. At low liquid fractions, most of the channel can be occupied by gas while the liquid remains close to the walls and in the corners.

With the baseline wall treatment, the resolved gas interface can approach the wall according to the standard wetting condition. When the physical film is smaller than the near-wall grid spacing, the numerical phase distribution can no longer represent the complete liquid layer between the gas and the solid.

The modified condition imposes

$$r_l = 1 \quad (3.41)$$

at the wall. Gas can still occupy the neighbouring cells, but not the wall face itself.

The pressure drop and morphology alone are not sufficient to evaluate the effect of this change. The wall shear and wall-normal velocity profiles are also compared. According to the mechanism discussed in Section 2.1.4, the liquid wall shear depends on the velocity gradient close to the solid boundary. If the velocity change between the stationary wall and the gas-dominated core is distributed over too large a numerical distance, this gradient can remain too small even when some liquid is present next to the wall.

A larger wall shear together with a steeper liquid velocity gradient would be consistent with the expected effect of keeping liquid at the solid boundary. If the phase distribution changes without a comparable change in the velocity profile, the boundary condition has mainly modified the local morphology rather than the missing near-wall momentum gradient.

The adaptive time step is monitored for the same cases. The liquid-film condition can produce sharper phase and velocity variations close to the refined wall cells, and the resulting numerical cost is included in the comparison.



## 4. Results

This chapter presents the numerical results for the three reference configurations in order of increasing proximity to an operating electrolyzer. The Cubaud and Ho [11] benchmark first provides a controlled hydraulic reference, the configuration of Sadeghi Lafmejani et al. [26] then examines distributed gas injection through a porous wall, and the Kim and Jung [23] case finally considers conditions derived from an operating PEM water electrolyzer. Across the three cases, pressure response, gas morphology, phase distribution, and near-wall behaviour are analysed using the quantities defined in Chapter 3.

### 4.1. Cubaud and Ho configuration

The Cubaud and Ho configuration provides the main quantitative benchmark for the horizontal two-phase-flow calculations. Since the channel is horizontal, gravity does not contribute directly to the streamwise pressure difference. The measured pressure response can therefore be compared directly with the hydraulic resistance generated by the gas–liquid flow.

The selected operating points span the bubbly, wedging, slug and gas-dominated regions of the experimental flow map. The analysis first considers the single-phase hydraulic resistance and the resulting flow morphologies. The normalized two-phase pressure drop is then compared with the two experimental scaling regimes reported by Cubaud and Ho. Particular attention is paid to the transition towards low liquid fractions, for which the remaining liquid becomes confined increasingly to wall films and corner regions.

#### 4.1.1. Single-phase reference

Before analysing the two-phase calculations, the hydraulic resistance of the numerical channel was verified using a liquid-only calculation. The pressure difference was evaluated between the two internal sampling planes at  $x_1 = 3$  mm and  $x_2 = 4$  mm, corresponding to a measurement length of 1 mm.

For the 200  $\mu\text{m}$  square channel and a liquid superficial velocity of  $J_L = 0.1$   $\text{m s}^{-1}$ , the liquid flow rate is

$$Q_L = J_L h^2 = 4.0 \times 10^{-9} \text{ m}^3 \text{ s}^{-1}. \quad (4.1)$$

Cubaud and Ho reported an experimental liquid resistance per unit length of approximately

$$\frac{R_{L,\text{exp}}}{L} = 1.64 \times 10^{13} \text{ kg m}^{-5} \text{ s}^{-1} \quad (4.2)$$

for the 200  $\mu\text{m}$  channel [11]. Over the present sampling distance this corresponds to approximately  $\Delta p_{L,\text{exp}} = 65.6 \text{ Pa}$ .

The analytical square-channel relation gives approximately  $\Delta p_{L,\text{th}} = 71.2 \text{ Pa}$ , whereas the numerical calculation gives  $\Delta p_{L,\text{num}} \simeq 68.6 \text{ Pa}$ .

The numerical value therefore lies between the experimental resistance and the analytical prediction, with deviations of only a few percent. The single-phase hydraulic resistance of the computational domain is therefore sufficiently accurate for the subsequent two-phase comparison.

#### 4.1.2. Flow morphology and regime transition

The numerical morphology is compared first with the experimental flow-regime classification. Figure 4.2 shows representative distributions of the total gas fraction,

$$r_g = r_{gC} + r_{gD}, \quad (4.3)$$

from gas-dominated conditions to bubbly flow. Representative experimental morphologies are shown in Fig. 4.1. These images provide the visual reference used to assess whether the numerical cases reproduce the characteristic phase connectivity and confinement of the different regimes.

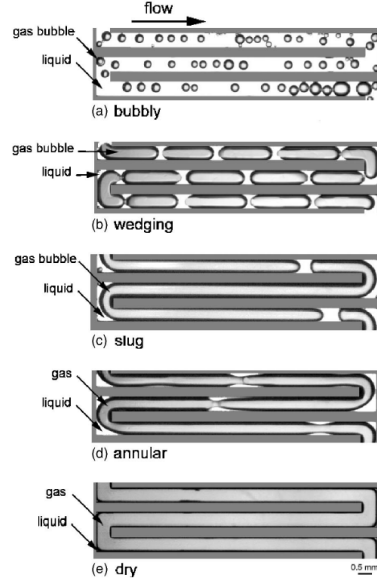


Figure 4.1: Experimental results from Cubaud and Ho. [11].

The lower-liquid-fraction cases were obtained with the refined mesh to improve the representation of the near-wall liquid regions. At high liquid fraction, the gas remains separated into individual structures surrounded by a continuous liquid phase. The numerical solution therefore reproduces the defining topology of the experimental bubbly regime.

As the gas flow rate increases, the bubbles become confined by the channel and progressively occupy a larger part of its cross-section. At lower liquid fractions, the model captures

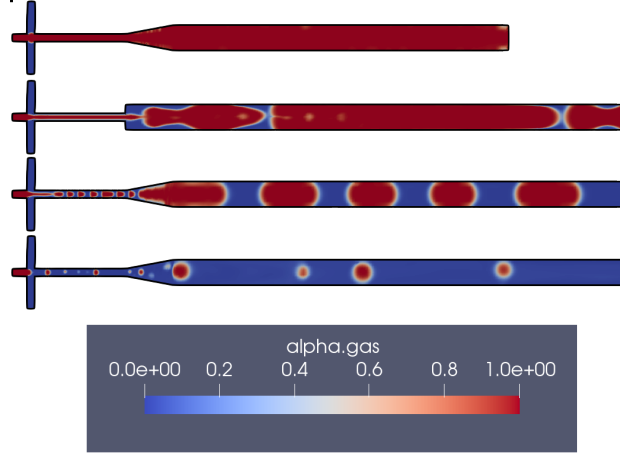


Figure 4.2: Numerical total gas-fraction fields for representative operating points spanning the flow-regime map of Cubaud and Ho. Gas is in red, water is in blue.

the transition from discrete bubbles towards channel-confined gas structures. The precise experimental contact-line geometry is not reproduced in detail, which is expected since the wedging regime is particularly sensitive to partial wetting and dewetting dynamics. Cubaud and Ho showed that wedging bubbles may contain dry, partially lubricated or fully lubricated wall regions depending on their velocity [11].

At lower liquid fraction, the numerical gas structures become strongly elongated and form slug-like morphologies. This agrees with the experimental observation that, below approximately  $\alpha_L = 0.2$ , gas bubbles become much longer than the channel width while remaining surrounded by liquid. Cubaud and Ho also report that the liquid film between such gas structures and the wall can become very thin and may locally dry out [11].

At the lowest liquid fractions considered numerically, the gas occupies almost the complete visible channel section. Only a small amount of liquid remains available close to the wall and corner regions. The numerical sequence therefore reproduces the main change in phase connectivity across the experimental regime map, from dispersed gas to channel-confined structures and finally to a gas-dominated state.

The model captures the large-scale morphology over a wide range of operating conditions, while the much smaller liquid film can remain unresolved, which becomes important for the pressure-drop comparison below.

#### 4.1.3. Pressure-drop validation on the baseline mesh

The hydraulic response is first evaluated using the baseline mesh. The two-phase pressure drop is normalized by the corresponding single-phase liquid pressure drop,

$$\Pi_{\Delta p} = \frac{\Delta p_{2P}}{\Delta p_L}. \quad (4.4)$$

Cubaud and Ho identified two different dependencies of this pressure multiplier on the homogeneous liquid fraction. In the bubbly and wedging regimes, corresponding to relatively high liquid fractions, the experimental results approximately follow

$$\Pi_{\Delta p} \simeq \alpha_L^{-1}. \quad (4.5)$$

At lower liquid fractions, once elongated gas structures occupy most of the channel and the liquid is transported increasingly through wall films and corner regions, the dependence becomes weaker,

$$\Pi_{\Delta p} \simeq \alpha_L^{-1/2}. \quad (4.6)$$

Figure 4.3 compares these two experimental trends with the numerical results obtained using the baseline mesh.

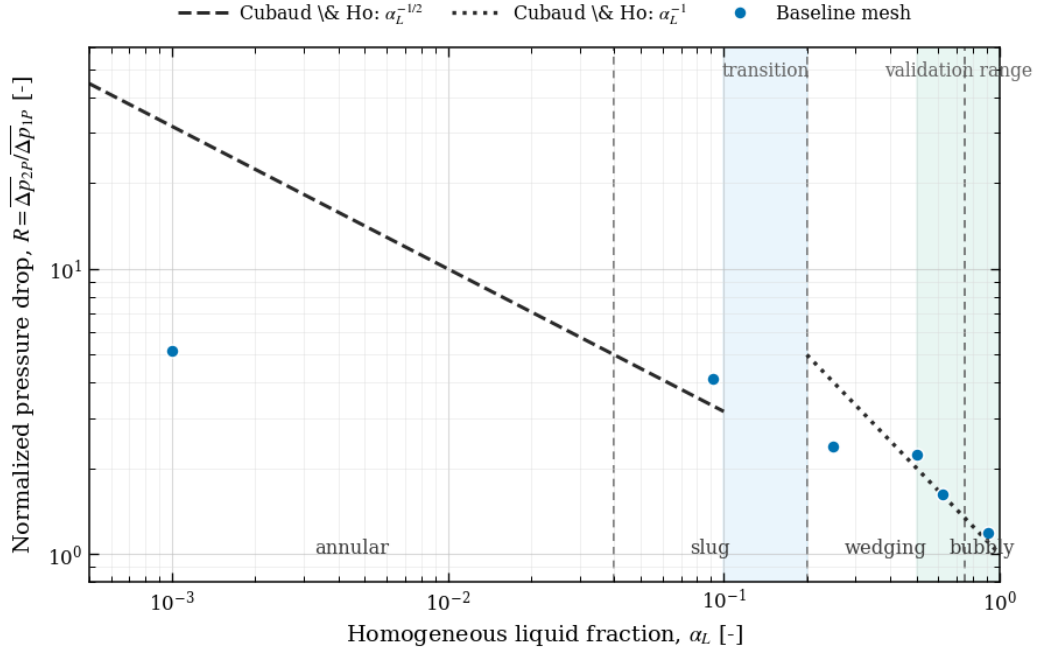


Figure 4.3: Normalized two-phase pressure drop obtained with the baseline mesh as a function of the homogeneous liquid fraction.[11].

In the high-liquid-fraction range, the numerical results reproduce the experimental behaviour well. The bubbly and wedging cases remain close to the  $\Pi_{\Delta p} \propto \alpha_L^{-1}$  relation, and both the magnitude and the increase in pressure drop with decreasing liquid fraction are captured.

The agreement deteriorates progressively when the liquid fraction is reduced. Around the transition towards slug flow, the numerical pressure multiplier begins to deviate from the experimental low-liquid-fraction behaviour. The discrepancy becomes substantially larger for the most gas-dominated cases. At high  $\alpha_L$ , a significant fraction of the channel cross-section remains occupied by liquid. At low  $\alpha_L$ , most of the gas is contained in large channel-spanning structures and the remaining liquid is increasingly restricted to thin wall films and corner regions. The characteristic length scale controlling the liquid-side dissipation therefore becomes much smaller than the channel width.

The baseline mesh still captures the large gas structures, but becomes too coarse to describe the increasingly thin near-wall liquid regions. This motivates the wall-normal refinement examined below.

#### 4.1.4. Influence of near-wall mesh refinement

The influence of spatial resolution was examined in more detail for the Cubaud operating point at  $\alpha_L = 0.25$ . Four successively refined meshes were used, ranging from approximately  $6 \times 10^4$  to  $4 \times 10^5$  computational cells. Since the refinement is non-uniform, the total number of cells alone does not provide a direct measure of the resolution available for the near-wall liquid film. The meshes are therefore also characterized using the film-resolution index  $N_f$  introduced in Eq. (3.31).

Table 4.1 summarizes the first wall-adjacent cell size, and the resulting value of  $N_f$ , together with the calculated pressure drop.

Table 4.1: Mesh sensitivity for the Cubaud case at  $\alpha_L = 0.25$ . The film-resolution index is defined as  $N_f = \delta_f / \Delta y_w$ , using an estimated film thickness of  $\delta_f \simeq 2.3 \mu\text{m}$ .

| Mesh          | $N_{\text{cells}}$ | $\Delta y_w$ [ $\mu\text{m}$ ] | $N_f$ | $\overline{\Delta p_{2P}}$ [Pa] | $\Pi_{\Delta p}$ |
|---------------|--------------------|--------------------------------|-------|---------------------------------|------------------|
| Baseline      | 56 000             | 20.00                          | 0.12  | 272.09                          | 3.955            |
| Mesh 1        | 113 500            | 6.67                           | 0.34  | 276.81                          | 4.024            |
| Mesh 2        | 218 500            | 3.41                           | 0.68  | 281.56                          | 4.093            |
| Finest stable | 365 500            | 2.29                           | 1.01  | 282.66                          | 4.109            |

Figure 4.4 shows the time-averaged two-phase pressure drop obtained with the four mesh levels.

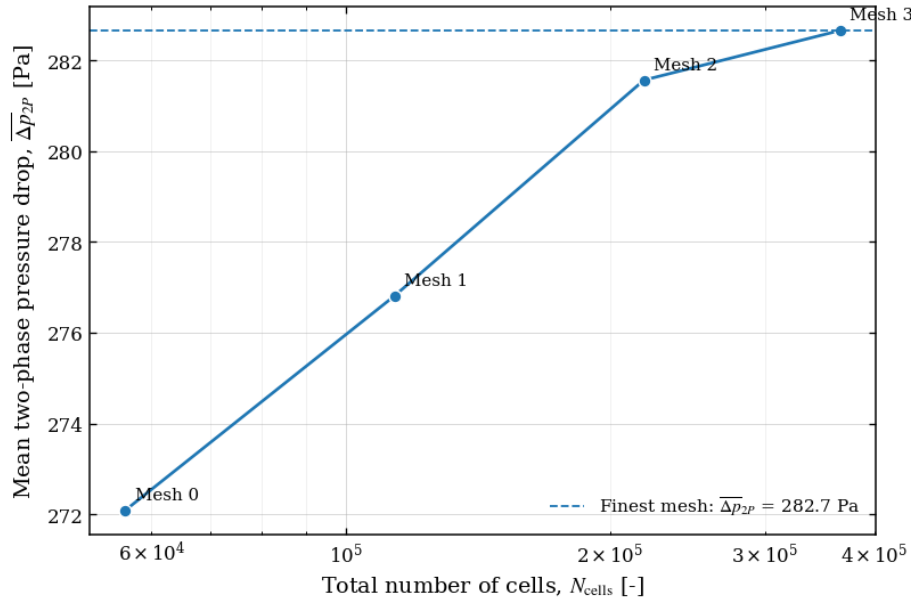


Figure 4.4: Mesh-sensitivity study for the Cubaud case at  $\alpha_L = 0.25$ . The dashed line indicates the pressure drop obtained with the finest stable mesh.

The calculated pressure drop increases systematically with refinement. The change remains appreciable between the first three meshes, indicating that the coarser discretizations do not yet provide a mesh-independent hydraulic response. Between the two finest meshes, however, the increase becomes much smaller. The finest stable mesh gives a mean pressure drop of approximately 282.7 Pa, while the preceding mesh already lies close to this value.

The reduction of the change between the last two refinement levels suggests that the solution is beginning to approach a mesh-insensitive pressure-drop value. This should not be interpreted as a formal demonstration of mesh convergence.

Such a refinement could not be obtained with the present numerical setup. Calculations performed on still finer meshes became unstable before a statistically stationary pressure signal was reached. The instability was associated with increasingly strong oscillations of the morphology-transfer process between the dispersed and continuous gas representations. These oscillations were not sufficiently damped and eventually caused the calculation to diverge.

This instability concerns the numerical transfer between  $g_D$  and  $g_C$ . The exact origin of the instability was not isolated in the present work. Refinement modifies the local phase-fraction gradients and the representation of structures entering the transfer criteria, while simultaneously reducing the characteristic cell size and admissible time step. The resulting coupling between interface resolution, morphology identification, and transfer source terms therefore becomes increasingly restrictive on the finest grids.

The four-mesh study consequently indicates a clear reduction of the discretization sensitivity at the highest available resolution, but a fully grid-converged solution cannot be claimed. The finest stable mesh is used in the following comparisons because it provides the smallest observed change in the pressure-drop prediction while remaining numerically stable.

Figure 4.5 compares the baseline results with calculations performed using this finest stable wall-refined mesh.

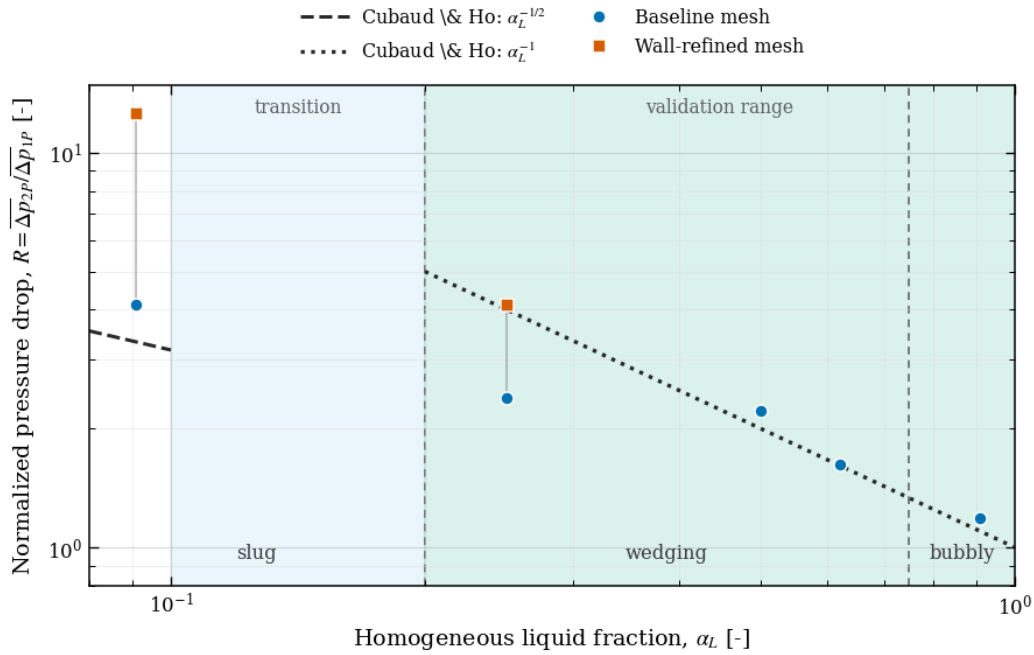


Figure 4.5: Influence of near-wall mesh refinement on the normalized two-phase pressure drop. [11].

The effect of refinement depends strongly on the liquid fraction. In the range where the liquid still occupies a substantial part of the cross-section, refinement improves the pressure-drop prediction. In particular, the refined result in the wedging range moves closer to the experimental  $\Pi_{\Delta p} \propto \alpha_L^{-1}$  trend. The shift towards the experimental trend shows that the baseline mesh already introduces measurable discretization error before film-dominated transport becomes dominant.

To interpret this improvement, it is useful to examine the gas–liquid distribution in a cross-section of the channel. Figure 4.6 compares the total gas-fraction field obtained on

the coarse and refined meshes. On the coarse mesh, the gas occupies almost the entire cross-section and the liquid-rich regions near the walls and corners remain only weakly represented. After refinement, the liquid is more clearly retained in the corner regions and the gas core becomes more distinctly separated from the solid boundaries.

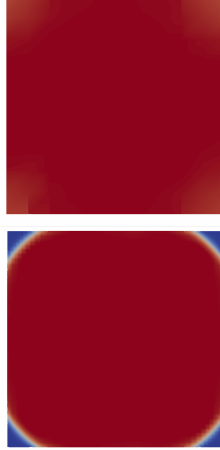


Figure 4.6: Cross-sectional comparison of the gas-fraction field for the coarse and refined meshes. Gas is in red, water is in blue.

In square microchannels, the remaining liquid becomes strongly non-uniform once the gas fraction is large. A significant part of the liquid transport occurs through thin wall-connected regions, especially in the corners. The refined mesh therefore improves the prediction not only because it is globally finer, but because it better captures the corner-flow paths through which the liquid continues to move.

The behaviour changes near the transition towards slug flow. At  $\alpha_L \simeq 0.1$ , the refined calculation no longer approaches the experimental low-liquid-fraction correlation. Instead, the predicted pressure multiplier increases strongly and lies above the  $\Pi_{\Delta p} \propto \alpha_L^{-1/2}$  trend.

The refined calculation is still not mesh-converged in the film-flow regime. The refinement improves the representation of the liquid distribution close to the walls and in the corners, but the characteristic thickness of the physical liquid film is already comparable with the local grid spacing. The numerical solution is consequently entering a regime in which the pressure drop becomes highly sensitive to the representation of the wall film.

For a physical film thickness  $\delta_f$ , the relevant wall shear is

$$\tau_w = \mu_l \left. \frac{\partial u_l}{\partial n} \right|_w. \quad (4.7)$$

If the liquid velocity changes by an amount of order  $\Delta U_l$  across the film, the physical gradient scales as

$$\left| \frac{\partial u_l}{\partial n} \right|_{\text{phys}} \sim \frac{\Delta U_l}{\delta_f}. \quad (4.8)$$

The required mesh resolution therefore becomes increasingly restrictive as  $\delta_f$  decreases. Resolving the large gas structure does not imply that the much thinner liquid film and its velocity gradient are resolved. In bubbly and wedging flow, improving the near-wall resolution produces a better quantitative pressure-drop prediction because the liquid distribution near the wall, and in particular the corner flow, is better represented. Once

the flow approaches the slug and film-dominated range, however, the liquid length scale becomes too small for the available mesh to provide a grid-independent representation of the near-wall dissipation.

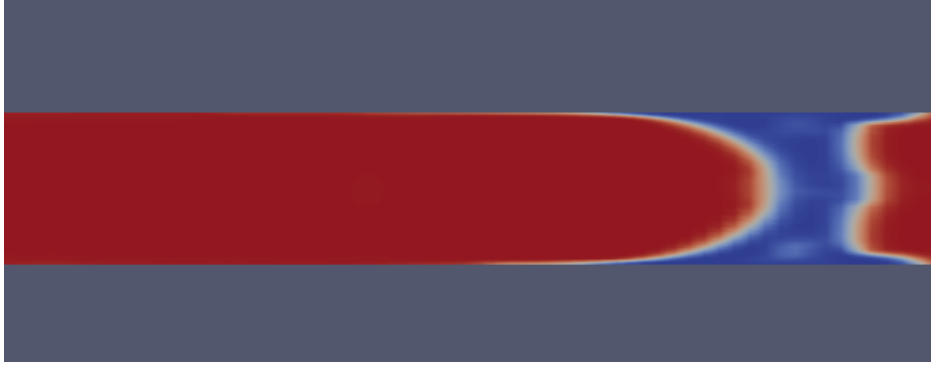


Figure 4.7: Close-up of the gas–liquid interface near the front of a slug for the finest stable mesh. Gas is in red, water is in blue.

The finest stable mesh used here provides only about one computational cell across the liquid-film thickness estimated from the Bretherton-type capillary-number relation. This resolution is not sufficient to resolve the film. Interface-resolving studies generally require several cells across the film thickness to obtain a mesh-independent prediction of the deposited film, the near-wall velocity gradient, and the associated wall shear stress.

With only one cell across the estimated film thickness, the calculation cannot reconstruct a converged velocity profile within the film and cannot represent the film as a stable lubricating layer along the wall. At most, the solution captures the onset of film deposition near the meniscus. The close-up in Fig. 4.7 shows a thin liquid layer forming at the bubble front, but it does not remain continuously established along the wall. Downstream, the gas region approaches the wall again, indicating that the deposited film is not stably resolved on the available mesh.

The pressure-drop results therefore remain resolution-sensitive because the dominant dissipative mechanism in this regime depends on a liquid film whose thickness and internal velocity gradient are below the reliable resolution of the present grid.

#### 4.1.5. Influence of the wall-wet boundary condition

The low-liquid-fraction Cubaud cases were repeated with the modified wall-wet phase-fraction boundary condition introduced in Section 3.4.4. In these calculations, the liquid phase is imposed at the solid boundary. The gas can still occupy the wall-adjacent cells, but the wall face itself remains liquid-covered.

Only the gas-rich operating points were recalculated. At larger  $\alpha_L$ , the liquid already occupies a significant fraction of the cross-section and remains connected to the wall without requiring the modified treatment. The boundary condition is expected to matter mainly once the liquid is restricted to films and corner regions, which is the range considered here.

Figure 4.8 shows that the wall-wet condition changes the hydraulic response in the gas-rich cases. The pressure multiplier increases as  $\alpha_L$  decreases, but the numerical slope remains weaker than the experimental  $\Pi_{\Delta p} \propto \alpha_L^{-1/2}$  trend. The modified boundary condition keeps liquid at the wall face but does not resolve the film inside the adjacent cells. On the finest stable mesh, the estimated film thickness remains of the order of one wall-normal cell. The calculation can therefore change the wetting topology without resolving the velocity profile across the physical liquid layer.

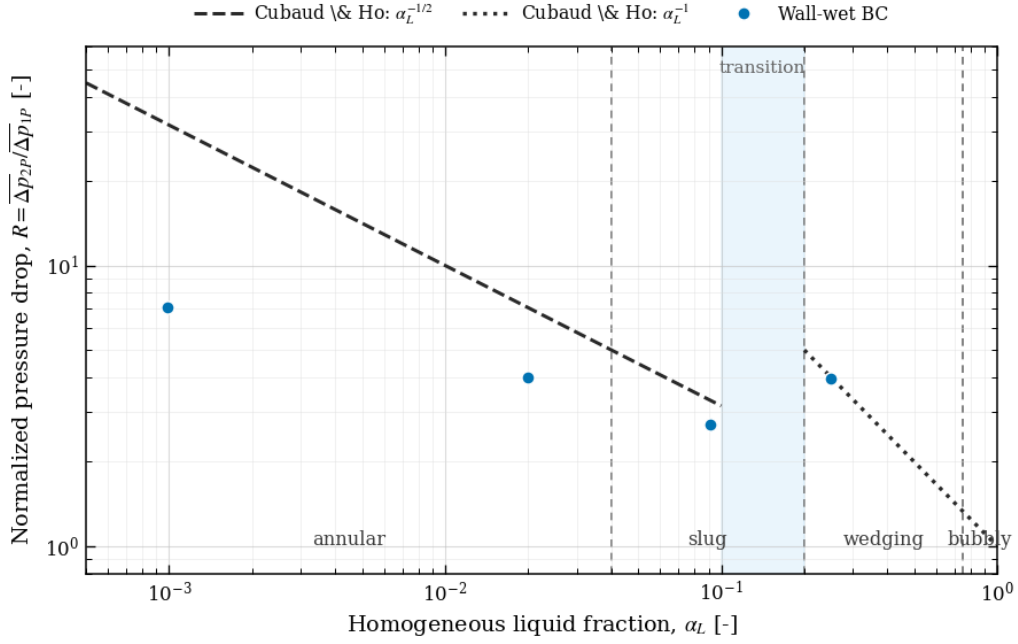


Figure 4.8: Normalized pressure drop obtained with the wall-wet boundary condition for the low-liquid-fraction Cubaud cases. The numerical results are compared with the empirical pressure-drop trends reported by Cubaud and Ho.

The wall-wet result supports the interpretation obtained from the mesh study: the low- $\alpha_L$  discrepancy cannot be removed only by imposing liquid at the boundary.

## 4.2. Lafmejani configuration

The Lafmejani configuration provides a different test of the MultiMorph model from the Cubaud. Gas is injected progressively through one wall rather than being introduced entirely at the channel inlet. Consequently, the local gas fraction increases in the streamwise direction and several gas morphologies can occur within the same channel.

The comparison is used primarily to assess the morphology-transfer mechanism. In particular, it shows how gas initially represented by the dispersed field  $g_D$  accumulates and is progressively transferred to the continuous field  $g_C$ .

### 4.2.1. Design-of-experiments setup

The Lafmejani configuration is used here primarily to assess the influence of the morphology-transfer parameters on the predicted gas distribution. Unlike the Cubaud benchmark, the objective is not first a direct pressure-drop validation, but rather a structured sensitivity study of the mechanisms controlling the transition from dispersed gas to continuous gas structures. A reference parameter set was defined and used as the starting point for the sensitivity analysis. The corresponding values are

$$d_{b,0} = 10^{-6} \text{ m}, \quad r_{C,\text{th},0} = 0.4, \quad \tau_{PT,0} = 0.1 \text{ s}. \quad (4.9)$$

Starting from this reference configuration, the three parameters were varied one at a time in order to isolate their individual influence on the predicted morphology. The resulting design of experiments is summarized in Table 4.2. This procedure is important because the three parameters do not act on the same part of the model. The threshold  $r_{C,\text{th}}$  affects the criterion for morphology transfer,  $\tau_{PT}$  controls the transfer rate, and  $d_b$  changes the

Table 4.2: Design of experiments used for the Lafmejani sensitivity study.

| Study                          | $d_b$       | $r_{C,th}$ | $\tau_{PT}$ |
|--------------------------------|-------------|------------|-------------|
| Reference configuration        | $10^{-6}$ m | 0.4        | 0.1 s       |
| Morphology-transfer time scale | $10^{-6}$ m | 0.4        | variable    |
| Continuous threshold           | $10^{-6}$ m | variable   | 0.1 s       |
| Dispersed bubble diameter      | variable    | 0.4        | 0.1 s       |

Euler–Euler momentum exchange of the dispersed gas. A one-factor-at-a-time variation is therefore chosen to identify which observed change in morphology is associated with which model parameter. The figures shown in the following subsections should consequently be interpreted as parameter-study results rather than as direct validation figures. The comparison with the experimental morphology is introduced only after a suitable parameter combination has been identified.

#### 4.2.2. Continuous threshold

The influence of `continuousThreshold` is shown in Fig. 4.9. For the present configuration, this parameter enters the criterion used by the `accumulationToContinuous` model when no separate accumulation threshold is prescribed.

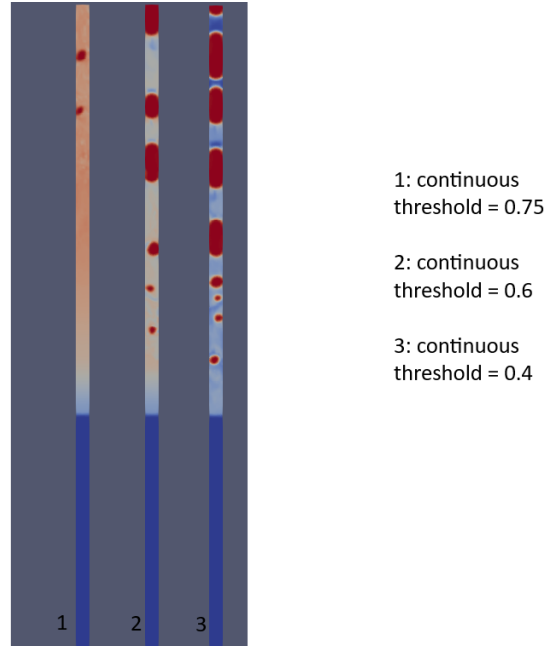


Figure 4.9: Influence of `continuousThreshold` on the distribution of continuous gas in the Lafmejani configuration. Gas is in red, water is in blue.

The transfer from  $g_D$  to  $g_C$  is activated when the local gas accumulation exceeds the prescribed threshold. Decreasing `continuousThreshold` therefore makes the transition to the continuous representation possible at a lower local gas fraction.

This behaviour is visible in Fig. 4.9. For `continuousThreshold`= 0.75, a larger fraction of the gas remains dispersed and only isolated continuous structures are formed. Reducing the threshold to 0.60 and subsequently to 0.40 produces continuous structures earlier and over a larger part of the channel.

### 4.2.3. Dispersed bubble diameter

Figure 4.10 shows the sensitivity to the bubble diameter used by the Euler–Euler closures.

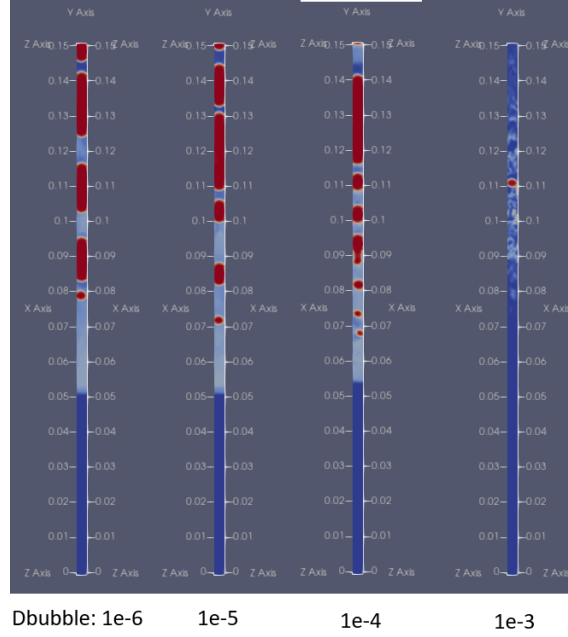


Figure 4.10: Influence of the dispersed-bubble diameter  $d_b$  on the Lafmejani configuration. Gas is in red, water is in blue.

Unlike  $\tau_{PT}$  and `continuousThreshold`, the bubble diameter does not enter the morphology-transfer criterion directly. It modifies the momentum exchange of the dispersed gas. For spherical dispersed structures, the interfacial area density scales as

$$a_i = \frac{6r_{gD}}{d_b}. \quad (4.10)$$

This change also affects the morphology transfer indirectly. For smaller values of  $d_b$ , the stronger gas–liquid momentum coupling reduces the relative motion of the dispersed gas with respect to the liquid. The gas therefore accumulates more readily close to the injection region, so that the local gas fraction reaches the accumulation criterion earlier in the channel. The transition from  $g_D$  to  $g_C$  consequently occurs closer to the gas-injection region.

This trend is visible in Fig. 4.10. Increasing  $d_b$  weakens the coupling of the dispersed gas with the liquid and shifts the formation of continuous gas structures further downstream. For  $d_b = 10^{-3}$  m, the dispersed gas remains strongly segregated over most of the channel. The local accumulation required for transfer to  $g_C$  is then strongly reduced, and the calculation eventually becomes numerically unstable.

The bubble diameter therefore affects not only the motion of the unresolved gas but also, indirectly, the spatial position of the numerical dispersed-to-continuous transition.

### 4.2.4. Morphology-transfer time scale

Figure 4.11 compares the gas morphology for

$$\tau_{PT} = 0.50, 0.10, 0.05, 0.01 \text{ s}. \quad (4.11)$$

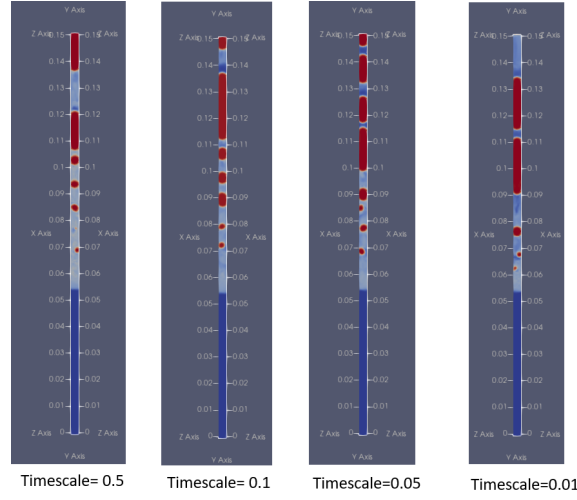


Figure 4.11: Influence of the morphology-transfer time scale  $\tau_{PT}$  on the gas morphology in the Lafmejani configuration. Gas is in red, water is in blue.

The largest value leaves most of the gas in the dispersed representation over a substantial part of the channel. Continuous structures become more prominent as  $\tau_{PT}$  is reduced, and the transition from dispersed to continuous gas occurs over a shorter streamwise distance. Among the investigated values,  $\tau_{PT} = 0.01$  s gives the closest morphology progression to the experimental visualisation and is therefore retained for the final parameter combination.

#### 4.2.5. Result with the selected parameter set

The sensitivity study above was used to identify a parameter combination that gives the most realistic morphology progression in the Lafmejani configuration. The selected parameter set is

$$d_b = 10^{-6} \text{ m}, \quad r_{C,th} = 0.2, \quad \tau_{PT} = 0.01 \text{ s}. \quad (4.12)$$

Figure 4.12 compares the resulting simulation with the experimental visualizations. With

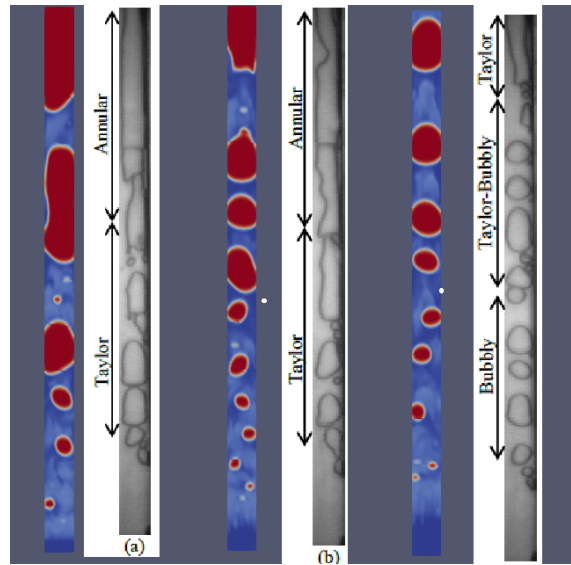


Figure 4.12: Comparison between the optimized numerical configuration and the experimental morphology reported by Sadeghi Lafmejani et al. [26].

this parameter set, the simulation reproduces the main sequence of morphological development observed experimentally. In the lower part of the channel, the gas first appears in a bubbly regime. Further downstream, the gas accumulates and a mixed bubbly–Taylor region develops. Larger Taylor-like structures are then formed higher in the channel. The agreement is therefore significantly improved compared with the reference parameter configuration. This confirms that the morphology transition in the Lafmejani case is highly sensitive to the numerical treatment of dispersed-to-continuous conversion and to the transport of the dispersed gas phase before this conversion takes place. The agreement nevertheless deteriorates once the gas structures become large enough to occupy most of the channel width and approach the wall. In the experiment, the upper part of the channel develops towards annular flow, whereas in the simulation the large gas structures are no longer sufficiently constrained by a stable liquid layer along the wall. The loss of annular behaviour is therefore not primarily a consequence of the morphology-transfer parameters anymore, but of the near-wall treatment once the continuous gas phase becomes dominant. This observation is consistent with the behaviour already identified in the Cubaud configuration. There as well, the numerical prediction deteriorated once the remaining liquid was confined to thin near-wall regions whose hydraulic effect depends strongly on the liquid-film representation.

### 4.3. Kim configuration

The Kim configuration is the most application-oriented case considered in this work and is therefore presented after the two more controlled reference configurations. Unlike the Cubaud benchmark, gas is not supplied together with the liquid at a common inlet. Instead, it is introduced progressively through the wall representing the porous transport layer.

The operating point considered here corresponds to the parallel-channel configuration at

$$j = 2 \text{ A cm}^{-2}, \quad (4.13)$$

for which Kim and Jung observed a slug-flow morphology containing Taylor bubbles with a width comparable to the channel width [23].

The numerical analysis is divided into a single-phase hydraulic check, followed by the development of the gas–liquid flow and the associated pressure response. Unless stated otherwise, time-averaged quantities are evaluated over

$$5 \text{ s} \leq t \leq 15 \text{ s}. \quad (4.14)$$

The first 5 s are excluded from the statistics to remove the initial filling transient.

#### 4.3.1. Single-phase hydraulic verification

Before introducing gas, the simplified Kim domain was simulated with water only. This separates errors associated with the hydraulic resistance of the numerical channel from those introduced later by the multiphase model.

The imposed liquid flow rate is  $Q_l = 2.39 \times 10^{-8} \text{ m}^3 \text{ s}^{-1}$ .

For the square channel of side length  $h = 3 \text{ mm}$ , this corresponds to a bulk velocity of

$$U_l = \frac{Q_l}{h^2} = 2.66 \times 10^{-3} \text{ m s}^{-1}. \quad (4.15)$$

Using  $\rho_l = 977.63 \text{ kg m}^{-3}$ ,  $\mu_l = 4.035 \times 10^{-4} \text{ Pa s}$ , the Reynolds number is

$$\text{Re} = \frac{\rho_l U_l h}{\mu_l} \simeq 19.3. \quad (4.16)$$

The single-phase flow is therefore well within the laminar regime.

For fully developed laminar flow through a square duct, the theoretical frictional pressure difference over the channel length  $L = 0.15 \text{ m}$  is

$$\Delta p_{\text{th}} = C_{\text{sq}} \frac{\mu_l L Q_l}{h^4}, \quad C_{\text{sq}} = 28.43. \quad (4.17)$$

The resulting analytical value is  $\Delta p_{\text{th}} = 0.508 \text{ Pa}$ .

This value is very small compared with the hydrostatic pressure difference over the vertical numerical channel,

$$\Delta p_{\text{grav,1P}} = \rho_l g H \simeq 1.439 \times 10^3 \text{ Pa}. \quad (4.18)$$

The hydraulic part of the numerical pressure difference was therefore evaluated both by subtracting the hydrostatic contribution from the static pressure field,

$$\Delta p_p = (p_{\text{in}} - p_{\text{out}}) - \rho_l g H, \quad (4.19)$$

and directly from the reduced-pressure field,

$$\Delta p_{p_{\text{rgh}}} = p_{\text{rgh,in}} - p_{\text{rgh,out}}. \quad (4.20)$$

Both evaluations converge to approximately  $\overline{\Delta p_{\text{1P}}} \simeq 0.489 \text{ Pa}$ .

The difference from the analytical square-duct solution is approximately

$$\frac{\overline{\Delta p_{\text{1P}}} - \Delta p_{\text{th}}}{\Delta p_{\text{th}}} \simeq -3.7\%. \quad (4.21)$$

A third check is obtained independently from the wall shear stress. For a fully developed square duct,

$$\Delta p_\tau = \frac{4L}{D_h} \bar{\tau}_w. \quad (4.22)$$

The theoretical wall shear stress is

$$\bar{\tau}_{w,\text{th}} = \frac{\Delta p_{\text{th}} D_h}{4L} = 2.54 \times 10^{-3} \text{ Pa}. \quad (4.23)$$

The numerical wall shear stress is approximately  $\bar{\tau}_{w,\text{CFD}} \simeq 2.45 \times 10^{-3} \text{ Pa}$ ,

which corresponds to  $\Delta p_\tau \simeq 0.491 \text{ Pa}$ .

The pressure difference reconstructed from the wall traction differs by less than 0.5% from the pressure-based numerical result. Figure 4.13 summarizes the three numerical evaluations of the single-phase pressure loss and compares them with the analytical square-duct

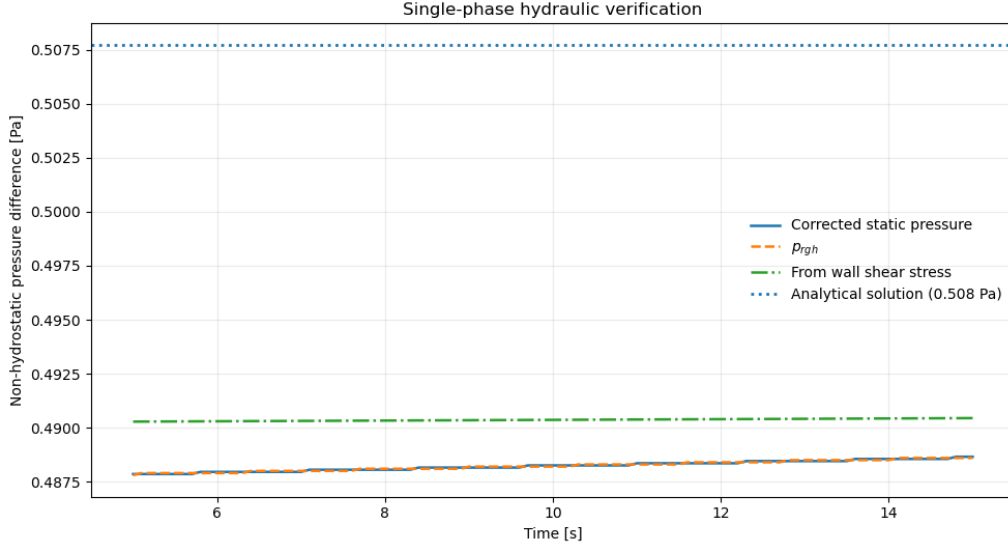


Figure 4.13: Single-phase pressure difference obtained from the corrected static pressure, the  $p_{rgh}$  field, and the wall-shear-stress balance, together with the analytical square-duct solution.

solution. The three numerical evaluations give nearly identical values. The pressure-based result is approximately 0.489 Pa, while the wall-shear reconstruction gives approximately 0.491 Pa. Both values remain within approximately 4% of the analytical square-duct prediction of 0.508 Pa. This agreement confirms that the hydraulic resistance of the simplified numerical channel is correctly represented before gas injection is introduced.

#### 4.3.2. Development of the two-phase flow

The two-phase calculation starts with a water-filled channel. Gas is then introduced progressively through the wall representing the porous transport layer. The channel gas content therefore increases during the initial part of the simulation before a statistically fluctuating state is reached.

The total gas fraction used for post-processing is

$$r_g = r_{gC} + r_{gD}, \quad (4.24)$$

and the volume-averaged gas hold-up is

$$\varepsilon_g(t) = \frac{1}{V} \int_V r_g(\mathbf{x}, t) dV. \quad (4.25)$$

Figure 4.14 shows the development of this quantity. After the initial filling period, the gas hold-up fluctuates around a value of order 0.4. The in-situ phase content is therefore substantially different from the homogeneous phase fraction calculated only from the imposed inlet flow rates.

The gas fraction increases rapidly during the initial filling of the channel and subsequently reaches a statistically fluctuating state. After approximately 5 s, no further systematic increase is observed, which motivates the use of the interval  $5 < t < 15$  s for the subsequent time averaging. The gas does not remain in the dispersed representation. As it accumulates along the injection wall, neighbouring structures merge and a large connected gas region

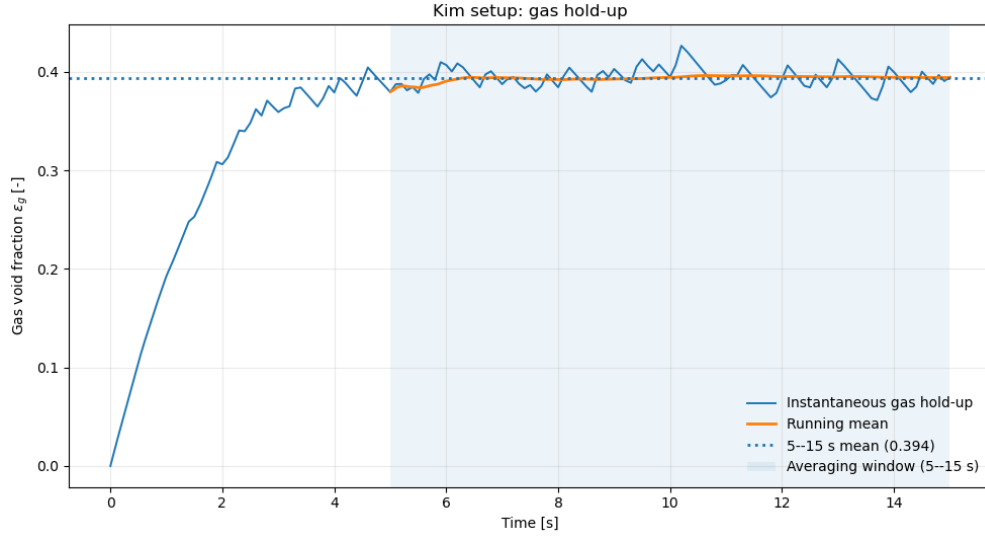


Figure 4.14: Evolution of the volume-averaged gas fraction in the Kim setup. The mean value is evaluated over  $5 \leq t \leq 15$  s.

develops. The resulting morphology occupies a substantial part of the channel cross-section while liquid remains in the remaining near-wall and corner regions.

Kim and Jung reported that the parallel flow field at  $2 \text{ A cm}^{-2}$  was dominated by slug flow, with Taylor bubbles whose width was approximately equal to the channel width [23]. The numerical result shown in Figure 4.15 reproduces the principal feature of this regime: gas structures become confined by the channel instead of remaining as isolated small bubbles.

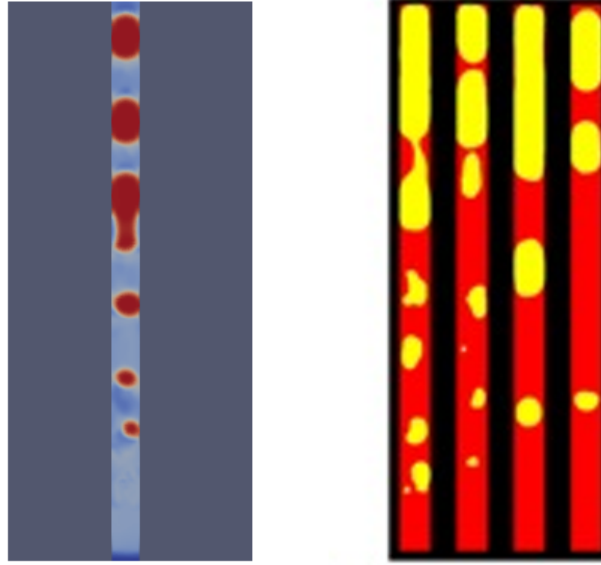


Figure 4.15: Gas morphology in the experimental parallel-flow configuration of Kim and Jung [23] and in the corresponding MultiMorph simulation. The numerical field is shown at  $t = 10$  s. In the left picture, gas is in red, water in blue, on the right picture, gas is in yellow and water in red.

The simplified numerical domain contains only one straight representative channel and does not reproduce the complete parallel flow field. Individual bubble lengths and discharge frequencies are therefore not expected to match the experiment directly. The comparison

is used as a qualitative check that the model reaches the same broad morphology under a gas-dominated PEM-like operating condition.

#### 4.3.3. Raw pressure response and gravitational contribution

The pressure response changes strongly once gas enters the vertical channel. Figure 4.16 compares the uncorrected pressure difference

$$\Delta p_{\text{raw}} = p_{\text{in}} - p_{\text{out}} \quad (4.26)$$

with the corresponding single-phase signal.

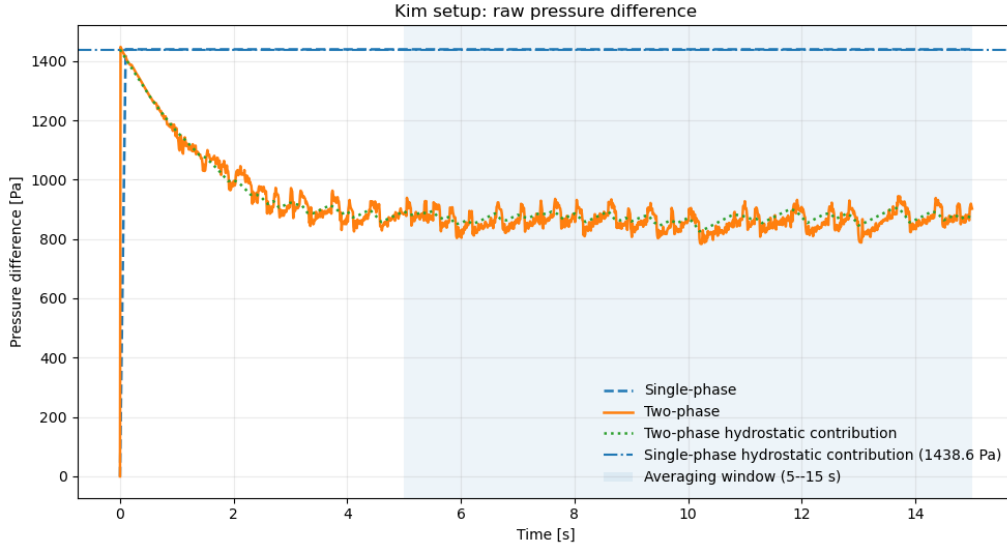


Figure 4.16: Evolution of the uncorrected pressure difference in the Kim setup, together with the single-phase reference and the hydrostatic contributions calculated from the liquid and mixture densities.

Figure 4.16 shows that the raw pressure difference decreases rapidly during the initial gas accumulation and subsequently reaches a fluctuating level well below the single-phase reference. At the same time, the calculated hydrostatic contribution follows the same overall decrease. The close evolution of these two quantities indicates that the reduction of the raw pressure signal is primarily associated with the decreasing weight of the fluid column rather than with a reduction of the hydraulic resistance.

For the two-phase flow, the instantaneous volume-averaged mixture density is estimated as

$$\rho_m(t) = (1 - \varepsilon_g(t)) \rho_l + \varepsilon_g(t) \rho_g. \quad (4.27)$$

The corresponding gravitational contribution is

$$\Delta p_{\text{grav}}(t) = \rho_m(t) g H. \quad (4.28)$$

Since  $\rho_g \ll \rho_l$ , the progressive replacement of water by gas reduces the mean density of the vertical column. The hydrostatic pressure difference consequently falls from approximately 1.44 kPa in the water-filled case to a value of order 0.9 kPa once the two-phase flow is established.

The magnitude of this change is much larger than the single-phase frictional pressure loss, which is only approximately 0.49 Pa. The raw pressure signal is therefore dominated by gravity in both the single-phase and two-phase calculations. The lower two-phase pressure difference cannot consequently be interpreted directly as a reduction of the hydraulic resistance.

A direct comparison with the experimental pressure-drop multiplier therefore requires the gravitational contribution to be considered separately. The different numerical approaches used for this purpose are examined in the following section.

#### 4.3.4. Reconstruction of the non-hydrostatic pressure response

Kim and Jung defined the experimental two-phase multiplier as

$$\Phi_w^2 = \frac{\overline{\Delta p_{2P}}}{\overline{\Delta p_w}}, \quad (4.29)$$

where the denominator corresponds to water-only operation of the flow field [23].

An important difference between the experiment and the present post-processing concerns the treatment of gravity. Additional information provided by the authors confirmed that the electrolyzer was mounted vertically and that the pressure difference was measured using tee-union connections at the inlet and outlet. No separate hydrostatic correction was applied to the reported experimental data, and the values used for the pressure multiplier were averaged over 600 s.<sup>1</sup>

The experimental pressure signal therefore corresponds to the differential pressure produced by the complete measurement arrangement. The available information is not sufficient to reconstruct the hydrostatic contribution of the pressure lines themselves, including their exact filling state and routing. A numerical subtraction of the fluid-column weight cannot therefore be assumed to reproduce exactly the quantity measured by the experimental pressure transducer.

Nevertheless, the gravitational contribution can be removed from the CFD solution to examine the remaining pressure response. The first approach uses the instantaneous mixture density,

$$\Delta p_{nh}(t) = \Delta p_{raw}(t) - \rho_m(t)gH. \quad (4.30)$$

A second representation uses the time-averaged mixture density,

$$\Delta p_{nh,\bar{\rho}}(t) = \Delta p_{raw}(t) - \bar{\rho}_m gH. \quad (4.31)$$

When both quantities are averaged over the same interval, their mean values are mathematically identical because the gravitational term is linear in  $\rho_m$ . Their instantaneous fluctuations differ, however, because the first expression follows the time variation of the gas hold-up.

The third diagnostic is the difference of the OpenFOAM reduced-pressure field,

$$\Delta p_{rgh} = p_{rgh,in} - p_{rgh,out}. \quad (4.32)$$

---

<sup>1</sup>S. Y. Jung, response to inquiry concerning the pressure measurement of Kim and Jung (2024), personal communication.

For the single-phase case, this quantity coincides with the manually hydrostatically corrected pressure difference. In the multiphase calculation the density varies in space and time, so  $\Delta p_{rgh}$  is retained as an additional pressure diagnostic rather than being interpreted directly as a pure frictional pressure loss.

Figure 4.17 compares the three representations over the statistically analysed interval.

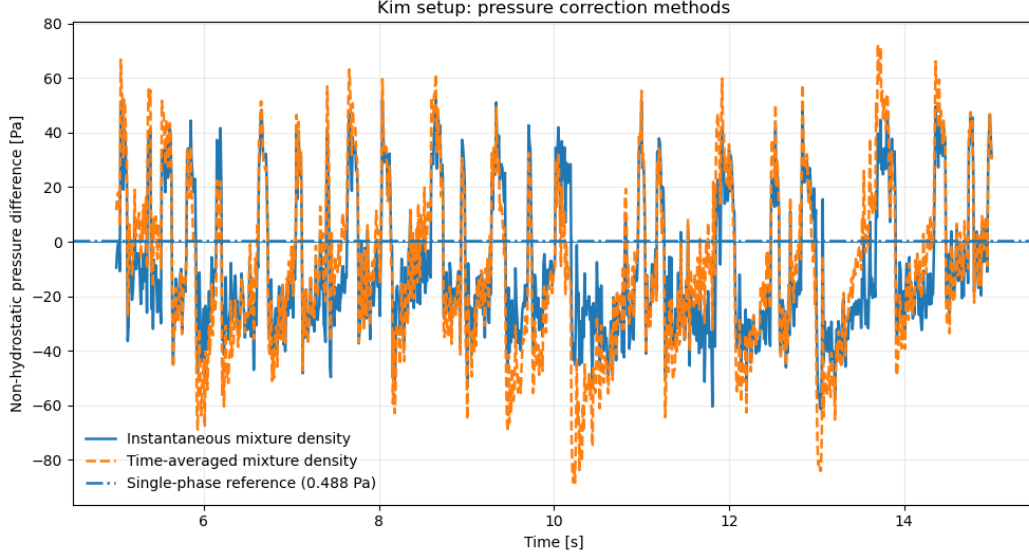


Figure 4.17: Numerical representations of the pressure difference after removal or reduction of the gravitational contribution in the Kim setup. The comparison is restricted to the statistical interval  $5 \leq t \leq 15$  s.

The resulting non-hydrostatic pressure signal is only of order 10 Pa, while the hydrostatic term is of order  $10^3$  Pa. The corrected quantity is consequently much more sensitive than the raw pressure difference.

This sensitivity can be quantified from

$$\frac{\partial \Delta p_{\text{grav}}}{\partial \varepsilon_g} = -(\rho_l - \rho_g) g H. \quad (4.33)$$

For the present channel,

$$\left| \frac{\partial \Delta p_{\text{grav}}}{\partial \varepsilon_g} \right| \simeq 1.44 \times 10^3 \text{ Pa}. \quad (4.34)$$

An absolute change in the mean gas fraction of only  $\Delta \varepsilon_g = 0.01$  therefore changes the reconstructed hydrostatic contribution by approximately 14 Pa.

This is already comparable to the complete non-hydrostatic residual.

A locally negative value of  $\Delta p_{\text{nh}}$  must therefore not be interpreted as a negative viscous resistance. After the gravitational contribution has been removed, the remaining pressure balance contains wall friction together with acceleration, gas-injection, capillary, and local interface contributions. The wall shear itself remains a dissipative quantity.

For this reason, the pressure reconstruction from the Kim configuration is used to analyse the numerical behaviour of the model, but it is not treated as the primary quantitative pressure-drop validation of MultiMorph. The horizontal Cubaud configuration is used for that purpose, since its streamwise pressure difference does not contain this dominant hydrostatic contribution.

### 4.3.5. Influence of the liquid-film boundary condition

The baseline *Kim setup* uses the standard phase-fraction treatment at the wall. A second calculation, denoted *Kim film BC*, imposes the liquid phase at the wall. The two cases use the same geometry and operating conditions, so that the effect of the wall treatment can be examined independently.

The distinction is particularly relevant in the gas-rich regime because the liquid viscosity is much larger than the gas viscosity. Where the wall remains wetted, the streamwise wall shear stress is mainly governed by the liquid velocity gradient,

$$\tau_w = \mu_l \left. \frac{\partial u_l}{\partial n} \right|_w. \quad (4.35)$$

Figure 4.18 compares the baseline and film-BC configurations. The morphology is shown at  $t = 10$  s, while the wall-shear-stress histories are displayed over the interval  $5 \leq t \leq 10$  s.

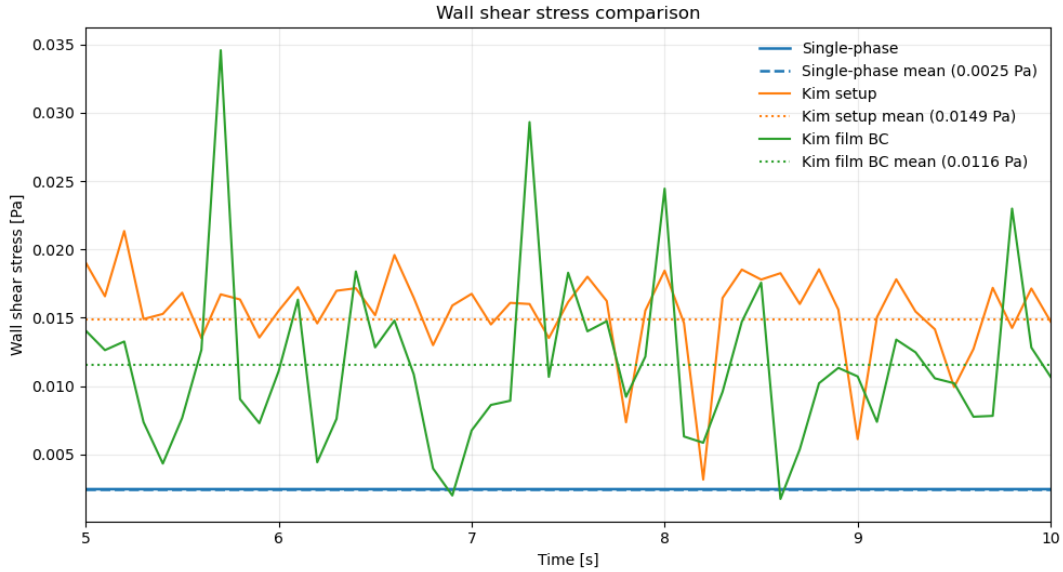


Figure 4.18: Comparison of the Kim setup and Kim film-BC configurations. The wall-shear-stress histories are shown over  $5 \leq t \leq 10$  s.

The two wall treatments lead to different near-wall phase distributions. In the baseline case, the continuous gas phase can approach the wall, whereas the film boundary condition preserves a liquid-covered boundary. This modifies both the local gas morphology and the liquid velocity gradients close to the solid surface.

The effect is also visible in the wall-shear-stress response. The baseline Kim case shows a mean wall shear stress of approximately  $1.49 \times 10^{-2}$  Pa, while the wall-wet case gives approximately  $1.16 \times 10^{-2}$  Pa over the considered interval. Both values remain substantially larger than the single-phase reference of approximately  $2.5 \times 10^{-3}$  Pa.

The reduction obtained with the film boundary condition shows that the predicted hydraulic response is sensitive to the treatment of the liquid remaining near the wall. The gas hold-up alone is therefore not sufficient to characterize the two-phase wall interaction; the distribution and connectivity of the liquid phase close to the solid boundary also influence the resulting wall shear stress.

To compare the near-wall morphology, the gas-liquid interface is visualized using the isosurface

$$r_g = r_{gC} + r_{gD} = 0.5. \quad (4.36)$$

Only this isosurface is displayed in Fig. 4.19; the complete gas-filled volume is not rendered. The apparently open gas structures therefore do not represent incomplete bubbles, but correspond to the visible part of the  $r_g = 0.5$  interface. The gas-dominated region lies on the side of this surface for which  $r_g > 0.5$ . With the wall-wet boundary condition,

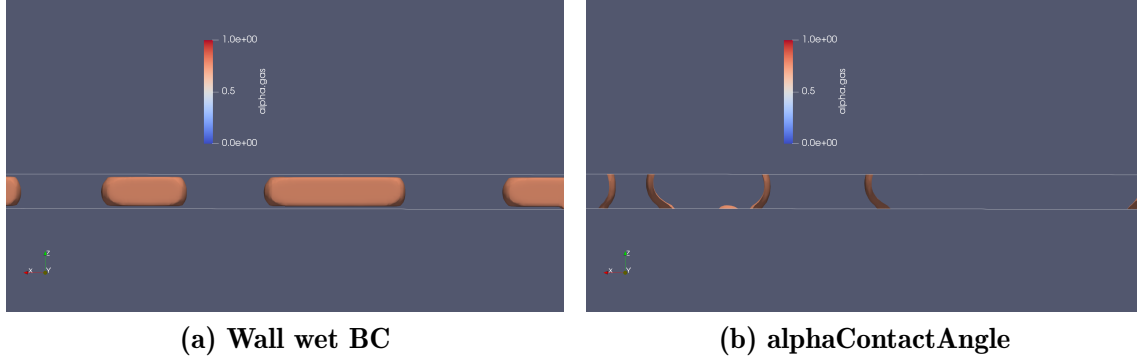


Figure 4.19: Near-wall gas morphology in the Kim configuration for (a) the wall-wet boundary condition and (b) the standard `alphaContactAngle` treatment. The gas-liquid interface is represented by the isosurface  $r_g = r_{gC} + r_{gD} = 0.5$ ; the complete gas volume is therefore not shown.

the  $r_g = 0.5$  interface remains detached from the solid wall. This is consistent with the imposed  $r_l = 1$  condition at the wall, which requires the gas fraction to decrease towards zero before reaching the solid boundary. A liquid-containing region therefore remains between the gas-dominated core and the wall.

With the `alphaContactAngle` treatment, the interface is instead allowed to approach and locally intersect the wall according to the prescribed wetting condition. The  $r_g = 0.5$  isosurface consequently reaches the solid boundary at several locations. This indicates a fundamentally different near-wall topology, even though the overall gas hold-up may remain comparable between the two calculations.

The liquid-containing region produced by the wall-wet boundary condition is not a directly resolved physical film. The boundary condition only enforces liquid at the wall face, whereas the `alphaContactAngle` treatment allows the near-wall phase distribution to emerge from the prescribed wetting condition and interface dynamics. Taken together, the Kim results verify the single-phase hydraulic resistance and reproduce the formation of channel-confined gas structures. The wall-BC comparison also shows the sensitivity of the gas-rich solution to the near-wall liquid treatment. Because the pressure response is strongly affected by gravity and by the experimental measurement arrangement, the Kim case is used mainly as a consistency test rather than as the quantitative pressure-drop benchmark.



## 5. Discussion

The agreement obtained with MultiMorph depends strongly on which quantity is considered. Across the three configurations, the large-scale gas morphology remains plausible over a wider range of conditions than the pressure-drop prediction. The difference becomes clear once the flow is gas-rich. Large gas structures can already be represented by the continuous field, while the remaining liquid is confined to much smaller wall films and corner regions.

This distinction was already anticipated from the numerical representation described in Sections 2.3.5 and 2.3.6. The results now show where it becomes relevant in practice. In particular, the Cubaud mesh study (Section 4.1.4) reaches approximately one cell across the estimated liquid-film thickness while the main gas structure extends over a much larger part of the channel. The gas morphology and the near-wall hydraulic scale are therefore not resolved at the same point.

### 5.1. Model Performance for Pressure-drop Prediction

The single-phase calculations provide a useful reference before discussing the two-phase results. The Cubaud liquid-only calculation in Section 4.1.1 lies between the analytical and experimental pressure losses. The Kim calculation gives a similar result: the pressure-based and wall-shear-based evaluations in Section 4.3.1 are both close to the analytical square-duct solution. The large discrepancies observed later are therefore not already present in the basic single-phase hydraulic resistance.

The Cubaud configuration gives the clearest quantitative assessment because the channel is horizontal and its streamwise pressure difference does not contain the large hydrostatic contribution present in Kim. On the baseline mesh, Section 4.1.3 shows good agreement with the experimental pressure multiplier in the bubbly and much of the wedging range. The calculated values follow the experimental  $\alpha_L^{-1}$  trend while a substantial fraction of the cross-section is still occupied by liquid.

The behaviour changes as the gas structures become elongated. The experimental transition towards the lower-liquid-fraction trend occurs in the same range in which the liquid is displaced towards the walls and corners [11]. The numerical pressure multiplier then departs from the experiment even though the gas structures remain qualitatively consistent with the expected regime.

The mesh study in Section 4.1.4 supports a near-wall origin for at least part of this discrepancy. Refinement changes the pressure drop systematically and gives a better representation of the corner liquid. The difference between the last two stable meshes becomes small, but a formal grid-converged solution was not obtained because further refinement became unstable. The finest mesh should therefore not be treated as a converged reference.

The effect of refinement is also not monotonic over the complete flow map. In the wedging range, the refined solution moves closer to the experimental trend. Around the transition towards the gas-rich regime, the pressure multiplier instead becomes too large. This matters because mesh refinement does more in MultiMorph than reduce a conventional discretization error. The continuous-interface momentum treatment depends on the local interface resolution, as described in Section 2.3.5, and the classification between continuous and dispersed gas is itself related to the phase distribution over the mesh. A finer grid can therefore change both the numerical resolution and the effective local treatment of the multiphase flow.

The Kim pressure results have a different limitation. The raw pressure response in Section 4.3.3 is dominated by the change in hydrostatic head as gas enters the vertical channel. After correction, the remaining pressure difference is only of order 10 Pa, compared with a hydrostatic contribution of order  $10^3$  Pa. As shown in Section 4.3.4, an absolute change of only 0.01 in mean gas fraction changes the reconstructed hydrostatic contribution by approximately 14 Pa. The experimental pressure measurement also includes an arrangement that cannot be reproduced exactly from the available information.

For this reason, the Kim case is more useful for the morphology and near-wall comparison than for a quantitative validation of frictional pressure loss. The Cubaud case remains the main hydraulic benchmark. Its results also show why pressure drop is more restrictive than visual agreement with the gas morphology: once most of the liquid is close to the wall, the hydraulic response depends on a length scale that is much smaller than the gas core itself.

## 5.2. Interpretation of Morphology Transitions

The morphology results are less restrictive than the pressure results. The Cubaud calculations in Section 4.1.2 recover the main sequence from dispersed bubbles to confined and elongated gas structures. The detailed wedging morphology is not reproduced at every operating point, but the main change in gas connectivity is captured.

The Lafmejani configuration tests the morphology-transfer mechanism more directly because gas enters the domain as  $g_D$  and larger structures must subsequently appear through transfer to  $g_C$ . The parameter studies in Sections 4.2.2, 4.2.3, and 4.2.4 show that this transition is sensitive to all three investigated parameters, but for different reasons. The threshold changes when accumulated gas can enter the continuous representation, the transfer time changes how quickly that conversion occurs, and  $d_b$  modifies the momentum exchange of the dispersed gas before the transition.

The final combination in Section 4.2.5 produces a morphology sequence close to the experimental visualisations: dispersed gas first dominates, followed by mixed bubbly–Taylor flow and larger continuous structures further downstream. This is a useful result because large structures emerge from gas that was initially introduced below the resolved scale.

The same result has to be interpreted with some caution. The final Lafmejani parameters were selected after examining their influence on the morphology. The comparison is therefore not an independent validation of  $d_b$ ,  $r_{C,th}$ , and  $\tau_{PT}$ . It shows that the available transfer mechanism can reproduce the observed progression after parameter adjustment.

A stronger test would require applying the same parameter set to additional operating conditions or geometries without retuning it.

The Kim result in Section 4.3.2 gives another example at a larger channel scale. Gas injected through the wall does not remain as a purely dispersed population. It accumulates into channel-confined structures comparable to the Taylor-bubble morphology reported by Kim and Jung for the parallel flow field [23].

The three configurations therefore do not indicate a general difficulty in forming large continuous gas structures. The main deviation appears after these structures occupy most of the channel and the remaining liquid is forced into the near-wall region. This is also where the pressure-drop prediction becomes strongly mesh-dependent.

### 5.3. Impact of Wall Boundary Conditions

The two wall treatments introduced in Section 3.4.4 act on the phase distribution at the solid boundary. The standard `alphaContactAngle` treatment allows the continuous interface to approach the wall according to the prescribed wetting condition. The modified wall-wet condition instead imposes liquid at the wall face. Neither treatment prescribes a physical film thickness.

The Cubaud results in Section 4.1.5 show that the modified condition changes the pressure response in the gas-rich cases. The pressure multiplier rises as  $\alpha_L$  is reduced, but the experimental low-liquid-fraction trend is still not recovered quantitatively. Keeping the wall liquid-covered therefore changes the near-wall topology without removing the underlying resolution problem.

The Kim comparison in Section 4.3.5 gives an additional result. The wall-wet condition keeps the  $r_g = 0.5$  interface detached from the solid boundary, whereas the interface can locally reach the wall with the standard treatment. The two cases clearly have different near-wall phase distributions.

The corresponding wall shear does not change simply because of the wall-wet boundary condition. At first, it can be expected that more liquid means more friction. The mean streamwise wall shear is  $1.49 \times 10^{-2}$  Pa for the baseline Kim case and  $1.16 \times 10^{-2}$  Pa with the wall-wet condition. Enforcing liquid at the boundary therefore reduces the mean wall shear in this case.

Changing the wall phase distribution also changes the local interface position and the liquid velocity field. The signed wall shear depends on the resulting velocity gradient, including regions of local reverse flow (recirculation), rather than only on whether liquid is present at the wall. The Kim result therefore argues against treating the wall-wet condition as a correction whose only purpose is to increase the hydraulic resistance.

Wettability remains another uncertainty. The Cubaud experiments involve partial wetting and moving contact lines [11], while the numerical treatment uses prescribed contact-angle values as described in Section 3.4.4. Some disagreement in the detailed wedging morphology can therefore also originate from the wetting description. The wall-wet calculation nevertheless shows that enforcing liquid-wall connectivity alone does not recover the measured pressure-drop trend.

The distinction is useful for the rest of the analysis. The boundary condition can control the numerical wetting topology, but it does not supply the missing momentum transport of a sub-grid film. Obtaining that transport with the present formulation still requires a sufficiently fine wall-normal mesh, which directly affects the computational cost.

## 5.4. Numerical Stability and Computational Cost

The calculations already require restrictive numerical settings before the film is fully resolved. The transient solver setup described in Section 3.3.1 uses small adaptive time steps, and the wall-refined cases add cells precisely in the region that becomes most restrictive as the film thickness decreases.

Refinement increases the cost in two ways. More cells have to be solved, and the smallest wall-normal spacing reduces the admissible global time step. A refinement applied only to a small physical region can therefore increase the cost of the complete transient calculation.

The Cubaud mesh sequence in Section 4.1.4 also reached a stability limit. Refinement beyond the finest reported mesh produced increasing oscillations between the continuous and dispersed gas representations, followed by divergence before a stationary pressure signal was obtained. The origin of these oscillations was not isolated in the present work, so they cannot be assigned to one specific part of the model. They nevertheless prevented further verification of the film-sensitive regime.

A related compromise appears in the morphology-transfer time scale. Section 4.2.4 shows that the smallest tested  $\tau_{PT}$  produces the morphology progression closest to the Lafmejani visualisations. From the transfer formulation in Section 2.3.4, reducing  $\tau_{PT}$  also strengthens the transfer source and makes the coupling more demanding numerically.

The resulting cost issue is directly connected to the limitation identified above. The unresolved gas representation is introduced to avoid resolving small bubbles individually. In the gas-rich Cubaud cases, however, the pressure-drop prediction pushes the wall mesh towards the much smaller liquid-film scale. The calculation then retains the additional dispersed field, its momentum closures and the morphology-transfer operations while also paying an increasingly large cost for near-wall resolution.

No direct wall-clock comparison with a pure VOF calculation was performed, so the present results do not establish which method is cheaper for the same accuracy. They do show that the expected computational advantage of MultiMorph becomes less clear once the wall-film scale controls the mesh. This point has to be considered when transferring the model from the validation channels to an electrolyzer geometry, where both sub-grid gas injection and gas-rich near-wall flow can occur in the same calculation.

## 5.5. Relevance for Electrolyzer Simulations

For electrolyzer applications, the two scale limitations appear at different stages of the same gas–liquid flow. At the PTL–channel interface, newly introduced gas structures can be much smaller than the channel mesh. After accumulation and coalescence, the gas becomes large enough to be represented continuously, but the remaining liquid can then be confined to a film much smaller than the gas structure. [34, 13].

The first situation is where the hybrid formulation is attractive. The Lafmejani and Kim setups described in Sections 3.2.2 and 3.2.3 both represent gas injection through a wall. In the numerical configurations, this gas enters through  $g_D$  and can subsequently be transferred to  $g_C$ . The Lafmejani results show that this approach can generate channel-scale continuous structures without resolving every injected gas structure individually.

The unresolved injection remains an idealisation of the PTL-scale process. A prescribed  $d_b$  replaces the actual distribution and evolution of bubble sizes. The sensitivity found in Section 4.2.3 shows that this assumption can propagate beyond the immediate injection region because it changes the transport of  $g_D$  before transfer to  $g_C$ . Predictive use of the

model would therefore require the dispersed-phase parameters to remain meaningful when the PTL and operating conditions change.

The liquid side of the problem becomes relevant once gas production is large. Kim and Jung report annular flow with a thin liquid film in the single-serpentine configuration, whereas the parallel configuration contains Taylor bubbles whose width is close to the channel width [23]. They associate the annular morphology with the possibility of removing gas while maintaining a liquid path towards the PTL. The near-wall liquid is therefore relevant not only to pressure loss but also to the availability of water close to the active region.

The present results suggest different levels of confidence depending on the quantity of interest. The transition from dispersed gas to large connected structures is reproduced qualitatively in the wall-injected cases, although its location remains parameter-sensitive. Quantitative pressure loss and wall shear are more demanding once the liquid is restricted to an under-resolved film. A gas morphology that looks plausible is not sufficient to establish that these near-wall quantities are correct.

For channel-scale electrolyzer calculations, retaining the unresolved gas description is therefore still useful, particularly for gas injection from the PTL. The missing part identified by the present validation is on the liquid side of the same multiscale problem. Addressing the hydraulic effect of liquid layers below the channel-mesh scale without requiring the present level of wall refinement is the corresponding direction for further model development. This point is taken up in the Outlook in Chapter 6.



## 6. Conclusion and Outlook

This chapter summarizes the main findings of the validation study and outlines the most relevant directions for further development of the MultiMorph model.

### 6.1. Main conclusions

This work assessed the MultiMorph Hybrid multiphase model for confined gas–liquid flows relevant to PEM water electrolysis. Three reference configurations were used for different purposes. The Cubaud and Ho microchannel provided the main quantitative pressure-drop benchmark, Lafmejani was used to examine morphology transfer under distributed wall gas injection, and the Kim configuration tested the model under conditions closer to an operating electrolyzer.

The results show that MultiMorph can reproduce the main changes in gas morphology over a wide range of conditions. Dispersed gas can accumulate and be transferred to the continuous representation, allowing large Taylor-like structures to form without resolving the initially injected bubbles individually. This behaviour was observed particularly clearly in the Lafmejani case. The morphology transition is nevertheless sensitive to the dispersed bubble diameter, the continuous threshold and the morphology-transfer time scale. Since the final Lafmejani parameter set was selected from this sensitivity study, the resulting agreement represents a successful calibration rather than a fully predictive validation of these parameters.

The pressure-drop results give a more restrictive assessment. The single-phase calculations reproduce the expected hydraulic resistance, and the Cubaud two-phase pressure drop is captured well while the liquid still occupies a sufficiently large part of the channel. The agreement deteriorates when elongated gas structures displace most of the liquid towards the walls and corners. Mesh refinement improves the representation of these regions, but the finest stable mesh reaches only approximately one cell across the estimated liquid-film thickness. Further refinement became numerically unstable before a grid-independent film solution could be demonstrated.

The main limitation identified in this work is therefore not the formation of large gas structures. It is the treatment of the much smaller liquid scale that remains between these structures and the wall. MultiMorph provides an unresolved representation for small gas bubbles but no equivalent sub-grid description of a thin wall-connected liquid

film. Consequently, the gas morphology can be represented while the near-wall velocity gradient controlling the hydraulic resistance remains under-resolved.

The modified wall-wet boundary condition confirms this distinction. Forcing liquid at the wall changes the wetting topology and affects both pressure drop and wall shear stress, but it does not prescribe the film thickness or its momentum transport. A boundary condition alone therefore does not remove the need to resolve the near-wall liquid scale.

This limitation also affects the expected computational benefits of the hybrid approach. In the gas-rich regime, obtaining a better pressure-drop prediction required increasingly fine wall resolution and smaller time steps, while the additional dispersed field and morphology-transfer models were still solved. No direct cost comparison with a pure VOF calculation was performed, so a general cost advantage of either approach cannot be concluded. For the cases considered here, however, the expected benefit of MultiMorph becomes less clear once the liquid film scale controls the required mesh.

## 6.2. Future work

The most direct extension of the present model is a dedicated unresolved liquid-film treatment. Refining the complete wall region until several cells are available across a micrometre-scale film conflicts with the multiscale motivation of MultiMorph. The film should instead be represented through a model that retains its hydraulic effect when its thickness is below the computational grid.

Such a model would require at least a local measure of the amount of liquid or film thickness and an equation for its transport along the wall. The resulting film state could then be coupled to the channel-scale momentum equations through an appropriate wall-shear closure. Exchange with resolved liquid regions and with the liquid remaining in the channel corners would also have to be treated consistently.

This would extend the existing MultiMorph concept to the liquid side of the problem. Small gas structures could remain in the Euler–Euler representation, large gas structures could continue to use the interface-resolved field, and thin liquid films would no longer have to be resolved explicitly to retain their effect on wall friction and liquid transport. Validation of such a model against controlled film and pressure drop measurements would be the next step before applying the approach to complete electrolyzer geometries.

# Symbols and Abbreviations

## Latin symbols

| Symbol                     | Description                                                               | Unit                             |
|----------------------------|---------------------------------------------------------------------------|----------------------------------|
| $A$                        | Channel cross-sectional area                                              | $\text{m}^2$                     |
| $A_{\text{act}}$           | Electrochemically active area                                             | $\text{m}^2$                     |
| $A_d$                      | Interfacial area-density measure used in the under-resolved drag model    | $\text{m}^{-1}$                  |
| $A_W$                      | Wall area used for wall-shear averaging                                   | $\text{m}^2$                     |
| $a_i$                      | Interfacial area per unit mixture volume                                  | $\text{m}^{-1}$                  |
| $C_d$                      | Drag-model coefficient                                                    | –                                |
| $C_{\text{sq}}$            | Geometrical coefficient for fully developed flow in a square duct         | –                                |
| $d_b$                      | Characteristic dispersed-bubble diameter                                  | $\text{m}$                       |
| $d_s$                      | Characteristic size of a gas structure                                    | $\text{m}$                       |
| $D_h$                      | Hydraulic diameter                                                        | $\text{m}$                       |
| $\mathbf{e}_x$             | Unit vector in the streamwise direction                                   | –                                |
| $F$                        | Faraday constant                                                          | $\text{C mol}^{-1}$              |
| $\mathcal{F}_{kj}$         | Function controlling morphology transfer from field $j$ to field $k$      | –                                |
| $f_D$                      | Darcy friction factor                                                     | –                                |
| $f_F$                      | Fanning friction factor                                                   | –                                |
| $\mathbf{f}_k^{MT}$        | Momentum-transfer force density acting on field $k$                       | $\text{N m}^{-3}$                |
| $g$                        | Gravitational acceleration                                                | $\text{m s}^{-2}$                |
| $G_l$                      | Local streamwise pressure-gradient contribution acting on the liquid film | $\text{Pa m}^{-1}$               |
| $h$                        | Side length of a square channel                                           | $\text{m}$                       |
| $H$                        | Vertical channel height                                                   | $\text{m}$                       |
| $I$                        | Electrical current                                                        | $\text{A}$                       |
| $I_c$                      | Continuous-structure indicator                                            | –                                |
| $\mathbf{i}_{\alpha\beta}$ | Interface vector between continuous fields $\alpha$ and $\beta$           | $\text{m}^{-1}$                  |
| $j$                        | Electrical current density                                                | $\text{A m}^{-2}$                |
| $J_k$                      | Superficial velocity of phase $k$                                         | $\text{m s}^{-1}$                |
| $J_m$                      | Mixture superficial velocity                                              | $\text{m s}^{-1}$                |
| $K_{\alpha\beta}^D$        | Drag coefficient between fields $\alpha$ and $\beta$                      | $\text{kg m}^{-3} \text{s}^{-1}$ |

| Symbol                 | Description                                                           | Unit                       |
|------------------------|-----------------------------------------------------------------------|----------------------------|
| $\bar{L}$              | Channel or characteristic length                                      | m                          |
| $L_p$                  | Distance between pressure-sampling sections                           | m                          |
| $L_{\alpha\beta}^{*I}$ | Interfacial viscous shear length scale                                | m                          |
| $\mathcal{M}_x$        | Streamwise phase-resolved momentum flux                               | Pa                         |
| $N_b$                  | Number of bubbles                                                     | –                          |
| $N_c$                  | Number of additional cells used by the continuous-structure indicator | –                          |
| $N_f$                  | Number of computational cells across the liquid-film thickness        | –                          |
| $N^I$                  | Maximum number of cells used for interface-information propagation    | –                          |
| $N_p$                  | Number of periods included in the averaging interval                  | –                          |
| $\mathbf{n}$           | Interface unit-normal vector                                          | –                          |
| $\mathbf{n}_{kj}$      | Interface-direction vector including interfacial area density         | $\text{m}^{-1}$            |
| $\mathbf{n}_w$         | Wall-normal unit vector                                               | –                          |
| $p$                    | Pressure                                                              | Pa                         |
| $p_{rgh}$              | Reduced pressure field used in OpenFOAM                               | Pa                         |
| $P_w$                  | Wetted perimeter                                                      | m                          |
| $Q_k$                  | Volumetric flow rate of phase $k$                                     | $\text{m}^3 \text{s}^{-1}$ |
| $R$                    | Hydraulic resistance                                                  | $\text{Pa s m}^{-3}$       |
| $R_t$                  | Tube radius                                                           | m                          |
| $R_1, R_2$             | Principal radii of interface curvature                                | m                          |
| $r_k$                  | Volume fraction of numerical field $k$                                | –                          |
| $r_{C,\text{th}}$      | Continuous-threshold parameter for morphology transfer                | –                          |
| $r_{\text{max},C}$     | Maximum local volume fraction among continuous fields                 | –                          |
| $S$                    | Gas-to-liquid phase slip ratio                                        | –                          |
| $\mathbf{S}_k$         | Strain-rate tensor of field $k$                                       | $\text{s}^{-1}$            |
| $t$                    | Time                                                                  | s                          |
| $\mathbf{u}_k$         | Velocity vector of numerical field $k$                                | $\text{m s}^{-1}$          |
| $U_b$                  | Bulk velocity                                                         | $\text{m s}^{-1}$          |
| $U_c$                  | Characteristic velocity                                               | $\text{m s}^{-1}$          |
| $\bar{U}_g, \bar{U}_l$ | Intrinsic mean gas and liquid velocities                              | $\text{m s}^{-1}$          |
| $U_i$                  | Gas–liquid interfacial velocity                                       | $\text{m s}^{-1}$          |
| $u_{\alpha\beta}^{*I}$ | Interfacial shear velocity                                            | $\text{m s}^{-1}$          |
| $V_m$                  | Measurement or averaging volume                                       | $\text{m}^3$               |
| $x, y$                 | Spatial coordinates                                                   | m                          |
| $y_{\alpha\beta}^{+I}$ | Dimensionless numerical-interface thickness                           | –                          |

## Greek symbols

| Symbol               | Description                                                          | Unit |
|----------------------|----------------------------------------------------------------------|------|
| $\alpha_{G,h}$       | Homogeneous inlet gas flow fraction                                  | –    |
| $\alpha_{L,h}$       | Homogeneous inlet liquid flow fraction                               | –    |
| $\alpha_L$           | Homogeneous liquid fraction used for the Cubaud and Ho configuration | –    |
| $\alpha_{\text{th}}$ | Accumulation threshold for dispersed-to-continuous gas transfer      | –    |

| Symbol                      | Description                                                                 | Unit                             |
|-----------------------------|-----------------------------------------------------------------------------|----------------------------------|
| $\bar{\Gamma}_{kj}^{PT}$    | Mass-transfer source from numerical field $j$ to field $k$                  | $\text{kg m}^{-3} \text{s}^{-1}$ |
| $\delta_f$                  | Physical liquid-film thickness                                              | m                                |
| $\Delta$                    | Local computational mesh size                                               | m                                |
| $\Delta_{\alpha\beta}^I$    | Estimated numerical-interface thickness                                     | m                                |
| $\Delta t$                  | Numerical time step                                                         | s                                |
| $\Delta y_w$                | Characteristic wall-normal size of the first computational cell             | m                                |
| $\Delta p$                  | Pressure difference                                                         | Pa                               |
| $\bar{\epsilon}_g$          | Time-averaged in-situ gas volume fraction                                   | —                                |
| $\bar{\epsilon}_l$          | Time-averaged in-situ liquid volume fraction                                | —                                |
| $\kappa$                    | Interface curvature                                                         | $\text{m}^{-1}$                  |
| $\lambda$                   | Dispersed-to-continuous viscosity ratio, $\mu_d/\mu_c$                      | —                                |
| $\lambda_{\text{st}}$       | Electro-stoichiometric ratio                                                | —                                |
| $\ell_c$                    | Capillary length                                                            | m                                |
| $\mu_k$                     | Dynamic viscosity of phase or field $k$                                     | $\text{Pa s}$                    |
| $\tilde{\nu}_{\alpha\beta}$ | Arithmetic-mean kinematic viscosity of fields $\alpha$ and $\beta$          | $\text{m}^2 \text{s}^{-1}$       |
| $\Pi_{\Delta p}$            | Normalized two-phase pressure drop                                          | —                                |
| $\rho_k$                    | Density of phase or field $k$                                               | $\text{kg m}^{-3}$               |
| $\bar{\rho}_m$              | Mean mixture density                                                        | $\text{kg m}^{-3}$               |
| $\sigma$                    | Gas-liquid surface tension                                                  | $\text{N m}^{-1}$                |
| $\tau_i$                    | Gas-liquid interfacial shear stress                                         | Pa                               |
| $\tau_w$                    | Wall shear stress                                                           | Pa                               |
| $\bar{\tau}_w$              | Mean wall shear stress                                                      | Pa                               |
| $\tau_{\alpha\beta}^I$      | Approximate interfacial shear stress                                        | Pa                               |
| $\tau_{PT}$                 | Morphology-transfer time scale                                              | s                                |
| $\tau_r$                    | Velocity-relaxation time used by the resolving drag model                   | s                                |
| $\Phi_w^2$                  | Two-phase pressure multiplier defined by Kim and Jung                       | —                                |
| $\chi_g$                    | Gas indicator function                                                      | —                                |
| $\psi_{\alpha\beta}^I$      | Interfacial shear-flow orientation indicator                                | —                                |
| $\xi_{\alpha\beta}^{UR}$    | Continuous-interface under-resolution indicator                             | —                                |
| $\theta$                    | Channel inclination angle relative to the horizontal                        | rad                              |
| $T_{\text{avg}}$            | Time interval used for statistical averaging                                | s                                |
| $\Delta U_l$                | Characteristic liquid velocity difference across the near-wall liquid layer | $\text{m s}^{-1}$                |
| $\rho_{\alpha\beta}$        | Mean density used for the interacting field pair $\alpha$ - $\beta$         | $\text{kg m}^{-3}$               |
| $\dot{n}_{\text{O}_2}$      | Molar oxygen production rate                                                | $\text{mols}^{-1}$               |

## Dimensionless numbers

| Symbol        | Description                                              | Definition                               |
|---------------|----------------------------------------------------------|------------------------------------------|
| Bo            | Bond number; ratio of gravitational to capillary effects | $\frac{(\rho_l - \rho_g)gD_h^2}{\sigma}$ |
| Ca            | Capillary number; ratio of viscous to capillary effects  | $\frac{\mu_l U_c}{\sigma}$               |
| $\text{Re}_k$ | Reynolds number of phase $k$                             | $\frac{\rho_k J_k D_h}{\mu_k}$           |

| Symbol | Description                                          | Definition                        |
|--------|------------------------------------------------------|-----------------------------------|
| We     | Weber number; ratio of inertial to capillary effects | $\frac{\rho_l U_c^2 D_h}{\sigma}$ |

## MultiMorph fields and indices

| Notation        | Description                                                                  |
|-----------------|------------------------------------------------------------------------------|
| $g_C$           | Continuous gas field; VOF-like representation of large gas structures        |
| $g_D$           | Dispersed gas field; Euler–Euler representation of unresolved gas structures |
| $l$             | Continuous liquid field                                                      |
| $\mathcal{K}$   | Set of numerical fields, $\mathcal{K} = \{g_C, g_D, l\}$                     |
| $k, j$          | Numerical-field indices                                                      |
| $\alpha, \beta$ | Generic pair of interacting numerical fields                                 |
| $c$             | Continuous donor field                                                       |
| $d$             | Dispersed donor field                                                        |
| $g, l$          | Gas and liquid phases                                                        |
| $w$             | Wall-related quantity                                                        |

## Abbreviations

| Abbreviation | Meaning                              |
|--------------|--------------------------------------|
| CFD          | Computational Fluid Dynamics         |
| CSF          | Continuum Surface Force              |
| DRIE         | Deep Reactive-Ion Etching            |
| HZDR         | Helmholtz-Zentrum Dresden-Rossendorf |
| PEM          | Proton Exchange Membrane             |
| PIV          | Particle Image Velocimetry           |
| PTL          | Porous Transport Layer               |
| SST          | Shear Stress Transport               |
| VOF          | Volume of Fluid                      |
| WSS          | Wall Shear Stress                    |

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# Appendix

## A. Additional Numerical Implementation Details

The main numerical setup is described in Chapter 3. This appendix provides additional implementation details that are useful for reproducing the calculations but are not required for the interpretation of the main results. The information includes the alternative Cubaud mesh constructions, mesh-quality indicators, discretization and solver settings, the wall-shear-stress post-processing, and the implementation of gas injection in the Kim configuration.

### A.1. Cubaud mesh constructions

Two different mesh constructions were used during the Cubaud study. The baseline configuration uses a continuous transition between the cross-shaped mixing region and the downstream test section. The section change is distributed over several cells, as shown in Fig. A.1.

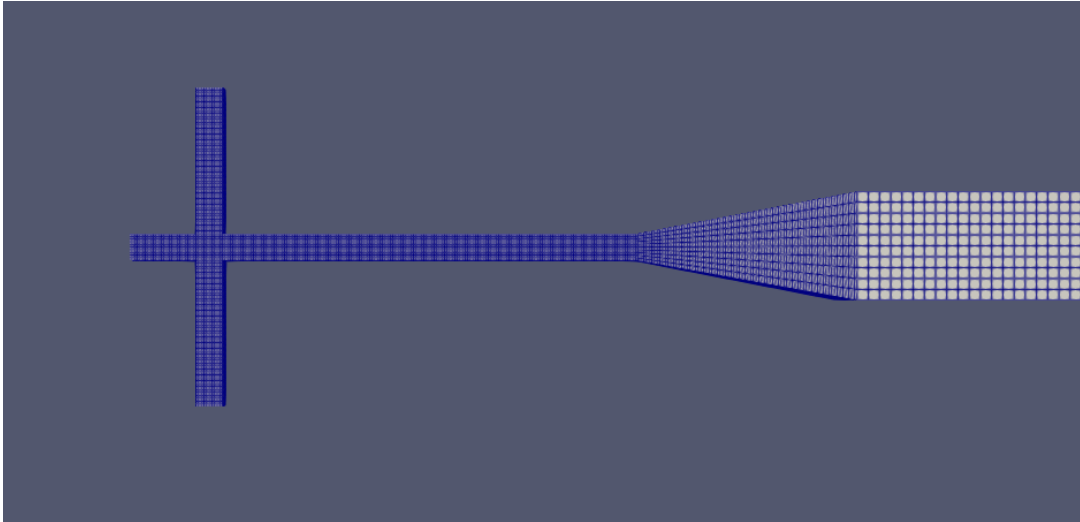


Figure A.1: Cubaud baseline mesh with a continuous transition between the mixing region and the downstream test section.

For the near-wall refinement study, a second mesh construction was used. The continuous transition was replaced by a discontinuous section change, while the downstream

test section was strongly graded towards the walls. The corresponding mesh is shown in Fig. A.2.

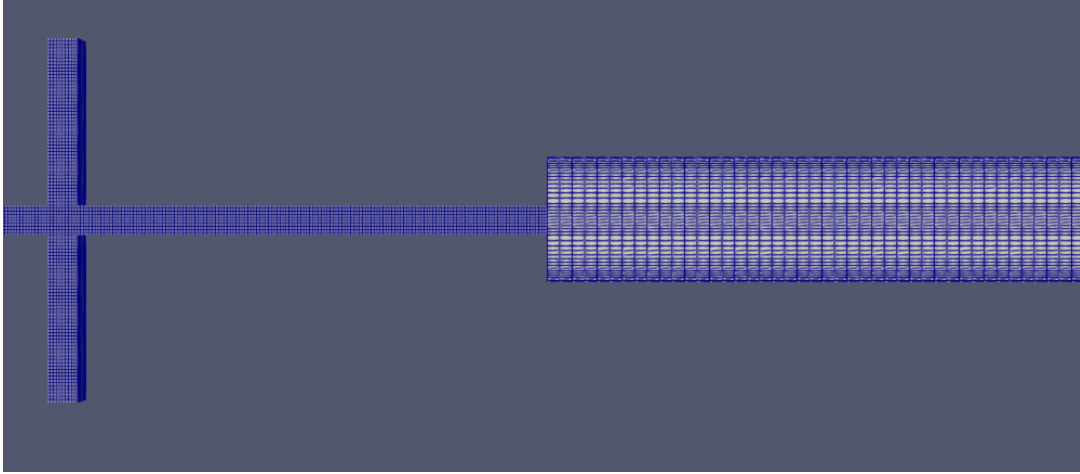


Figure A.2: Cubaud wall-refined mesh with a discontinuous section change and wall-normal grading in the downstream test section.

The second construction allows the smallest cells to be concentrated close to the walls in the region used for the pressure-drop analysis. The fine wall-normal resolution therefore does not have to be extended through the complete mixing section.

The two meshes are not strict geometrically identical refinements of each other because their section transitions are different. The pressure difference is evaluated further downstream, inside the constant-section part of the channel, as described in Section 3.3.2.

## A.2. Mesh-quality assessment

The principal mesh-quality indicators were obtained using `checkMesh`. All four meshes consist exclusively of hexahedral cells and passed the OpenFOAM topology and geometry checks. The main values are listed in Table A.1.

Table A.1: Mesh size and quality indicators for the main numerical configurations.

| Configuration   | Cells   | Max. AR | Max. non-orth. | Mean non-orth. | Max. skew.             |
|-----------------|---------|---------|----------------|----------------|------------------------|
| Cubaud baseline | 56 000  | 4.32    | 13.42°         | 2.56°          | 0.468                  |
| Cubaud refined  | 365 500 | 8.75    | 0°             | 0°             | $1.04 \times 10^{-12}$ |
| Lafmejani       | 45 360  | 2.09    | 0°             | 0°             | $3.34 \times 10^{-13}$ |
| Kim             | 50 000  | 1.00    | 0°             | 0°             | $5.55 \times 10^{-13}$ |

The Kim and Lafmejani meshes are fully orthogonal. The Kim mesh is also uniform, with a maximum aspect ratio of one. The Lafmejani mesh reaches a maximum aspect ratio of 2.09.

The continuous section change of the Cubaud baseline mesh introduces a small amount of non-orthogonality. The maximum value is 13.42°, with an average of 2.56°, while the maximum skewness is 0.468. The complete mesh still passes the `checkMesh` quality checks.

The refined Cubaud construction is orthogonal. Its maximum aspect ratio increases to 8.75 because the cells close to the wall are reduced mainly in the wall-normal direction. This anisotropy is introduced deliberately to increase the resolution of the liquid film without applying the same refinement in the streamwise direction.

The total cell count increases from  $5.6 \times 10^4$  for the baseline Cubaud mesh to  $3.655 \times 10^5$  for the finest stable mesh, corresponding to a factor of approximately 6.5. The smaller wall-normal cells also reduce the admissible time step through the Courant-number constraint. Wall refinement therefore increases the computational cost through both the number of cells and the number of time steps required to reach the same physical simulation time.

### A.3. Numerical discretization and solution settings

#### A.3.1. Spatial and temporal discretization

The discretization schemes used in the calculations are summarized in Table A.2. Time integration uses the Euler scheme. The phase-fraction equations use van Leer convection schemes, while the momentum equations use linear-upwind and limited-linear discretizations.

Table A.2: Main discretization schemes used in the calculations.

| Term                               | Scheme                   |
|------------------------------------|--------------------------|
| Time derivative                    | Euler                    |
| Gradient                           | Gauss linear             |
| Phase-fraction convection          | Gauss vanLeer            |
| Relative phase-fraction convection | Gauss vanLeer            |
| Momentum convection                | Gauss linearUpwind       |
| Velocity-flux convection           | Gauss limitedLinear 1    |
| Turbulence-variable convection     | Gauss linearUpwind       |
| Laplacian terms                    | Gauss linear uncorrected |
| Interpolation                      | linear                   |

The Laplacian discretization is uncorrected. No additional correction for mesh non-orthogonality is therefore applied at this level. Wall distance is calculated using the Poisson method specified in the `wallDist` dictionary.

#### A.3.2. Linear solvers and PIMPLE coupling

The phase fractions and velocity fields are solved with `smoothSolver`, while the pressure equation uses GAMG. The corresponding absolute and relative tolerances are listed in Table A.3.

Table A.3: Linear-solver settings used for the principal numerical fields.

| Field                   | Solver                    | Smoother                    | Tolerance | relTol |
|-------------------------|---------------------------|-----------------------------|-----------|--------|
| <code>alpha.*</code>    | <code>smoothSolver</code> | <code>symGaussSeidel</code> | $10^{-5}$ | 0.01   |
| <code>p_rgh</code>      | GAMG                      | <code>DICGaussSeidel</code> | $10^{-7}$ | 0.01   |
| <code>p_rghFinal</code> | GAMG                      | <code>DICGaussSeidel</code> | $10^{-7}$ | 0      |
| <code>U.*</code>        | <code>smoothSolver</code> | <code>symGaussSeidel</code> | $10^{-7}$ | 0.01   |

For GAMG, the coarsest level is limited to 20 cells. The final pressure solution uses `relTol` = 0, so that the final solve is continued to the specified absolute tolerance.

The phase-fraction and pressure-velocity correction settings are listed in Table A.4.

Two sub-cycles and two correction steps are therefore performed for the phase-fraction equations. The PIMPLE loop uses one outer corrector and two pressure correctors. No additional non-orthogonal pressure-correction step is used.

Table A.4: Phase-fraction and PIMPLE correction settings.

| Setting                               | Value |
|---------------------------------------|-------|
| <code>nAlphaSubCycles</code>          | 2     |
| <code>nAlphaCorr</code>               | 2     |
| <code>splitPhaseFlux</code>           | true  |
| <code>MULESCorr</code>                | yes   |
| <code>nLimiterIter</code>             | 3     |
| <code>nOuterCorrectors</code>         | 1     |
| <code>nCorrectors</code>              | 2     |
| <code>nNonOrthogonalCorrectors</code> | 0     |
| <code>momentumPredictor</code>        | true  |

### A.3.3. Adaptive time-step control

The simulations use an adaptive time step controlled by the Courant number. The corresponding `controlDict` settings are given in Table A.5.

Table A.5: Adaptive time-step settings used in the calculations.

| Setting                     | Value       |
|-----------------------------|-------------|
| Initial $\Delta t$          | $10^{-6}$ s |
| <code>adjustTimeStep</code> | yes         |
| Maximum Courant number      | 0.3         |
| Maximum $\Delta t$          | $10^{-4}$ s |

The specified value of  $\Delta t = 10^{-6}$  s is the initial time step. During the simulation, the time step is adjusted to satisfy

$$Co \leq 0.3.$$

The adaptive procedure is limited by

$$\Delta t_{\max} = 10^{-4} \text{ s}.$$

No lower bound on  $\Delta t$  is prescribed in the `controlDict`. The actual time step can therefore become much smaller when the local velocity or the smallest cells impose a more restrictive Courant condition.

For a characteristic local velocity  $U$  and cell size  $\Delta x$ , the Courant number scales as

$$Co \sim \frac{U \Delta t}{\Delta x}.$$

Reducing the wall-normal cell size consequently reduces the admissible time step when the local velocity remains of the same order. This effect becomes relevant for the wall-refined Cubaud meshes, where the smallest cells are introduced precisely in the near-wall region.

## A.4. Wall-shear-stress post-processing

Wall shear stress was evaluated using a coded OpenFOAM function object. The post-processing reads the velocity and phase-fraction fields of the continuous gas  $g_C$ , dispersed gas  $g_D$ , and liquid  $l$ .

For each numerical field  $k$ , the wall-normal velocity gradient returned by `snGrad()` is first projected onto the plane tangent to the wall. With  $\mathbf{n}$  the wall-normal vector,

$$\mathbf{g}_{k,t} = \frac{\partial \mathbf{u}_k}{\partial n} - \left( \frac{\partial \mathbf{u}_k}{\partial n} \cdot \mathbf{n} \right) \mathbf{n}.$$

The wall-shear vector used for post-processing is then calculated as

$$\boldsymbol{\tau}_{w,\text{num}} = \sum_{k \in \{g_C, g_D, l\}} r_k \mu_k \mathbf{g}_{k,t}.$$

Its magnitude is stored separately as

$$\tau_{w,\text{mag}} = |\boldsymbol{\tau}_{w,\text{num}}|.$$

For the Kim calculations, the dynamic viscosities used by the post-processing function are

$$\mu_{g_C} = \mu_{g_D} = 1.8 \times 10^{-5} \text{ Pa s}, \quad \mu_l = 4.035 \times 10^{-4} \text{ Pa s}.$$

The liquid viscosity is therefore consistent with the water properties used for the Kim calculations at 70°C.

On a wall face for which  $r_l = 1$  and the two gas fractions vanish, the expression reduces to the liquid contribution,

$$\boldsymbol{\tau}_{w,\text{num}} = \mu_l \mathbf{g}_{l,t}.$$

For the standard wetting treatment, gas can locally reach the wall and its viscous contribution is retained through the phase-fraction weighting.

The central part of the implementation is

```
gradAirC  -= (gradAirC  & n) * n;
gradAirD  -= (gradAirD  & n) * n;
gradWater -= (gradWater & n) * n;

vector tAirC  = muAirC  * gradAirC;
vector tAirD  = muAirD  * gradAirD;
vector tWater = muWater * gradWater;

tauPatch[facei] =
    aAirC[facei] * tAirC
  + aAirD[facei] * tAirD
  + aWater[facei] * tWater;

tauMagPatch[facei] = mag(tauPatch[facei]);
```

Both the vector field `wallShearStress` and its magnitude `wallShearStressMag` are written at every selected output time.

### A.5. Kim wall-injection implementation

The numerical Kim configuration does not model the electrochemical reaction directly. Gas production is represented by injection through the `reaction` patch. The dispersed gas velocity is imposed only over the active section

$$0.05 \leq x \leq 0.10 \text{ m.}$$

Outside this interval, the imposed dispersed-gas velocity is zero. Inside the active region, the boundary value is

$$\mathbf{u}_{gD} = (0, 0, 0.0024) \text{ m s}^{-1}.$$

The active wall section has a length of 0.05 m and a width of 0.003 m. Its area is therefore

$$A_{\text{reaction}} = 0.05 \times 0.003 = 1.5 \times 10^{-4} \text{ m}^2.$$

The product of the imposed normal velocity and this area is

$$U_{gD} A_{\text{reaction}} = 0.0024 \times 1.5 \times 10^{-4} = 3.60 \times 10^{-7} \text{ m}^3 \text{ s}^{-1}.$$

This value is close to the target oxygen volumetric flow rate of approximately  $3.64 \times 10^{-7} \text{ m}^3 \text{ s}^{-1}$  derived from the experimental operating condition. The phase flux also depends on the dispersed-gas fraction imposed by the corresponding phase-fraction boundary condition.

The velocity condition is implemented using `codedFixedValue`. The position of the centre of each boundary face is checked before the velocity is assigned:

```
const vectorField& pos = patch().Cf();
vectorField& field = *this;

forAll(field, i)
{
    scalar x = pos[i].x();

    if (x < 5e-2 || x > 10e-2)
    {
        field[i] = vector(0, 0, 0);
    }
    else
    {
        field[i] = vector(0, 0, 0.0024);
    }
}
```

Gas consequently enters the domain through the dispersed field  $g_D$ . The continuous gas field  $g_C$  is not prescribed at this boundary. Continuous gas structures appearing further inside the channel result from the accumulation and morphology-transfer mechanisms of MultiMorph.